

Curriculum Vitae: updated August 30, 2026

I. IDENTIFYING DATA

Name M. Brandon Westover, MD, PhD
Current Position Emily Fisher Landau Professor of Neurology, Harvard Medical School, Beth Israel Deaconess Medical Center (BIDMC)
Current Affiliation Beth Israel Deaconess Medical Center (BIDMC)

II. EDUCATION HISTORY

Colleges and Universities Attended

7/1999 BS in Physics, Brigham Young University
7/2006 PhD, Physics, Washington University in St. Louis
7/2006 MD, Washington University School of Medicine

Residency and Fellowship Training

2006 – 2007 Internship, Barnes Jewish Hospital, Washington University School of Medicine, St. Louis, MO
2007 – 2010 Residency, Partners Neurology Program (Massachusetts General Hospital and Brigham and Women's Hospital), Boston, MA
2010 – 2012 Fellowship, Epilepsy and Clinical Neurophysiology, Massachusetts General Hospital, Boston, MA

Board Certification

2011 American Board of Psychiatry and Neurology (No. 56586)

III. EMPLOYMENT

Academic Appointments:

07/12-05/14 Instructor in Neurology, Harvard Medical School, Boston, MA
05/14-05/18 Assistant Professor in Neurology, Harvard Medical School, Boston, MA
05/18-10/22 Associate Professor
10/22 – present Professor of Neurology, Harvard Medical School, Boston, MA
11/22 – present Emily Fisher Landau Professor of Neurology, Harvard Medical School, Boston, MA

Other Appointments:

2022 to present Director, Critical Care EEG Monitoring Service, BIDMC

IV. HONORS AND AWARDS

2011 Best Paper Award Statistical Literacy (www.statlit.org) Statistical Education
2012 Best Paper Award Association for Computing Machinery: Special Interest Group on Knowledge Discovery and Data Mining (SIGKDD) Basic Research
2014 Rappaport Fellowship Massachusetts General Hospital Nominated award
2018 #1 Most Disruptive AI Innovator Partners Healthcare Partners World Innovation Forum
2018 Innovation Discovery Grant Partners Innovation Office Artificial Intelligence Research / Development
2018 MGH-MIT Grand Challenge Winner ECOR Artificial Intelligence Research
2018 Best Neurology Clinical Research Abstract MGH Neurology Department Clinical Research
2018 Best Paper Award CyberWorlds 2018, Singapore Computer Science

2019	Disruptive Dozen AI Innovator	Massachusetts General Hospital Partners HealthcareWorld Innovation Forum
2019	Innovation Discovery Award (IDG)	Partners Healthcare World Innovation Forum
2019	Innovative Applications in Analytics Award (IAAA)	INFORMS Conference on Business Analytics & Operations ResearchFor achievements featuring novel combinations of descriptive, predictive and prescriptive analytics.
2019	Best Oral Abstract: Translational American Delirium Society (ADS)	ADS annual meeting
2019	Derek Denny-Brown Young Neurological Scholar Award	American Neurological Association
	ANA's Highest award. For outstanding scientific advances toward the prevention, diagnosis, treatment, and cure of neurological diseases	

V. BIBLIOGRAPHY

Peer-reviewed original research (370 total)

1. Eliasmith C, **Westover MB**, Anderson CH. "A general framework for neurobiological modeling: an application to the vestibular system." *Neurocomputing* (Amst). 2002 Jun;44-46:1071-6. doi: 10.1016/s0925-2312(02)00418-6. PMCID: PMC6788744.
2. **Westover MB**, Eliasmith C, Anderson CH. "Linearly decodable functions from neural population codes." *Neurocomputing* (Amst). 2002 Jun;44-46:691-696. doi: 10.1016/s0925-2312(02)00459-9. Epub 2002 Mar 26. PMCID: PMC7062372.
3. Fischer BJ, **Westover MB**. "The neural multiple access channel." *Neurocomputing* (Amst). 2003 Jun;52-54:511-518. doi: 10.1016/s0925-2312(02)00762-2. Epub 2003 Jan 29. PMCID: PMC7062373.
4. **Westover MB**, Anderson CH. "Layer 4C in monkey V1 may linearize the output of the LGN." *Neurocomputing* (Amst). 2003 Jun;52-54:671-676. doi: 10.1016/s0925-2312(02)00863-9. Epub 2003 Jan 28. PMCID: PMC7062374.
5. **Westover MB**, O'Sullivan JA. "Achievable Rates for Pattern Recognition." *IEEE Trans Inf Theory*. 2008 Jan;54(1):299-320. doi: 10.1109/tit.2007.911296. Epub 2008 Jan 04. PMCID: PMC7062371.
6. **Westover MB**. "Asymptotic Geometry of Multiple Hypothesis Testing." *IEEE Trans Inf Theory*. 2008 Jul;54(7):3327-3329. doi: 10.1109/TIT.2008.924656. Epub 2008 Jun 17. PMCID: PMC6788803.
7. Woeltje KF, Butler AM, Goris AJ, Tutlam NT, Doherty JA, **Westover MB**, Ferris V, Bailey TC. "Automated surveillance for central line-associated bloodstream infection in intensive care units." *Infect Control Hosp Epidemiol*. 2008 Sep;29(9):842-6. doi: 10.1086/590261. PMCID: PMC6788749.
8. Mueen A, Keogh E, Zhu Q, Cash S, **Westover B**. "Exact Discovery of Time Series Motifs." *Proc SIAM Int Conf Data Min*. 2009;2009:473-484. doi: 10.1137/1.9781611972795.41. PMCID: PMC6814436.
9. Chu-Shore J, **Westover MB**, Bianchi MT. "Power law versus exponential state transition dynamics: application to sleep-wake architecture." *PLoS One*. 2010 Dec 02;5(12):e14204. doi: 10.1371/journal.pone.0014204. PMCID: PMC2996311.
10. Mueen A, Keogh E, Zhu Q, Cash SS, **Westover MB**, Bigdely-Shamlo N. "A disk-aware algorithm for time series motif discovery." *Data Min Knowl Discov*. 2011 Jan;22(1-2):73-105. doi: 10.1007/s10618-010-0176-8. Epub 2010 Apr 18. PMCID: PMC7062370.
11. Zhang WJ, **Westover MB**, Keary CJ. "Premortem diagnosis of sporadic Creutzfeldt-Jakob disease aided by positron-emission tomography imaging." *AJNR Am J Neuroradiol*. 2011 Jan;32(1):E18. doi: 10.3174/ajnr.A2292. Epub 2010 Nov 11. PMCID: PMC6788750.
12. **Westover MB**, Bianchi MT, Eckman MH, Greenberg SM. "Statin use following intracerebral hemorrhage: a decision analysis." *Arch Neurol*. 2011 May;68(5):573-9. doi: 10.1001/archneurol.2010.356. Epub 2011 Jan 10. PMCID: PMC3158138.
 - Featured article with accompanying editorial: *Arch Neurol*. 2011;68(5):565-566. "Statins After Intracerebral Hemorrhage: To Treat or Not to Treat".
 - Covered in over 150 news feeds, including CBS News, ABC News, US News and World Report, and Medscape Medical News.

13. **Westover MB, Westover KD**, Bianchi MT. "Significance testing as perverse probabilistic reasoning." BMC Med. 2011 Feb 28;9:20. doi: 10.1186/1741-7015-9-20. PMCID: PMC3058025.
14. **Westover KD, Westover MB**, Winer EP, Richardson AL, Iglehart JD, Punglia RS. "Should a sentinel node biopsy be performed in patients with high-risk breast cancer?." Int J Breast Cancer. 2011;2011:973245. doi: 10.4061/2011/973245. Epub 2011 Aug 10. PMCID: PMC3262582.
15. Eiseman NA, **Westover MB**, Mietus JE, Thomas RJ, Bianchi MT. "Classification algorithms for predicting sleepiness and sleep apnea severity." J Sleep Res. 2012 Feb;21(1):101-12. doi: 10.1111/j.1365-2869.2011.00935.x. Epub 2011 Jul 14. PMCID: PMC3698244.
16. Pascual-Leone A, Freitas C, Oberman L, Horvath JC, Halko M, Eldaief M, Bashir S, Vernet M, Shafi M, **Westover B**, Vahabzadeh-Hagh AM, Rotenberg A. "Characterizing brain cortical plasticity and network dynamics across the age-span in health and disease with TMS-EEG and TMS-fMRI." Brain Topogr. 2011 Oct;24(3-4):302-15. doi: 10.1007/s10548-011-0196-8. Epub 2011 Aug 14. PMCID: PMC3374641.
17. Thakur KT, **Westover MB**. "Cerebral infarction due to smoker's polycythemia." BMJ Case Rep. 2011 Oct 28;2011. doi: 10.1136/bcr.08.2011.4714. PMCID: PMC3207785.
18. **Westover MB**, Cormier J, Bianchi MT, Shafi M, Kilbride R, Cole AJ, Cash SS. "Revising the "Rule of Three" for inferring seizure freedom." Epilepsia. 2012 Feb;53(2):368-76. doi: 10.1111/j.1528-1167.2011.03355.x. Epub 2011 Dec 22. PMCID: PMC3267849.
19. **Westover MB**, Eiseman NA, Bianchi MT. "Should risky treatments be reserved for secondary prevention? Theoretical considerations regarding risk-benefit tradeoffs." J Clin Epidemiol. 2012 Aug;65(8):877-86. doi: 10.1016/j.jclinepi.2012.02.011. Epub 2012 May 27. PMCID: PMC6794097.
20. Chu CJ, Kramer MA, Pathmanathan J, Bianchi MT, **Westover MB**, Wison L, Cash SS. "Emergence of stable functional networks in long-term human electroencephalography." J Neurosci. 2012 Feb 22;32(8):2703-13. doi: 10.1523/JNEUROSCI.5669-11.2012. PMCID: PMC3361717.
21. Shafi MM, **Westover MB**, Fox MD, Pascual-Leone A. "Exploration and modulation of brain network interactions with noninvasive brain stimulation in combination with neuroimaging." Eur J Neurosci. 2012 Mar;35(6):805-25. doi: 10.1111/j.1460-9568.2012.08035.x. PMCID: PMC3313459.
22. Rakthanmanon T, Campana B, Mueen A, Batista G, **Westover B**, Zhu Q, Zakaria J, Keogh E. "Searching and Mining Trillions of Time Series Subsequences under Dynamic Time Warping." KDD. 2012 Aug;2012:262-270. doi: 10.1145/2339530.2339576. PMCID: PMC6816304.
 • Best Paper Award at Association for Computing Machinery's annual SIGKDD (Special Interest Group on Knowledge Discovery and Data Mining) meeting, 2012.
23. Eiseman NA, **Westover MB**, Ellenbogen JM, Bianchi MT. "The impact of body posture and sleep stages on sleep apnea severity in adults." J Clin Sleep Med. 2012 Dec 15;8(6):655-66A. doi: 10.5664/jcsm.2258. PMCID: PMC3501662.
24. Van Esbroeck A, **Westover B**. "Data-driven modeling of sleep states from EEG." Annu Int Conf IEEE Eng Med Biol Soc. 2012;2012:5090-3. doi: 10.1109/EMBC.2012.6347138. PMCID: PMC6788787.
25. Shafi MM, **Westover MB**, Cole AJ, Kilbride RD, Hoch DB, Cash SS. "Absence of early epileptiform abnormalities predicts lack of seizures on continuous EEG." Neurology. 2012 Oct 23;79(17):1796-801. doi: 10.1212/WNL.0b013e3182703fbc. Epub 2012 Oct 10. PMCID: PMC3475619.
26. Srinivasan P, **Westover MB**, Bianchi MT. "Propagation of uncertainty in Bayesian diagnostic test interpretation." South Med J. 2012 Sep;105(9):452-9. doi: 10.1097/SMJ.0b013e3182621a2c. PMCID: PMC6785978.
Westover MB, Eiseman NA, Cash SS, Bianchi MT. "Information theoretic quantification of diagnostic uncertainty." Open Med Inform J. 2012;6:36-50. doi: 10.2174/1874431101206010036. Epub 2012 Dec 14. PMCID: PMC3537080.
27. **Westover MB**, Cohen AB. "Reversible vasoconstriction syndrome with bilateral basal ganglia hemorrhages." J Neuroimaging. 2013 Jan;23(1):122-5. doi: 10.1111/j.1552-6569.2011.00645.x. Epub 2011 Sep 07. PMCID: PMC6788746.
28. **Westover MB**, Bianchi MT, Shafi M, Hoch DB, Cole AJ, Chiappa K, Cash SS. "Inferring seizure frequency from brief EEG recordings." J Clin Neurophysiol. 2013 Apr;30(2):174-7. doi: 10.1097/WNP.0b013e3182767c35. PMCID: PMC3616271.

29. Ferguson M, Bianchi MT, Sutter R, Rosenthal ES, Cash SS, Kaplan PW, **Westover MB**. "Calculating the risk benefit equation for aggressive treatment of non-convulsive status epilepticus." *Neurocrit Care*. 2013 Apr;18(2):216-27. doi: 10.1007/s12028-012-9785-y. PMID: PMC3767472.
30. Rakthanmanon T, Campana B, Mueen A, Batista G, **Westover B**, Zhu Q, Zakaria J, Keogh E. "Addressing Big Data Time Series: Mining Trillions of Time Series Subsequences Under Dynamic Time Warping." *ACM Trans Knowl Discov Data*. 2013 Sep;7(3). PMID: PMC6790126.
31. **Westover MB**, Ching S, Shafi MM, Cash SS, Brown EN. "Real-time segmentation and tracking of brain metabolic state in ICU EEG recordings of burst suppression." *Annu Int Conf IEEE Eng Med Biol Soc*. 2013;2013:7108-11. doi: 10.1109/EMBC.2013.6611196. PMID: PMC3939432.
Ching S, Liberman MY, Chemali JJ, **Westover MB**, Kenny JD, Solt K, Purdon PL, Brown EN. "Real-time closed-loop control in a rodent model of medically induced coma using burst suppression." *Anesthesiology*. 2013 Oct;119(4):848-60. doi: 10.1097/ALN.0b013e31829d4ab4. PMID: PMC3857134.
32. **Westover MB**, Bianchi MT, Yang C, Schneider JA, Greenberg SM. "Estimating cerebral microinfarct burden from autopsy samples." *Neurology*. 2013 Apr 09;80(15):1365-9. doi: 10.1212/WNL.0b013e31828c2f52. Epub 2013 Mar 13. PMID: PMC3662273.
33. Featured article with accompanying editorial: *Neurology*. 2013 Apr 9;80(15):1358-9, "Cerebral Microinfarcts: Enumerating the Innumerable".
34. Recognized as work of special significance by Faculty of 1000: Baird A and Adamski M: F1000Prime Recommendation of [Westover MB et al., *Neurology* 2013, 80(15):1365-9]. In F1000Prime, 30 Apr 2013; DOI: 10.3410/f.718000104.793474863. F1000Prime.com/718000104#eval793474863.
 - Brandon **Westover M**, Shafi MM, Ching S, Chemali JJ, Purdon PL, Cash SS, Brown EN. "Real-time segmentation of burst suppression patterns in critical care EEG monitoring." *J Neurosci Methods*. 2013 Sep 30;219(1):131-41. doi: 10.1016/j.jneumeth.2013.07.003. Epub 2013 Jul 23. PMID: PMC3939433.
 - Ng MC, **Westover MB**, Cole AJ. "Treating seizures in Creutzfeldt-Jakob disease." *Epilepsy Behav Case Rep*. 2014;2:75-9. doi: 10.1016/j.ebcr.2014.01.004. Epub 2014 Feb 07. PMID: PMC4308028.
35. Williamson CA, Wahlster S, Shafi MM, **Westover MB**. "Sensitivity of compressed spectral arrays for detecting seizures in acutely ill adults." *Neurocrit Care*. 2014 Feb;20(1):32-9. doi: 10.1007/s12028-013-9912-4. PMID: PMC6794096.
36. Eiseman NA, Bianchi MT, **Westover MB**. "The information theoretic perspective on medical diagnostic inference." *Hosp Pract (1995)*. 2014 Apr;42(2):125-38. doi: 10.3810/hp.2014.04.1110. PMID: PMC6993929.
37. Bai Z, Huang GB, Wang D, Wang H, **Westover MB**. "Sparse extreme learning machine for classification." *IEEE Trans Cybern*. 2014 Oct;44(10):1858-70. doi: 10.1109/TCYB.2014.2298235. PMID: PMC4883115.
38. Shafi MM, Brandon **Westover M**, Oberman L, Cash SS, Pascual-Leone A. "Modulation of EEG functional connectivity networks in subjects undergoing repetitive transcranial magnetic stimulation." *Brain Topogr*. 2014 Jan;27(1):172-91. doi: 10.1007/s10548-013-0277-y. Epub 2013 Mar 08. PMID: PMC4026010.
39. Kheder A, Bianchi MT, **Westover MB**. "Burst suppression in sleep in a routine outpatient EEG." *Epilepsy Behav Case Rep*. 2014;2:71-4. doi: 10.1016/j.ebcr.2014.01.003. Epub 2014 Feb 12. PMID: PMC4308090.
40. Bianchi MT, **Westover MB**. "Insomnia and morning motor vehicle accidents: a decision analysis of the risk of hypnotics versus the risk of untreated insomnia." *J Clin Psychopharmacol*. 2014 Jun;34(3):400-2. doi: 10.1097/JCP.0000000000000134. PMID: PMC6794095.
41. Moura LM, Shafi MM, Ng M, Pati S, Cash SS, Cole AJ, Hoch DB, Rosenthal ES, **Westover MB**. "Spectrogram screening of adult EEGs is sensitive and efficient." *Neurology*. 2014 Jul 01;83(1):56-64. doi: 10.1212/WNL.0000000000000537. Epub 2014 May 23. PMID: PMC4114174.
42. Reviewed in *Epilepsy Currents*, official journal of the American Epilepsy Society: Epilepsy
43. Carlson C. "Can We Screen EEGs More Efficiently? Spectrographic Review of EEG Data." *Epilepsy Curr*. 2015;15(1):24-5. doi: 10.5698/1535-7597-15.1.24. PMID: PMC4320952.
 - Berkowitz AL, **Westover MB**, Bianchi MT, Chou SH. "Aspirin for acute stroke of unknown etiology in

- resource-limited settings: a decision analysis.” *Neurology*. 2014 Aug 26;83(9):787-93. doi: 10.1212/WNL.0000000000000730. Epub 2014 Jul 23. PMID: PMC4155044.
- Gaspard N, Hirsch LJ, LaRoche SM, Hahn CD, **Westover MB**, Critical Care EEG Monitoring Research Consortium. “Interrater agreement for Critical Care EEG Terminology.” *Epilepsia*. 2014 Sep;55(9):1366-73. doi: 10.1111/epi.12653. Epub 2014 Jun 02. PMID: PMC4879939.
44. Akeju O, Pavone KJ, **Westover MB**, Vazquez R, Prerau MJ, Harrell PG, Hartnack KE, Rhee J, Sampson AL, Habeeb K, Gao L, Pierce ET, Walsh JL, Brown EN, Purdon PL. “A comparison of propofol- and dexmedetomidine-induced electroencephalogram dynamics using spectral and coherence analysis.” *Anesthesiology*. 2014 Nov;121(5):978-89. doi: 10.1097/ALN.0000000000000419. PMID: PMC4304638.
 - Bianchi MT, Lipoma T, Darling C, Alameddine Y, **Westover MB**. “Automated sleep apnea quantification based on respiratory movement.” *Int J Med Sci*. 2014;11(8):796-802. doi: 10.7150/ijms.9303. Epub 2014 May 30. PMID: PMC4057486.
 45. Jing Jin, Dauwels J, Cash S, **Westover MB**. “SpikeGUI: software for rapid interictal discharge annotation via template matching and online machine learning.” *Annu Int Conf IEEE Eng Med Biol Soc*. 2014;2014:4435-8. doi: 10.1109/EMBC.2014.6944608. PMID: PMC4416962.
 46. Akeju O, **Westover MB**, Pavone KJ, Sampson AL, Hartnack KE, Brown EN, Purdon PL. “Effects of sevoflurane and propofol on frontal electroencephalogram power and coherence.” *Anesthesiology*. 2014 Nov;121(5):990-8. doi: 10.1097/ALN.0000000000000436. PMID: PMC4206606.
 47. Akama-Garren EH, Bianchi MT, Leveroni C, Cole AJ, Cash SS, **Westover MB**. “Weighing the value of memory loss in the surgical evaluation of left temporal lobe epilepsy: a decision analysis.” *Epilepsia*. 2014 Nov;55(11):1844-53. doi: 10.1111/epi.12790. Epub 2014 Sep 19. PMID: PMC4877127.
 48. Bianchi MT, Hershman S, Bahadoran M, Ferguson M, **Westover MB**. “The challenge of undiagnosed sleep apnea in low-risk populations: a decision analysis.” *Mil Med*. 2014 Aug;179(8 Suppl):47-54. doi: 10.7205/MILMED-D-13-00483. PMID: PMC6788752.
 49. Y W, Mb W, C S, G S, Mt B, Kd W. “Applications of a Capacitor-Based Respiratory Position Sensing Device: Implications for Radiation Therapy.” *Austin J Med Oncol*. 2014;1(2). Epub 2014 Oct 24. PMID: PMC6956860.
 50. **Westover MB**, Shafi MM, Bianchi MT, Moura LM, O'Rourke D, Rosenthal ES, Chu CJ, Donovan S, Hoch DB, Kilbride RD, Cole AJ, Cash SS. “The probability of seizures during EEG monitoring in critically ill adults.” *Clin Neurophysiol*. 2015 Mar;126(3):463-71. doi: 10.1016/j.clinph.2014.05.037. Epub 2014 Jul 11. PMID: PMC4289643.
 51. Featured article with accompanying editorial: “Early epileptiform discharges and the yield of prolonged EEG monitoring.” Dangayach N, Claassen J, *Clinical Neurophysiology* (2014), doi: <http://dx.doi.org/10.1016/j.clinph.2014.07.002>.
 52. Berkowitz AL, **Westover MB**, Bianchi MT, Chou SH. “Aspirin for secondary prevention after stroke of unknown etiology in resource-limited settings.” *Neurology*. 2014 Sep 09;83(11):1004-11. doi: 10.1212/WNL.0000000000000779. Epub 2014 Aug 13. PMID: PMC4162302.
 53. Alvarez V, **Westover MB**, Drislane FW, Dworetzky BA, Curley D, Lee JW, Rossetti AO. “Evaluation of a clinical tool for early etiology identification in status epilepticus.” *Epilepsia*. 2014 Dec;55(12):2059-2068. doi: 10.1111/epi.12852. Epub 2014 Nov 10. PMID: PMC4870016.
 - O'Connor KL, **Westover MB**, Phillips MT, Iftimia NA, Buckley DA, Ogilvy CS, Shafi MM, Rosenthal ES. “High risk for seizures following subarachnoid hemorrhage regardless of referral bias.” *Neurocrit Care*. 2014 Dec;21(3):476-82. doi: 10.1007/s12028-014-9974-y. PMID: PMC4878846.
 54. Kenny JD, **Westover MB**, Ching S, Brown EN, Solt K. “Propofol and sevoflurane induce distinct burst suppression patterns in rats.” *Front Syst Neurosci*. 2014;8:237. doi: 10.3389/fnsys.2014.00237. Epub 2014 Dec 18. PMID: PMC4270179.
 55. Ng MC, Gaspard N, Cole AJ, Hoch DB, Cash SS, Bianchi M, O'Rourke DA, Rosenthal ES, Chu CJ, **Westover MB**. “The standardization debate: A conflation trap in critical care electroencephalography.” *Seizure*. 2015 Jan;24:52-8. doi: 10.1016/j.seizure.2014.09.017. Epub 2014 Oct 16. PMID: PMC4465375.
 56. de Gusmão CM, Berkowitz AL, Hung AY, **Westover MB**. “Cerebrospinal fluid shunt-induced chorea: case report and review of the literature on shunt-related movement disorders.” *Pract Neurol*. 2015

- Feb;15(1):42-4. doi: 10.1136/practneurol-2014-000913. Epub 2014 Jul 04. PMID: PMC4870013.
57. Shafi MM, Vernet M, Klooster D, Chu CJ, Boric K, Barnard ME, Romatoski K, **Westover MB**, Christodoulou JA, Gabrieli JD, Whitfield-Gabrieli S, Pascual-Leone A, Chang BS. "Physiological consequences of abnormal connectivity in a developmental epilepsy." *Ann Neurol*. 2015 Mar;77(3):487-503. doi: 10.1002/ana.24343. Epub 2015 Jan 29. PMID: PMC4394240.
 58. Dranias MR, **Westover MB**, Cash S, VanDongen AM. "Stimulus information stored in lasting active and hidden network states is destroyed by network bursts." *Front Integr Neurosci*. 2015;9:14. doi: 10.3389/fnint.2015.00014. Epub 2015 Feb 23. PMID: PMC4337383.
 59. **Westover MB**, Kim SE, Ching S, Purdon PL, Brown EN. "Robust control of burst suppression for medical coma." *J Neural Eng*. 2015 Aug;12(4):046004. doi: 10.1088/1741-2560/12/4/046004. Epub 2015 May 28. PMID: PMC4517835.
 60. Alvarez V, Drislane FW, **Westover MB**, Dworetzky BA, Lee JW. "Characteristics and role in outcome prediction of continuous EEG after status epilepticus: A prospective observational cohort." *Epilepsia*. 2015 Jun;56(6):933-41. doi: 10.1111/epi.12996. Epub 2015 May 07. PMID: PMC4878827.
 61. Alvarez V, Lee JW, Drislane FW, **Westover MB**, Novy J, Dworetzky BA, Rossetti AO. "Practice variability and efficacy of clonazepam, lorazepam, and midazolam in status epilepticus: A multicenter comparison." *Epilepsia*. 2015 Aug;56(8):1275-85. doi: 10.1111/epi.13056. Epub 2015 Jul 03. PMID: PMC4877129.
 62. Auriel E, **Westover MB**, Bianchi MT, Reijmer Y, Martinez-Ramirez S, Ni J, Van Etten E, Frosch MP, Fotiadis P, Schwab K, Vashkevich A, Boulouis G, Younger AP, Johnson KA, Sperling RA, Hedden T, Gurol ME, Viswanathan A, Greenberg SM. "Estimating Total Cerebral Microinfarct Burden From Diffusion-Weighted Imaging." *Stroke*. 2015 Aug;46(8):2129-35. doi: 10.1161/STROKEAHA.115.009208. Epub 2015 Jul 09. PMID: PMC4519384.
 63. Biswal S, Nip Z, Moura Junior V, Bianchi MT, Rosenthal ES, **Westover MB**. "Automated information extraction from free-text EEG reports." *Annu Int Conf IEEE Eng Med Biol Soc*. 2015;2015:6804-7. doi: 10.1109/EMBC.2015.7319956. PMID: PMC4872711.
 64. Zhou DW, **Westover MB**, McClain LM, Nagaraj SB, Bajwa EK, Quraishi SA, Akeju O, Cobb JP, Purdon PL. "Clustering analysis to identify distinct spectral components of encephalogram burst suppression in critically ill patients." *Annu Int Conf IEEE Eng Med Biol Soc*. 2015;2015:7258-61. doi: 10.1109/EMBC.2015.7320067. PMID: PMC4870011.
 65. An J, Jonnalagadda D, Moura V, Purdon PL, Brown EN, **Westover MB**. "Spatial variation in automated burst suppression detection in pharmacologically induced coma." *Annu Int Conf IEEE Eng Med Biol Soc*. 2015;2015:7430-3. doi: 10.1109/EMBC.2015.7320109. PMID: PMC4876722.
 66. Ghassemi MM, Amorim E, Pati SB, Mark RG, Brown EN, Purdon PL, **Westover MB**. "An enhanced cerebral recovery index for coma prognostication following cardiac arrest." *Annu Int Conf IEEE Eng Med Biol Soc*. 2015;2015:534-7. doi: 10.1109/EMBC.2015.7318417. PMID: PMC4870018.
 67. Struck AF, **Westover MB**. "Variability in clinical assessment of neuroimaging in temporal lobe epilepsy." *Seizure*. 2015 Aug;30:132-5. doi: 10.1016/j.seizure.2015.06.011. Epub 2015 Jul 02. PMID: PMC4887849.
 68. **Westover MB**, Ching S, Kumaraswamy VM, Akeju SO, Pierce E, Cash SS, Kilbride R, Brown EN, Purdon PL. "The human burst suppression electroencephalogram of deep hypothermia." *Clin Neurophysiol*. 2015 Oct;126(10):1901-1914. doi: 10.1016/j.clinph.2014.12.022. Epub 2015 Jan 16. PMID: PMC4504839.
 69. Gaspard N, Foreman BP, Alvarez V, Cabrera Kang C, Probasco JC, Jongeling AC, Meyers E, Espinera A, Haas KF, Schmitt SE, Gerard EE, Gofton T, Kaplan PW, Lee JW, Legros B, Szaflarski JP, **Westover BM**, LaRoche SM, Hirsch LJ, Critical Care EEG Monitoring Research Consortium (CCEMRC). "New-onset refractory status epilepticus: Etiology, clinical features, and outcome." *Neurology*. 2015 Nov 03;85(18):1604-13. doi: 10.1212/WNL.0000000000001940. Epub 2015 Aug 21. PMID: PMC4642147.
 70. Struck AF, Cole AJ, Cash SS, **Westover MB**. "The number of seizures needed in the EMU." *Epilepsia*. 2015 Nov;56(11):1753-9. doi: 10.1111/epi.13090. Epub 2015 Jul 27. PMID: PMC4877132.
 71. Hermans MC, **Westover MB**, van Putten MJAM, Hirsch LJ, Gaspard N. "Quantification of EEG reactivity

- in comatose patients.” Clin Neurophysiol. 2016 Jan;127(1):571-580. doi: 10.1016/j.clinph.2015.06.024. Epub 2015 Jul 02. PMID: PMC4885124.
72. Ustun B, **Westover MB**, Rudin C, Bianchi MT. “Clinical Prediction Models for Sleep Apnea: The Importance of Medical History over Symptoms.” J Clin Sleep Med. 2016 Feb;12(2):161-8. doi: 10.5664/jcsm.5476. PMID: PMC4751423.
 73. Moro M, **Westover MB**, Kelly J, Bianchi MT. “Decision Modeling in Sleep Apnea: The Critical Roles of Pretest Probability, Cost of Untreated Obstructive Sleep Apnea, and Time Horizon.” J Clin Sleep Med. 2016 Mar;12(3):409-18. doi: 10.5664/jcsm.5596. PMID: PMC4773629.
 74. Bagheri E, Jin J, Dauwels J, Cash S, **Westover MB**. “FAST AND EFFICIENT REJECTION OF BACKGROUND WAVEFORMS IN INTERICTAL EEG.” Proc IEEE Int Conf Acoust Speech Signal Process. 2016 Mar;2016:744-748. doi: 10.1109/ICASSP.2016.7471774. Epub 2016 May 19. PMID: PMC5835012.
 75. Johansen AR, Jin J, Maszczyk T, Dauwels J, Cash SS, **Westover MB**. “EPILEPTIFORM SPIKE DETECTION VIA CONVOLUTIONAL NEURAL NETWORKS.” Proc IEEE Int Conf Acoust Speech Signal Process. 2016 Mar;2016:754-758. doi: 10.1109/ICASSP.2016.7471776. Epub 2016 May 19. PMID: PMC5842703.
 76. Thomas J, Jin J, Dauwels J, Cash SS, **Westover MB**. “CLUSTERING OF INTERICTAL SPIKES BY DYNAMIC TIME WARPING AND AFFINITY PROPAGATION.” Proc IEEE Int Conf Acoust Speech Signal Process. 2016 Mar;2016:749-753. doi: 10.1109/ICASSP.2016.7471775. Epub 2016 May 19. PMID: PMC5842698.
 77. Fürbass F, Herta J, Koren J, **Westover MB**, Hartmann MM, Gruber A, Baumgartner C, Kluge T. “Monitoring burst suppression in critically ill patients: Multi-centric evaluation of a novel method.” Clin Neurophysiol. 2016 Apr;127(4):2038-46. doi: 10.1016/j.clinph.2016.02.001. Epub 2016 Feb 09. PMID: PMC4879619.
 78. Jing J, Dauwels J, Rakthanmanon T, Keogh E, Cash SS, **Westover MB**. “Rapid annotation of interictal epileptiform discharges via template matching under Dynamic Time Warping.” J Neurosci Methods. 2016 Dec 01;274:179-190. doi: 10.1016/j.jneumeth.2016.02.025. Epub 2016 Mar 02. PMID: PMC5519352.
 79. Lee JW, LaRoche S, Choi H, Rodriguez Ruiz AA, Fertig E, Politsky JM, Herman ST, Loddenkemper T, Sansevere AJ, Korb PJ, Abend NS, Goldstein JL, Sinha SR, Dombrowski KE, Ritzl EK, **Westover MB**, Gavvala JR, Gerard EE, Schmitt SE, Szaflarski JP, Ding K, Haas KF, Buchsbaum R, Hirsch LJ, Wusthoff CJ, Hopp JL, Hahn CD, Critical Care EEG Monitoring Research Consortium (CCEMRC). “Development and Feasibility Testing of a Critical Care EEG Monitoring Database for Standardized Clinical Reporting and Multicenter Collaborative Research.” J Clin Neurophysiol. 2016 Apr;33(2):133-40. doi: 10.1097/WNP.0000000000000230. PMID: PMC4878836.
 80. O'Rourke D, Chen PM, Gaspard N, Foreman B, McClain L, Karakis I, Mahulikar A, **Westover MB**. “Response Rates to Anticonvulsant Trials in Patients with Triphasic-Wave EEG Patterns of Uncertain Significance.” Neurocrit Care. 2016 Apr;24(2):233-9. doi: 10.1007/s12028-015-0151-8. PMID: PMC4870012.
 81. Struck AF, **Westover MB**, Hall LT, Deck GM, Cole AJ, Rosenthal ES. “Metabolic Correlates of the Ictal-Interictal Continuum: FDG-PET During Continuous EEG.” Neurocrit Care. 2016 Jun;24(3):324-31. doi: 10.1007/s12028-016-0245-y. PMID: PMC5478419.
 82. Haider HA, Esteller R, Hahn CD, **Westover MB**, Halford JJ, Lee JW, Shafi MM, Gaspard N, Herman ST, Gerard EE, Hirsch LJ, Ehrenberg JA, LaRoche SM, Critical Care EEG Monitoring Research Consortium. “Sensitivity of quantitative EEG for seizure identification in the intensive care unit.” Neurology. 2016 Aug 30;87(9):935-44. doi: 10.1212/WNL.0000000000003034. Epub 2016 Jul 27. PMID: PMC5035158.
 83. Nagaraj SB, Ramaswamy SM, Biswal S, Boyle EJ, Zhou DW, McClain LM, Rosenthal ES, Purdon PL, **Westover MB**. “Heart rate variability as a biomarker for sedation depth estimation in ICU patients.” Annu Int Conf IEEE Eng Med Biol Soc. 2016 Aug;2016:6397-6400. doi: 10.1109/EMBC.2016.7592192. PMID: PMC5478422.
 84. Zafar SF, **Westover MB**, Gaspard N, Gilmore EJ, Foreman BP, O'Connor KL, Rosenthal ES. “Interrater Agreement for Consensus Definitions of Delayed Ischemic Events After Aneurysmal Subarachnoid

- Hemorrhage.” *J Clin Neurophysiol*. 2016 Jun;33(3):235-40. doi: 10.1097/WNP.0000000000000276. PMID: PMC4894325.
85. Wickerling E, Gaspard N, Zafar S, Moura VJ, Biswal S, Bechek S, O'Connor K, Rosenthal ES, **Westover MB**. “Automation of Classical QEEG Trending Methods for Early Detection of Delayed Cerebral Ischemia: More Work to Do.” *J Clin Neurophysiol*. 2016 Jun;33(3):227-34. doi: 10.1097/WNP.0000000000000278. PMID: PMC4894333.
 86. Muniz CF, Shenoy AV, O'Connor KL, Bechek SC, Boyle EJ, Guanci MM, Tehan TM, Zafar SF, Cole AJ, Patel AB, **Westover MB**, Rosenthal ES. “Clinical Development and Implementation of an Institutional Guideline for Prospective EEG Monitoring and Reporting of Delayed Cerebral Ischemia.” *J Clin Neurophysiol*. 2016 Jun;33(3):217-26. doi: 10.1097/WNP.0000000000000281. PMID: PMC4907266.
 87. Hotan GC, Struck AF, Bianchi MT, Eskandar EN, Cole AJ, **Westover MB**. “Decision analysis of intracranial monitoring in non-lesional epilepsy.” *Seizure*. 2016 Aug;40:59-70. doi: 10.1016/j.seizure.2016.06.010. Epub 2016 Jun 18. PMID: PMC4967015.
 88. Nagaraj SB, McClain LM, Zhou DW, Biswal S, Rosenthal ES, Purdon PL, **Westover MB**. “Automatic Classification of Sedation Levels in ICU Patients Using Heart Rate Variability.” *Crit Care Med*. 2016 Sep;44(9):e782-9. doi: 10.1097/CCM.0000000000001708. PMID: PMC4987179.
 89. Alvarez V, Lee JW, **Westover MB**, Drislane FW, Novy J, Faouzi M, Marchi NA, Dworetzky BA, Rossetti AO. “Therapeutic coma for status epilepticus: Differing practices in a prospective multicenter study.” *Neurology*. 2016 Oct 18;87(16):1650-1659. doi: 10.1212/WNL.0000000000003224. Epub 2016 Sep 24. PMID: PMC5085074.
 90. Nelson S, Edlow BL, Wu O, Rosenthal ES, **Westover MB**, Rordorf G. “Default Mode Network Perfusion in Aneurysmal Subarachnoid Hemorrhage.” *Neurocrit Care*. 2016 Oct;25(2):237-42. doi: 10.1007/s12028-016-0244-z. PMID: PMC4958026.
 91. Amorim E, Rittenberger JC, Zheng JJ, **Westover MB**, Baldwin ME, Callaway CW, Popescu A, Post Cardiac Arrest Service. “Continuous EEG monitoring enhances multimodal outcome prediction in hypoxic-ischemic brain injury.” *Resuscitation*. 2016 Dec;109:121-126. doi: 10.1016/j.resuscitation.2016.08.012. Epub 2016 Aug 21. PMID: PMC5124407.
 92. Kim JA, Rosenthal ES, Biswal S, Zafar S, Shenoy AV, O'Connor KL, Bechek SC, Valdery Moura J, Shafi MM, Patel AB, Cash SS, **Westover MB**. “Epileptiform abnormalities predict delayed cerebral ischemia in subarachnoid hemorrhage.” *Clin Neurophysiol*. 2017 Jun;128(6):1091-1099. doi: 10.1016/j.clinph.2017.01.016. Epub 2017 Jan 29. PMID: PMC5476529.
 93. Sinha N, Dauwels J, Kaiser M, Cash SS, Brandon **Westover M**, Wang Y, Taylor PN. “Predicting neurosurgical outcomes in focal epilepsy patients using computational modelling.” *Brain*. 2017 Feb;140(2):319-332. doi: 10.1093/brain/aww299. Epub 2016 Dec 23. PMID: PMC5278304.
 94. Moura LM, Carneiro TS, Kwasnik D, Moura VF, Blodgett CS, Cohen J, McKenna Guanci M, Hoch DB, Hsu J, Cole AJ, **Westover MB**. “cEEG electrode-related pressure ulcers in acutely hospitalized patients.” *Neurol Clin Pract*. 2017 Feb;7(1):15-25. doi: 10.1212/CPJ.0000000000000312. PMID: PMC5310208.
 95. Bargiela D, Bianchi MT, **Westover MB**, Chibnik LB, Healy BC, De Jager PL, Xia Z. “Selection of first-line therapy in multiple sclerosis using risk-benefit decision analysis.” *Neurology*. 2017 Feb 14;88(7):677-684. doi: 10.1212/WNL.00000000000003612. Epub 2017 Jan 13. PMID: PMC5317380.
 96. Kempfle JS, BuSaba NY, Dobrowski JM, **Westover MB**, Bianchi MT. “A cost-effectiveness analysis of nasal surgery to increase continuous positive airway pressure adherence in sleep apnea patients with nasal obstruction.” *Laryngoscope*. 2017 Apr;127(4):977-983. doi: 10.1002/lary.26257. Epub 2016 Sep 22. PMID: PMC5483184.
 97. Dreier JP, Fabricius M, Ayata C, Sakowitz OW, Shuttleworth CW, Dohmen C, Graf R, Vajkoczy P, Helbok R, Suzuki M, Schiefecker AJ, Major S, Winkler MK, Kang EJ, Milakara D, Oliveira-Ferreira AI, Reiffurth C, Revankar GS, Sugimoto K, Dengler NF, Hecht N, Foreman B, Feyen B, Kondziella D, Friberg CK, Piilgaard H, Rosenthal ES, **Westover MB**, Maslarova A, Santos E, Hertle D, Sánchez-Porras R, Jewell SL, Balança B, Platz J, Hinzman JM, Lückl J, Schoknecht K, Schöll M, Drenckhahn C, Feuerstein D, Eriksen N, Horst V, Bretz JS, Jahnke P, Scheel M, Bohner G, Rostrup E, Pakkenberg B, Heinemann U, Claassen J, Carlson AP, Kowoll CM, Lublinsky S, Chassidim Y, Shelef I, Friedman A,

- Brinker G, Reiner M, Kirov SA, Andrew RD, Farkas E, Güresir E, Vatter H, Chung LS, Brennan KC, Lieutaud T, Marinesco S, Maas AI, Sahuquillo J, Dahlem MA, Richter F, Herreras O, Boutelle MG, Okonkwo DO, Bullock MR, Witte OW, Martus P, van den Maagdenberg AM, Ferrari MD, Dijkhuizen RM, Shutter LA, Andaluz N, Schulte AP, MacVicar B, Watanabe T, Woitzik J, Lauritzen M, Strong AJ, Hartings JA. "Recording, analysis, and interpretation of spreading depolarizations in neurointensive care: Review and recommendations of the COSBID research group." *J Cereb Blood Flow Metab.* 2017 May;37(5):1595-1625. doi: 10.1177/0271678X16654496. Epub 2016 Jan 01. PMID: PMC5435289.
98. Amorim E, Williamson CA, Moura LMVR, Shafi MM, Gaspard N, Rosenthal ES, Guanci MM, Rajajee V, **Westover MB**. "Performance of Spectrogram-Based Seizure Identification of Adult EEGs by Critical Care Nurses and Neurophysiologists." *J Clin Neurophysiol.* 2017 Jul;34(4):359-364. doi: 10.1097/WNP.0000000000000368. PMID: PMC5482787.
99. Moura LM, **Westover MB**, Kwasnik D, Cole AJ, Hsu J. "Causal inference as an emerging statistical approach in neurology: an example for epilepsy in the elderly." *Clin Epidemiol.* 2017;9:9-18. doi: 10.2147/CLEP.S121023. Epub 2016 Dec 30. PMID: PMC5221551.
100. Jingzhi An, Purdon PL, Solt K, Sims NM, Brown EN, **Westover MB**. "Design, implementation, and evaluation of a physiological closed-loop control device for medically-induced coma." *Annu Int Conf IEEE Eng Med Biol Soc.* 2017 Jul;2017:4313-4316. doi: 10.1109/EMBC.2017.8037810. PMID: PMC5835010.
101. Nagaraj SB, Biswal S, Boyle EJ, Zhou DW, McClain LM, Bajwa EK, Quraishi SA, Akeju O, Barbieri R, Purdon PL, **Westover MB**. "Patient-Specific Classification of ICU Sedation Levels From Heart Rate Variability." *Crit Care Med.* 2017 Jul;45(7):e683-e690. doi: 10.1097/CCM.0000000000002364. PMID: PMC5474145.
- Thomas J, Jing Jin, Dauwels J, Cash SS, **Westover MB**. "Automated epileptiform spike detection via affinity propagation-based template matching." *Annu Int Conf IEEE Eng Med Biol Soc.* 2017 Jul;2017:3057-3060. doi: 10.1109/EMBC.2017.8037502. PMID: PMC5835014.
102. Struck AF, Osman G, Rampal N, Biswal S, Legros B, Hirsch LJ, **Westover MB**, Gaspard N. "Time-dependent risk of seizures in critically ill patients on continuous electroencephalogram." *Ann Neurol.* 2017 Aug;82(2):177-185. doi: 10.1002/ana.24985. Epub 2017 Jul 19. PMID: PMC5842678.
103. Bagheri E, Dauwels J, Dean BC, Waters CG, **Westover MB**, Halford JJ. "Interictal epileptiform discharge characteristics underlying expert interrater agreement." *Clin Neurophysiol.* 2017 Oct;128(10):1994-2005. doi: 10.1016/j.clinph.2017.06.252. Epub 2017 Jul 18. PMID: PMC5842710.
104. Pati S, McClain L, Moura L, Fan Y, **Westover MB**. "Accuracy of Limited-Montage Electroencephalography in Monitoring Postanoxic Comatose Patients." *Clin EEG Neurosci.* 2017 Nov;48(6):422-427. doi: 10.1177/1550059417715389. Epub 2017 Jun 22. PMID: PMC5835011.
105. Romero K, Goparaju B, Russo K, **Westover MB**, Bianchi MT. "Alternative remedies for insomnia: a proposed method for personalized therapeutic trials." *Nat Sci Sleep.* 2017;9:97-108. doi: 10.2147/NSS.S128095. Epub 2017 Mar 17. PMID: PMC5364017.
106. Sun H, Jia J, Goparaju B, Huang GB, Sourina O, Bianchi MT, **Westover MB**. "Large-Scale Automated Sleep Staging." *Sleep.* 2017 Oct 01;40(10). doi: 10.1093/sleep/zsx139. PMID: PMC6251659.
107. Struck AF, Ustun B, Ruiz AR, Lee JW, LaRoche SM, Hirsch LJ, Gilmore EJ, Vlachy J, Haider HA, Rudin C, **Westover MB**. "Association of an Electroencephalography-Based Risk Score With Seizure Probability in Hospitalized Patients." *JAMA Neurol.* 2017 Dec 01;74(12):1419-1424. doi: 10.1001/jamaneurol.2017.2459. PMID: PMC5822188.
108. Rebergen DJ, Nagaraj SB, Rosenthal ES, Bianchi MT, van Putten MJAM, **Westover MB**. "ADARRI: a novel method to detect spurious R-peaks in the electrocardiogram for heart rate variability analysis in the intensive care unit." *J Clin Monit Comput.* 2018 Feb;32(1):53-61. doi: 10.1007/s10877-017-9999-9. Epub 2017 Feb 16. PMID: PMC5559344.
109. Bagheri E, Jin J, Dauwels J, Cash S, **Westover MB**. "CLASSIFIER CASCADE TO AID IN DETECTION OF EPILEPTIFORM TRANSIENTS IN INTERICTAL EEG." *Proc IEEE Int Conf Acoust Speech Signal Process.* 2018 Apr;2018:970-974. doi: 10.1109/ICASSP.2018.8461992. Epub 2018 Sep 13. PMID: PMC6775762.

110. Shelton KT, Qu J, Bilotta F, Brown EN, Cudemus G, D'Alessandro DA, Deng H, DiBiasio A, Gitlin JA, Hahm EY, Hobbs LE, Houle TT, Ibalá R, Loggia ML, Pavone KJ, Shaefi S, Tolis G, **Westover MB**, Akeju O. "Minimizing ICU Neurological Dysfunction with Dexmedetomidine-induced Sleep (MINDDS): protocol for a randomised, double-blind, parallel-arm, placebo-controlled trial." *BMJ Open*. 2018 Apr 20;8(4):e020316. doi: 10.1136/bmjopen-2017-020316. PMID: PMC5914725.
111. Kim JA, Boyle EJ, Wu AC, Cole AJ, Staley KJ, Zafar S, Cash SS, **Westover MB**. "Epileptiform activity in traumatic brain injury predicts post-traumatic epilepsy." *Ann Neurol*. 2018 Apr;83(4):858-862. doi: 10.1002/ana.25211. Epub 2018 Apr 10. PMID: PMC5912971.
112. Zafar SF, Postma EN, Biswal S, Fleuren L, Boyle EJ, Bechek S, O'Connor K, Shenoy A, Jonnalagadda D, Kim J, Shafi MS, Patel AB, Rosenthal ES, **Westover MB**. "Electronic Health Data Predict Outcomes After Aneurysmal Subarachnoid Hemorrhage." *Neurocrit Care*. 2018 Apr;28(2):184-193. doi: 10.1007/s12028-017-0466-8. PMID: PMC5886829.
113. Rosenthal ES, Biswal S, Zafar SF, O'Connor KL, Bechek S, Shenoy AV, Boyle EJ, Shafi MM, Gilmore EJ, Foreman BP, Gaspard N, Leslie-Mazwi TM, Rosand J, Hoch DB, Ayata C, Cash SS, Cole AJ, Patel AB, **Westover MB**. "Continuous electroencephalography predicts delayed cerebral ischemia after subarachnoid hemorrhage: A prospective study of diagnostic accuracy." *Ann Neurol*. 2018 May;83(5):958-969. doi: 10.1002/ana.25232. Epub 2018 May 16. PMID: PMC6021198.
114. **Westover AM**, Shapiro D, **Westover MB**, Bianchi MT. "Rule of 100: a litmus test for informationless diagnostic tests." *Postgrad Med J*. 2018 Jun;94(1112):364-366. doi: 10.1136/postgradmedj-2018-135555. Epub 2018 Mar 07. PMID: PMC6771257.
115. Thomas J, Comoretto L, Jin J, Dauwels J, Cash SS, **Westover MB**. "EEG Classification Via Convolutional Neural Network-Based Interictal Epileptiform Event Detection." *Annu Int Conf IEEE Eng Med Biol Soc*. 2018 Jul;2018:3148-3151. doi: 10.1109/EMBC.2018.8512930. PMID: PMC6775768.
116. Wasterlain CG, Stöhr T, Matagne A. "The acute and chronic effects of the novel anticonvulsant lacosamide in an experimental model of status epilepticus." *Epilepsy Res*. 2011 Mar;94(1-2):10-7. doi: 10.1016/j.epilepsyres.2010.12.014. Epub 2011 Jan 28. PMID: PMC3055953.
117. Sun H, Nagaraj SB, Akeju O, Purdon PL, **Westover BM**. "Brain Monitoring of Sedation in the Intensive Care Unit Using a Recurrent Neural Network." *Annu Int Conf IEEE Eng Med Biol Soc*. 2018 Jul;2018:1-4. doi: 10.1109/EMBC.2018.8513185. PMID: PMC6775766.
118. Herlopian A, Dietrich J, Abramson JS, Cole AJ, **Westover MB**. "EEG findings in CAR T-cell therapy-related encephalopathy." *Neurology*. 2018 Jul 31;91(5):227-229. doi: 10.1212/WNL.0000000000005910. Epub 2018 Jun 29. PMID: PMC6093761.
119. Song J, Sun H, Jing J, Carlos L, Chao L, Cash SS, Zhang R, **Westover MB**. "A Mean Field Model of Acute Hepatic Encephalopathy." *Annu Int Conf IEEE Eng Med Biol Soc*. 2018 Jul;2018:2366-2369. doi: 10.1109/EMBC.2018.8512786. PMID: PMC7088433.
120. Jing J, d'Angremont E, Zafar S, Rosenthal ES, Tabaeizadeh M, Ebrahim S, Dauwels J, **Westover MB**. "Rapid Annotation of Seizures and Interictal-ictal Continuum EEG Patterns." *Annu Int Conf IEEE Eng Med Biol Soc*. 2018 Jul;2018:3394-3397. doi: 10.1109/EMBC.2018.8513059. PMID: PMC6776236.
121. Shenoy ES, Rosenthal ES, Shao YP, Biswal S, Ghanta M, Ryan EE, Suslak D, Swanson N, Junior VM, Hooper DC, **Westover MB**. "Real-Time, Automated Detection of Ventilator-Associated Events: Avoiding Missed Detections, Misclassifications, and False Detections Due to Human Error." *Infect Control Hosp Epidemiol*. 2018 Jul;39(7):826-833. doi: 10.1017/ice.2018.97. Epub 2018 May 17. PMID: PMC6776240.
122. Halford JJ, **Westover MB**, LaRoche SM, Macken MP, Kutluay E, Edwards JC, Bonilha L, Kalamangalam GP, Ding K, Hopp JL, Arain A, Dawson RA, Martz GU, Wolf BJ, Waters CG, Dean BC. "Interictal Epileptiform Discharge Detection in EEG in Different Practice Settings." *J Clin Neurophysiol*. 2018 Sep;35(5):375-380. doi: 10.1097/WNP.0000000000000492. PMID: PMC6126936.
123. Zafar SF, Postma EN, Biswal S, Boyle EJ, Bechek S, O'Connor K, Shenoy A, Kim J, Shafi MS, Patel AB, Rosenthal ES, **Westover MB**. "Effect of epileptiform abnormality burden on neurologic outcome and antiepileptic drug management after subarachnoid hemorrhage." *Clin Neurophysiol*. 2018 Nov;129(11):2219-2227. doi: 10.1016/j.clinph.2018.08.015. Epub 2018 Sep 01. PMID: PMC6478499.

124. Bao EL, Chao LY, Ni P, Moura LMVR, Cole AJ, Cash SS, Hoch DB, Bianchi MT, **Westover MB**. "Antiepileptic drug treatment after an unprovoked first seizure: A decision analysis." *Neurology*. 2018 Oct 09;91(15):e1429-e1439. doi: 10.1212/WNL.0000000000006319. Epub 2018 Sep 12. PMID: PMC6177278.
125. Amorim E, Gilmore EJ, Abend NS, Hahn CD, Gaspard N, Herman ST, Hirsch LJ, Lee JW, Cash SS, **Westover MB**. "EEG Reactivity Evaluation Practices for Adult and Pediatric Hypoxic-Ischemic Coma Prognostication in North America." *J Clin Neurophysiol*. 2018 Nov;35(6):510-514. doi: 10.1097/WNP.000000000000517. PMID: PMC6424574.
126. An J, Jonnalagadda D, Moura V, Purdon PL, Brown EN, **Westover MB**. "Variability in pharmacologically-induced coma for treatment of refractory status epilepticus." *PLoS One*. 2018;13(10):e0205789. doi: 10.1371/journal.pone.0205789. Epub 2018 Oct 31. PMID: PMC6209214.
127. Osman G, Rahangdale R, Britton JW, Gilmore EJ, Haider HA, Hantus S, Herlopian A, Hocker SE, Woo Lee J, Legros B, Mendoza M, Punia V, Rampal N, Szaflarski JP, Wallace AD, **Westover MB**, Hirsch LJ, Gaspard N. "Bilateral independent periodic discharges are associated with electrographic seizures and poor outcome: A case-control study." *Clin Neurophysiol*. 2018 Nov;129(11):2284-2289. doi: 10.1016/j.clinph.2018.07.025. Epub 2018 Aug 29. PMID: PMC6785981.
128. An S, Malhotra K, Dilley C, Han-Burgess E, Valdez JN, Robertson J, Clark C, **Westover MB**, Sun J. "Predicting drug-resistant epilepsy - A machine learning approach based on administrative claims data." *Epilepsy Behav*. 2018 Dec;89:118-125. doi: 10.1016/j.yebeh.2018.10.013. Epub 2018 Nov 07. PMID: PMC6461470.
129. Herlopian A, Struck AF, Rosenthal E, **Westover BM**. "Neuroimaging Correlates of Periodic Discharges." *J Clin Neurophysiol*. 2018 Jul;35(4):279-294. doi: 10.1097/WNP.0000000000000466. PMID: PMC9371611.
130. Amorim E, Ghassemi MM, Lee JW, Greer DM, Kaplan PW, Cole AJ, Cash SS, Bianchi MT, **Westover MB**. "Estimating the False Positive Rate of Absent Somatosensory Evoked Potentials in Cardiac Arrest Prognostication." *Crit Care Med*. 2018 Dec;46(12):e1213-e1221. doi: 10.1097/CCM.0000000000003436. PMID: PMC6424571.
131. Biswal S, Sun H, Goparaju B, **Westover MB**, Sun J, Bianchi MT. "Expert-level sleep scoring with deep neural networks." *J Am Med Inform Assoc*. 2018 Dec 01;25(12):1643-1650. doi: 10.1093/jamia/ocy131. PMID: PMC6289549.
132. Fadayomi AB, Ibala R, Bilotta F, **Westover MB**, Akeju O. "A Systematic Review and Meta-Analysis Examining the Impact of Sleep Disturbance on Postoperative Delirium." *Crit Care Med*. 2018 Dec;46(12):e1204-e1212. doi: 10.1097/CCM.0000000000003400. PMID: PMC6274586.
133. Nagaraj SB, McClain LM, Boyle EJ, Zhou DW, Ramaswamy SM, Biswal S, Akeju O, Purdon PL, **Westover MB**. "Electroencephalogram Based Detection of Deep Sedation in ICU Patients Using Atomic Decomposition." *IEEE Trans Biomed Eng*. 2018 Dec;65(12):2684-2691. doi: 10.1109/TBME.2018.2813265. Epub 2018 Mar 07. PMID: PMC6424570.
134. Subramaniam T, Jain A, Hall LT, Cole AJ, **Westover MB**, Rosenthal ES, Struck AF. "Lateralized periodic discharges frequency correlates with glucose metabolism." *Neurology*. 2019 Feb 12;92(7):e670-e674. doi: 10.1212/WNL.0000000000006903. Epub 2019 Jan 11. PMID: PMC6382363.
135. Sun H, Paixao L, Oliva JT, Goparaju B, Carvalho DZ, van Leeuwen KG, Akeju O, Thomas RJ, Cash SS, Bianchi MT, **Westover MB**. "Brain age from the electroencephalogram of sleep." *Neurobiol Aging*. 2019 Feb;74:112-120. doi: 10.1016/j.neurobiolaging.2018.10.016. Epub 2018 Oct 19. PMID: PMC6478501.
136. Zhuo Ding J, Mallick R, Carpentier J, McBain K, Gaspard N, Brandon **Westover M**, Fantaneanu TA. "Resident training and interrater agreements using the ACNS critical care EEG terminology." *Seizure*. 2019 Mar;66:76-80. doi: 10.1016/j.seizure.2019.02.013. Epub 2019 Feb 20. PMID: PMC6778405.
137. Song JL, Li Q, Pan M, Zhang B, **Westover MB**, Zhang R. "Seizure tracking of epileptic EEGs using a model-driven approach." *J Neural Eng*. 2020 Jan 06;17(1):016024. doi: 10.1088/1741-2552/ab2409. PMID: PMC6874715.
138. Struck AF, Rodriguez-Ruiz AA, Osman G, Gilmore EJ, Haider HA, Dhakar MB, Schrettnner M, Lee JW,

- Gaspard N, Hirsch LJ, **Westover MB**, Critical Care EEG Monitoring Research Consortium (CCERMRC). "Comparison of machine learning models for seizure prediction in hospitalized patients." *Ann Clin Transl Neurol*. 2019 Jul;6(7):1239-1247. doi: 10.1002/acn3.50817. Epub 2019 Jun 27. PMID: PMC6649418.
139. Ghassemi MM, Amorim E, Alhanai T, Lee JW, Herman ST, Sivaraju A, Gaspard N, Hirsch LJ, Scirica BM, Biswal S, Moura Junior V, Cash SS, Brown EN, Mark RG, **Westover MB**, Critical Care Electroencephalogram Monitoring Research Consortium. "Quantitative Electroencephalogram Trends Predict Recovery in Hypoxic-Ischemic Encephalopathy." *Crit Care Med*. 2019 Oct;47(10):1416-1423. doi: 10.1097/CCM.0000000000003840. PMID: PMC6746597.
 140. Amorim E, van der Stoel M, Nagaraj SB, Ghassemi MM, Jing J, O'Reilly UM, Scirica BM, Lee JW, Cash SS, **Westover MB**. "Quantitative EEG reactivity and machine learning for prognostication in hypoxic-ischemic brain injury." *Clin Neurophysiol*. 2019 Oct;130(10):1908-1916. doi: 10.1016/j.clinph.2019.07.014. Epub 2019 Jul 25. PMID: PMC6751020.
 141. Kimchi EY, Neelagiri A, Whitt W, Sagi AR, Ryan SL, Gadbois G, Groothuysen D, **Westover MB**. "Clinical EEG slowing correlates with delirium severity and predicts poor clinical outcomes." *Neurology*. 2019 Sep 24;93(13):e1260-e1271. doi: 10.1212/WNL.00000000000008164. Epub 2019 Aug 29. PMID: PMC7011865.
 142. Sun H, Kimchi E, Akeju O, Nagaraj SB, McClain LM, Zhou DW, Boyle E, Zheng WL, Ge W, **Westover MB**. "Automated tracking of level of consciousness and delirium in critical illness using deep learning." *NPJ Digit Med*. 2019;2:89. doi: 10.1038/s41746-019-0167-0. Epub 2019 Sep 09. PMID: PMC6733797.
 143. Song JL, Paixao L, Li Q, Li SH, Zhang R, **Westover MB**. "A novel neural computational model of generalized periodic discharges in acute hepatic encephalopathy." *J Comput Neurosci*. 2019 Dec;47(2-3):109-124. doi: 10.1007/s10827-019-00727-3. Epub 2019 Sep 11. PMID: PMC6881550.
 144. Bagheri E, Jin J, Dauwels J, Cash S, **Westover MB**. "A fast machine learning approach to facilitate the detection of interictal epileptiform discharges in the scalp electroencephalogram." *J Neurosci Methods*. 2019 Oct 01;326:108362. doi: 10.1016/j.jneumeth.2019.108362. Epub 2019 Jul 13. PMID: PMC6993930.
 145. Zheng WL, Sun H, Akeju O, **Westover MB**. "Adaptive Sedation Monitoring From EEG in ICU Patients With Online Learning." *IEEE Trans Biomed Eng*. 2020 Jun;67(6):1696-1706. doi: 10.1109/TBME.2019.2943062. Epub 2019 Sep 23. PMID: PMC7085963.
 146. Jing J, Herlopian A, Karakis I, Ng M, Halford JJ, Lam A, Maus D, Chan F, Dolatshahi M, Muniz CF, Chu C, Sacca V, Pathmanathan J, Ge W, Sun H, Dauwels J, Cole AJ, Hoch DB, Cash SS, **Westover MB**. "Interrater Reliability of Experts in Identifying Interictal Epileptiform Discharges in Electroencephalograms." *JAMA Neurol*. 2020 Jan 01;77(1):49-57. doi: 10.1001/jamaneurol.2019.3531. PMID: PMC6806666.
 147. Jing J, Sun H, Kim JA, Herlopian A, Karakis I, Ng M, Halford JJ, Maus D, Chan F, Dolatshahi M, Muniz C, Chu C, Sacca V, Pathmanathan J, Ge W, Dauwels J, Lam A, Cole AJ, Cash SS, **Westover MB**. "Development of Expert-Level Automated Detection of Epileptiform Discharges During Electroencephalogram Interpretation." *JAMA Neurol*. 2020 Jan 01;77(1):103-108. doi: 10.1001/jamaneurol.2019.3485. PMID: PMC6806668.
 148. Abou Jaoude M, Jing J, Sun H, Jacobs CS, Pellerin KR, **Westover MB**, Cash SS, Lam AD. "Detection of mesial temporal lobe epileptiform discharges on intracranial electrodes using deep learning." *Clin Neurophysiol*. 2020 Jan;131(1):133-141. doi: 10.1016/j.clinph.2019.09.031. Epub 2019 Nov 11. PMID: PMC6879011.
 149. Worden LT, Chinappen DM, Stoyell SM, Gold J, Paixao L, Krishnamoorthy K, Kramer MA, **Westover MB**, Chu CJ. "The probability of seizures during continuous EEG monitoring in high-risk neonates." *Epilepsia*. 2019 Dec;60(12):2508-2518. doi: 10.1111/epi.16387. Epub 2019 Nov 19. PMID: PMC7083278.
 150. Reyna MA, Josef CS, Jeter R, Shashikumar SP, **Westover MB**, Nemati S, Clifford GD, Sharma A. "Early Prediction of Sepsis From Clinical Data: The PhysioNet/Computing in Cardiology Challenge 2019." *Crit Care Med*. 2020 Feb;48(2):210-217. doi: 10.1097/CCM.0000000000004145. PMID:

PMC6964870.

151. Moise AM, Karakis I, Herlopian A, Dhakar M, Hirsch LJ, Cotsonis G, LaRoche S, Cabrera Kang CM, **Westover B**, Rodriguez A. "Continuous EEG Findings in Autoimmune Encephalitis." *J Clin Neurophysiol*. 2021 Mar 01;38(2):124-129. doi: 10.1097/WNP.0000000000000654. PMID: PMC7263965.
152. Struck AF, Tabaeizadeh M, Schmitt SE, Ruiz AR, Swisher CB, Subramaniam T, Hernandez C, Kaleem S, Haider HA, Cissé AF, Dhakar MB, Hirsch LJ, Rosenthal ES, Zafar SF, Gaspard N, **Westover MB**. "Assessment of the Validity of the 2HELPS2B Score for Inpatient Seizure Risk Prediction." *JAMA Neurol*. 2020 Apr 01;77(4):500-507. doi: 10.1001/jamaneurol.2019.4656. PMID: PMC6990873.
153. Kashkooli K, Polk SL, Chamadia S, Hahm E, Ethridge B, Gitlin J, Ibalá R, Mekonnen J, Pedemonte J, Murphy JM, Sun H, **Westover MB**, Akeju O. "Drug-Specific Models Improve the Performance of an EEG-based Automated Brain-State Prediction System." *Annu Int Conf IEEE Eng Med Biol Soc*. 2019 Jul;2019:5808-5811. doi: 10.1109/EMBC.2019.8856935. PMID: PMC7077760.
154. Polk SL, Kashkooli K, Nagaraj SB, Chamadia S, Murphy JM, Sun H, **Westover MB**, Barbieri R, Akeju O. "Automatic Detection of General Anesthetic-States using ECG-Derived Autonomic Nervous System Features." *Annu Int Conf IEEE Eng Med Biol Soc*. 2019 Jul;2019:2019-2022. doi: 10.1109/EMBC.2019.8857704. PMID: PMC7077759.
155. Paixao L, Sikka P, Sun H, Jain A, Hogan J, Thomas R, **Westover MB**. "Excess brain age in the sleep electroencephalogram predicts reduced life expectancy." *Neurobiol Aging*. 2020 Apr;88:150-155. doi: 10.1016/j.neurobiolaging.2019.12.015. Epub 2019 Dec 23. PMID: PMC7085452.
156. **Westover MB**, Gururangan K, Markert MS, Blond BN, Lai S, Benard S, Bickel S, Hirsch LJ, Parvizi J. "Diagnostic Value of Electroencephalography with Ten Electrodes in Critically Ill Patients." *Neurocrit Care*. 2020 Oct;33(2):479-490. doi: 10.1007/s12028-019-00911-4. PMID: PMC7416437.
157. Li Q, Song JL, Li SH, **Westover MB**, Zhang R. "Effects of Cholinergic Neuromodulation on Thalamocortical Rhythms During NREM Sleep: A Model Study." *Front Comput Neurosci*. 2019;13:100. doi: 10.3389/fncom.2019.00100. Epub 2020 Jan 23. PMID: PMC6990259.
158. Goldstein CA, Berry RB, Kent DT, Kristo DA, Seixas AA, Redline S, **Westover MB**, Abbasi-Feinberg F, Aurora RN, Carden KA, Kirsch DB, Malhotra RK, Martin JL, Olson EJ, Ramar K, Rosen CL, Rowley JA, Shelgikar AV. "Artificial intelligence in sleep medicine: an American Academy of Sleep Medicine position statement." *J Clin Sleep Med*. 2020 Apr 15;16(4):605-607. doi: 10.5664/jcsm.8288. PMID: PMC7161449.
159. Vink JJT, Klooster DCW, Ozdemir RA, **Westover MB**, Pascual-Leone A, Shafi MM. "EEG Functional Connectivity is a Weak Predictor of Causal Brain Interactions." *Brain Topogr*. 2020 Mar;33(2):221-237. doi: 10.1007/s10548-020-00757-6. Epub 2020 Feb 24. PMID: PMC7191224.
160. Goldstein CA, Berry RB, Kent DT, Kristo DA, Seixas AA, Redline S, **Westover MB**. "Artificial intelligence in sleep medicine: background and implications for clinicians." *J Clin Sleep Med*. 2020 Apr 15;16(4):609-618. doi: 10.5664/jcsm.8388. PMID: PMC7161463.
161. Hogan J, Sun H, Aboul Nour H, Jing J, Tabaeizadeh M, Shoukat M, Javed F, Kassa S, Edhi MM, Bordbar E, Gallagher J, Junior VM, Ghanta M, Shao YP, Akeju O, Cole AJ, Rosenthal ES, Zafar S, **Westover MB**. "Burst Suppression: Causes and Effects on Mortality in Critical Illness." *Neurocrit Care*. 2020 Oct;33(2):565-574. doi: 10.1007/s12028-020-00932-4. PMID: PMC7483190.
162. Elmer J, Coppler PJ, Solanki P, **Westover MB**, Struck AF, Baldwin ME, Kurz MC, Callaway CW. "Sensitivity of Continuous Electroencephalography to Detect Ictal Activity After Cardiac Arrest." *JAMA Netw Open*. 2020 Apr 01;3(4):e203751. doi: 10.1001/jamanetworkopen.2020.3751. PMID: PMC7189220.
163. Lissak IA, Zafar SF, **Westover MB**, Schleicher RL, Kim JA, Leslie-Mazwi T, Stapleton CJ, Patel AB, Kimberly WT, Rosenthal ES. "Soluble ST2 Is Associated With New Epileptiform Abnormalities Following Nontraumatic Subarachnoid Hemorrhage." *Stroke*. 2020 Apr;51(4):1128-1134. doi: 10.1161/STROKEAHA.119.028515. Epub 2020 Mar 11. PMID: PMC7123848.
164. Rubin DB, Angelini B, Shoukat M, Chu CJ, Zafar SF, **Westover MB**, Cash SS, Rosenthal ES. "Electrographic predictors of successful weaning from anaesthetics in refractory status epilepticus." *Brain*.

- 2020 Apr 01;143(4):1143-1157. doi: 10.1093/brain/awaa069. PMCID: PMC7174057.
165. Pedemonte JC, Plummer GS, Chamadia S, Locascio JJ, Hahm E, Ethridge B, Gitlin J, Ibala R, Mekonnen J, Colon KM, **Westover MB**, D'Alessandro DA, Tolis G, Houle T, Shelton KT, Qu J, Akeju O. "Electroencephalogram Burst-suppression during Cardiopulmonary Bypass in Elderly Patients Mediates Postoperative Delirium." *Anesthesiology*. 2020 Aug;133(2):280-292. doi: 10.1097/ALN.0000000000003328. PMCID: PMC7365754.
 166. Kashkooli K, Polk SL, Hahm EY, Murphy J, Ethridge BR, Gitlin J, Ibala R, Mekonnen J, Pedemonte JC, Sun H, **Westover MB**, Barbieri R, Akeju O, Chamadia S. "Improved tracking of sevoflurane anesthetic states with drug-specific machine learning models." *J Neural Eng*. 2020 Aug 04;17(4):046020. doi: 10.1088/1741-2552/ab98da. PMCID: PMC7540939.
 167. Tabaeizadeh M, Aboul Nour H, Shoukat M, Sun H, Jin J, Javed F, Kassa S, Edhi M, Bordbar E, Gallagher J, Moura VJ, Ghanta M, Shao YP, Cole AJ, Rosenthal ES, **Westover MB**, Zafar SF. "Burden of Epileptiform Activity Predicts Discharge Neurologic Outcomes in Severe Acute Ischemic Stroke." *Neurocrit Care*. 2020 Jun;32(3):697-706. doi: 10.1007/s12028-020-00944-0. PMCID: PMC7416505.
 168. Beuchat I, Sivaraju A, Amorim E, Gilmore EJ, Dunet V, Rossetti AO, **Westover MB**, Hsu L, Scirica BM, Silva D, Tang K, Lee JW. "MRI-EEG correlation for outcome prediction in postanoxic myoclonus: A multicenter study." *Neurology*. 2020 Jul 28;95(4):e335-e341. doi: 10.1212/WNL.0000000000009610. Epub 2020 Jun 01. PMCID: PMC7455317.
 169. Amorim E, Mo SS, Palacios S, Ghassemi MM, Weng WH, Cash SS, Bianchi MT, **Westover MB**. "Cost-effectiveness analysis of multimodal prognostication in cardiac arrest with EEG monitoring." *Neurology*. 2020 Aug 04;95(5):e563-e575. doi: 10.1212/WNL.0000000000009916. Epub 2020 Jul 13. PMCID: PMC7455344.
 170. Sun H, Ganglberger W, Panneerselvam E, Leone MJ, Quadri SA, Goparaju B, Tesh RA, Akeju O, Thomas RJ, **Westover MB**. "Sleep staging from electrocardiography and respiration with deep learning." *Sleep*. 2020 Jul 13;43(7). doi: 10.1093/sleep/zsz306. PMCID: PMC7355395.
 171. Amorim E, McGraw CM, **Westover MB**. "A Theoretical Paradigm for Evaluating Risk-Benefit of Status Epilepticus Treatment." *J Clin Neurophysiol*. 2020 Sep;37(5):385-392. doi: 10.1097/WNP.0000000000000753. PMCID: PMC7516305.
 172. Zafar SF, Amorim E, Williamsom CA, Jing J, Gilmore EJ, Haider HA, Swisher C, Struck A, Rosenthal ES, Ng M, Schmitt S, Lee JW, Brandon **Westover M**. "A standardized nomenclature for spectrogram EEG patterns: Inter-rater agreement and correspondence with common intensive care unit EEG patterns." *Clin Neurophysiol*. 2020 Sep;131(9):2298-2306. doi: 10.1016/j.clinph.2020.05.032. Epub 2020 Jun 24. PMCID: PMC7461156.
 173. Foy BH, Carlson JCT, Reinertsen E, Padros I Valls R, Pallares Lopez R, Palanques-Tost E, Mow C, **Westover MB**, Aguirre AD, Higgins JM. "Association of Red Blood Cell Distribution Width With Mortality Risk in Hospitalized Adults With SARS-CoV-2 Infection." *JAMA Netw Open*. 2020 Sep 01;3(9):e2022058. doi: 10.1001/jamanetworkopen.2020.22058. PMCID: PMC7512057.
 174. Prasanth T, Thomas J, Yuvaraj R, Jing J, Cash SS, Chaudhari R, Leng TY, Rathakrishnan R, Rohit S, Saini V, **Westover BM**, Dauwels J. "Deep Learning for Interictal Epileptiform Spike Detection from scalp EEG frequency sub bands." *Annu Int Conf IEEE Eng Med Biol Soc*. 2020 Jul;2020:3703-3706. doi: 10.1109/EMBC44109.2020.9175644. PMCID: PMC7545315.
 175. Ye E, Sun H, Leone MJ, Paixao L, Thomas RJ, Lam AD, **Westover MB**. "Association of Sleep Electroencephalography-Based Brain Age Index With Dementia." *JAMA Netw Open*. 2020 Sep 01;3(9):e2017357. doi: 10.1001/jamanetworkopen.2020.17357. PMCID: PMC7522697.
 176. Sun H, Jain A, Leone MJ, Alabsi HS, Brenner LN, Ye E, Ge W, Shao YP, Boutros CL, Wang R, Tesh RA, Magdamo C, Collens SI, Ganglberger W, Bassett IV, Meigs JB, Kalpathy-Cramer J, Li MD, Chu JT, Dougan ML, Stratton LW, Rosand J, Fischl B, Das S, Mukerji SS, Robbins GK, **Westover MB**. "CoVA: An Acuity Score for Outpatient Screening that Predicts Coronavirus Disease 2019 Prognosis." *J Infect Dis*. 2021 Jan 04;223(1):38-46. doi: 10.1093/infdis/jiaa663. PMCID: PMC7665643.
 177. Thomas J, Jin J, Thangavel P, Bagheri E, Yuvaraj R, Dauwels J, Rathakrishnan R, Halford JJ, Cash SS, **Westover B**. "Automated Detection of Interictal Epileptiform Discharges from Scalp

- Electroencephalograms by Convolutional Neural Networks.” *Int J Neural Syst*. 2020 Nov;30(11):2050030. doi: 10.1142/S0129065720500306. Epub 2020 Aug 19. PMID: PMC7606586.
178. Abou Jaoude M, Sun H, Pellerin KR, Pavlova M, Sarkis RA, Cash SS, **Westover MB**, Lam AD. “Expert-level automated sleep staging of long-term scalp electroencephalography recordings using deep learning.” *Sleep*. 2020 Nov 12;43(11). doi: 10.1093/sleep/zsaa112. PMID: PMC7686563.
 179. Vespa PM, Olson DM, John S, Hobbs KS, Gururangan K, Nie K, Desai MJ, Markert M, Parvizi J, Bleck TP, Hirsch LJ, **Westover MB**. “Evaluating the Clinical Impact of Rapid Response Electroencephalography: The DECIDE Multicenter Prospective Observational Clinical Study.” *Crit Care Med*. 2020 Sep;48(9):1249-1257. doi: 10.1097/CCM.0000000000004428. PMID: PMC7735649.
 180. Ayub N, Cohen J, Jing J, Jain A, Tesh R, Mukerji SS, Zafar SF, **Westover MB**, Kimchi EY. “Clinical Electroencephalography Findings and Considerations in Hospitalized Patients with Coronavirus SARS-CoV-2.” *medRxiv*. 2020 Jul 15. doi: 10.1101/2020.07.13.20152207. PMID: PMC7373142.
 181. Thorn EL, Ostrowski LM, Chinappen DM, Jing J, **Westover MB**, Stufflebeam SM, Kramer MA, Chu CJ. “Persistent abnormalities in Rolandic thalamocortical white matter circuits in childhood epilepsy with centrotemporal spikes.” *Epilepsia*. 2020 Nov;61(11):2500-2508. doi: 10.1111/epi.16681. Epub 2020 Sep 18. PMID: PMC7722074.
 182. Goldenholz DM, Goldenholz SR, Romero J, Moss R, Sun H, **Westover B**. “Development and Validation of Forecasting Next Reported Seizure Using e-Diaries.” *Ann Neurol*. 2020 Sep;88(3):588-595. doi: 10.1002/ana.25812. Epub 2020 Jul 09. PMID: PMC7720795.
 183. Lam AD, Sarkis RA, Pellerin KR, Jing J, Dworetzky BA, Hoch DB, Jacobs CS, Lee JW, Weisholtz DS, Zepeda R, **Westover MB**, Cole AJ, Cash SS. “Association of epileptiform abnormalities and seizures in Alzheimer disease.” *Neurology*. 2020 Oct 20;95(16):e2259-e2270. doi: 10.1212/WNL.0000000000010612. Epub 2020 Aug 06. PMID: PMC7713786.
 184. Hogan J, Sun H, Paixao L, Westmeijer M, Sikka P, Jin J, Tesh R, Cardoso M, Cash SS, Akeju O, Thomas R, **Westover MB**. “Night-to-night variability of sleep electroencephalography-based brain age measurements.” *Clin Neurophysiol*. 2021 Jan;132(1):1-12. doi: 10.1016/j.clinph.2020.09.029. Epub 2020 Oct 29. PMID: PMC7855943.
 185. Jing J, d'Angremont E, Ebrahim S, Tabaeizadeh M, Ng M, Herlopian A, Dauwels J, Brandon **Westover M**. “Rapid annotation of seizures and interictal-ictal-injury continuum EEG patterns.” *J Neurosci Methods*. 2021 Jan 01;347:108956. doi: 10.1016/j.jneumeth.2020.108956. Epub 2020 Oct 21. PMID: PMC7744406.
 186. Shashikumar SP, Wardi G, Paul P, Carlile M, Brenner LN, Hibbert KA, North CM, Mukerji SS, Robbins GK, Shao YP, **Westover MB**, Nemati S, Malhotra A. “Development and Prospective Validation of a Deep Learning Algorithm for Predicting Need for Mechanical Ventilation.” *Chest*. 2021 Jun;159(6):2264-2273. doi: 10.1016/j.chest.2020.12.009. Epub 2020 Dec 17. PMID: PMC8027289.
 187. Zafar SF, Khozein RJ, LaRoche SM, **Westover MB**, Gilmore EJ. “Impact of the COVID-19 Pandemic on Continuous EEG Utilization.” *J Clin Neurophysiol*. 2022 Nov 01;39(7):567-574. doi: 10.1097/WNP.0000000000000802. Epub 2020 Dec 22. PMID: PMC8217411.
 188. Lissak IA, Locascio JJ, Zafar SF, Schleicher RL, Patel AB, Leslie-Mazwi T, Stapleton CJ, Koch MJ, Kim JA, Anderson K, Rosand J, **Westover MB**, Kimberly WT, Rosenthal ES. “Electroencephalography, Hospital Complications, and Longitudinal Outcomes After Subarachnoid Hemorrhage.” *Neurocrit Care*. 2021 Oct;35(2):397-408. doi: 10.1007/s12028-020-01177-x. Epub 2021 Jan 22. PMID: PMC7822587.
 189. Conklin J, Frosch MP, Mukerji SS, Rapalino O, Maher MD, Schaefer PW, Lev MH, Gonzalez RG, Das S, Champion SN, Magdamo C, Sen P, Harrold GK, Alabsi H, Normandin E, Shaw B, Lemieux JE, Sabeti PC, Branda JA, Brown EN, **Westover MB**, Huang SY, Edlow BL. “Susceptibility-weighted imaging reveals cerebral microvascular injury in severe COVID-19.” *J Neurol Sci*. 2021 Feb 15;421:117308. doi: 10.1016/j.jns.2021.117308. Epub 2021 Jan 15. PMID: PMC7832284.
 190. Fernandes M, Sun H, Jain A, Alabsi HS, Brenner LN, Ye E, Ge W, Collens SI, Leone MJ, Das S, Robbins GK, Mukerji SS, **Westover MB**. “Classification of the Disposition of Patients Hospitalized with COVID-19: Reading Discharge Summaries Using Natural Language Processing.” *JMIR Med Inform*. 2021 Feb 10;9(2):e25457. doi: 10.2196/25457. PMID: PMC7879729.

191. Janocko NJ, Jing J, Fan Z, Teagarden DL, Villarreal HK, Morton ML, Groover O, Loring DW, Drane DL, **Westover MB**, Karakis I. "DDESVSFS: A simple, rapid and comprehensive screening tool for the Differential Diagnosis of Epileptic Seizures VS Functional Seizures." *Epilepsy Res.* 2021 Mar;171:106563. doi: 10.1016/j.epilepsyres.2021.106563. Epub 2021 Jan 21. PMID: PMC8092190.
192. Ozdemir RA, Tadayon E, Boucher P, Sun H, Momi D, Ganglberger W, **Westover MB**, Pascual-Leone A, Santarnecchi E, Shafi MM. "Cortical responses to noninvasive perturbations enable individual brain fingerprinting." *Brain Stimul.* 2021;14(2):391-403. doi: 10.1016/j.brs.2021.02.005. Epub 2021 Feb 12. PMID: PMC8108003.
193. Lin L, Al-Faraj A, Ayub N, Bravo P, Das S, Ferlini L, Karakis I, Lee JW, Mukerji SS, Newey CR, Pathmanathan J, Abdennadher M, Casassa C, Gaspard N, Goldenholz DM, Gilmore EJ, Jing J, Kim JA, Kimchi EY, Ladha HS, Tobochnik S, Zafar S, Hirsch LJ, **Westover MB**, Shafi MM. "Electroencephalographic Abnormalities are Common in COVID-19 and are Associated with Outcomes." *Ann Neurol.* 2021 May;89(5):872-883. doi: 10.1002/ana.26060. Epub 2021 Mar 24. PMID: PMC8104061.
194. Hirsch LJ, Fong MWK, Leitinger M, LaRoche SM, Beniczky S, Abend NS, Lee JW, Wusthoff CJ, Hahn CD, **Westover MB**, Gerard EE, Herman ST, Haider HA, Osman G, Rodriguez-Ruiz A, Maciel CB, Gilmore EJ, Fernandez A, Rosenthal ES, Claassen J, Husain AM, Yoo JY, So EL, Kaplan PW, Nuwer MR, van Putten M, Sutter R, Drislane FW, Trinka E, Gaspard N. "American Clinical Neurophysiology Society's Standardized Critical Care EEG Terminology: 2021 Version." *J Clin Neurophysiol.* 2021 Jan 01;38(1):1-29. doi: 10.1097/WNP.0000000000000806. PMID: PMC8135051.
195. Kramer MA, Stoyell SM, Chinappen D, Ostrowski LM, Spencer ER, Morgan AK, Emerton BC, Jing J, **Westover MB**, Eden UT, Stickgold R, Manoach DS, Chu CJ. "Focal Sleep Spindle Deficits Reveal Focal Thalamocortical Dysfunction and Predict Cognitive Deficits in Sleep Activated Developmental Epilepsy." *J Neurosci.* 2021 Feb 24;41(8):1816-1829. doi: 10.1523/JNEUROSCI.2009-20.2020. Epub 2021 Jan 19. PMID: PMC8115887.
196. Gaspard N, **Westover MB**, Hirsch LJ. "Assessment of a Study of Continuous vs Repeat-Spot Electroencephalography in Patients With Critical Illness." *JAMA Neurol.* 2021 Mar 01;78(3):369. doi: 10.1001/jamaneurol.2020.5348. PMID: PMC8136812.
197. Ge W, Jing J, An S, Herlopian A, Ng M, Struck AF, Appavu B, Johnson EL, Osman G, Haider HA, Karakis I, Kim JA, Halford JJ, Dhakar MB, Sarkis RA, Swisher CB, Schmitt S, Lee JW, Tabaeizadeh M, Rodriguez A, Gaspard N, Gilmore E, Herman ST, Kaplan PW, Pathmanathan J, Hong S, Rosenthal ES, Zafar S, Sun J, Brandon **Westover M**. "Deep active learning for Interictal Ictal Injury Continuum EEG patterns." *J Neurosci Methods.* 2021 Mar 01;351:108966. doi: 10.1016/j.jneumeth.2020.108966. Epub 2020 Oct 22. PMID: PMC8135050.
198. Li Q, **Westover MB**, Zhang R, Chu CJ. "Computational Evidence for a Competitive Thalamocortical Model of Spikes and Spindle Activity in Rolandic Epilepsy." *Front Comput Neurosci.* 2021;15:680549. doi: 10.3389/fncom.2021.680549. Epub 2021 Jun 18. PMID: PMC8249809.
199. Mukerji SS, Das S, Alabsi H, Brenner LN, Jain A, Magdamo C, Collens SI, Ye E, Keller K, Boutros CL, Leone MJ, Newhouse A, Foy B, Li MD, Lang M, Anahtar MN, Shao YP, Ge W, Sun H, Triant VA, Kalpathy-Cramer J, Higgins J, Rosand J, Robbins GK, **Westover MB**. "Prolonged Intubation in Patients With Prior Cerebrovascular Disease and COVID-19." *Front Neurol.* 2021;12:642912. doi: 10.3389/fneur.2021.642912. Epub 2021 Apr 09. PMID: PMC8062773.
200. Jones FJS, Sanches PR, Smith JR, Zafar SF, Blacker D, Hsu J, Schwamm LH, Newhouse JP, **Westover MB**, Moura LMVR. "Seizure Prophylaxis After Spontaneous Intracerebral Hemorrhage." *JAMA Neurol.* 2021 Sep 01;78(9):1128-1136. doi: 10.1001/jamaneurol.2021.2249. PMID: PMC8314179.
201. Pedemonte JC, Sun H, Franco-Garcia E, Zhou C, Heng M, Quraishi SA, **Westover B**, Akeju O. "Postoperative delirium mediates 180-day mortality in orthopaedic trauma patients." *Br J Anaesth.* 2021 Jul;127(1):102-109. doi: 10.1016/j.bja.2021.03.033. Epub 2021 May 29. PMID: PMC8258970.
202. Yang JC, Harid NM, Nascimento FA, Kokkinos V, Shaughnessy A, Lam AD, **Westover MB**, Leslie-Mazwi TM, Hochberg LR, Rosenthal ES, Cole AJ, Richardson RM, Cash SS. "Responsive

- neurostimulation for focal motor status epilepticus.” *Ann Clin Transl Neurol*. 2021 Jun;8(6):1353-1361. doi: 10.1002/acn3.51318. Epub 2021 May 06. PMID: PMC8164849.
203. Thomas J, Thangavel P, Peh WY, Jing J, Yuvaraj R, Cash SS, Chaudhari R, Karia S, Rathakrishnan R, Saini V, Shah N, Srivastava R, Tan YL, **Westover B**, Dauwels J. “Automated Adult Epilepsy Diagnostic Tool Based on Interictal Scalp Electroencephalogram Characteristics: A Six-Center Study.” *Int J Neural Syst*. 2021 May;31(5):2050074. doi: 10.1142/S0129065720500744. Epub 2021 Jan 12. PMID: PMC9343226.
 204. Leone MJ, Sun H, Boutros CL, Liu L, Ye E, Sullivan L, Thomas RJ, Robbins GK, Mukerji SS, **Westover MB**. “HIV increases sleep-based brain age despite antiretroviral therapy.” *Sleep*. 2021 Aug 13;44(8). doi: 10.1093/sleep/zsab058. PMID: PMC8361332.
 205. Jones FJS, Sanches PR, Smith JR, Zafar SF, Hernandez-Diaz S, Blacker D, Hsu J, Schwamm LH, **Westover MB**, Moura LMVR. “Anticonvulsant Primary and Secondary Prophylaxis for Acute Ischemic Stroke Patients: A Decision Analysis.” *Stroke*. 2021 Aug;52(9):2782-2791. doi: 10.1161/STROKEAHA.120.033299. Epub 2021 Jun 15. PMID: PMC8384723.
 206. Kim JA, Zheng WL, Elmer J, Jing J, Zafar SF, Ghanta M, Moura V, Gilmore EJ, Hirsch LJ, Patel A, Rosenthal E, **Westover MB**. “High epileptiform discharge burden predicts delayed cerebral ischemia after subarachnoid hemorrhage.” *Clin Neurophysiol*. 2022 Sep;141:139-146. doi: 10.1016/j.clinph.2021.01.022. Epub 2021 Mar 10. PMID: PMC8429508.
 207. Ligtstein S, Song J, Jin J, Sun H, Paixao L, Zafar S, **Westover MB**. “Do Triphasic Waves and Nonconvulsive Status Epilepticus Arise From Similar Mechanisms? A Computational Model.” *J Clin Neurophysiol*. 2021 Sep 01;38(5):366-375. doi: 10.1097/WNP.0000000000000719. PMID: PMC8429048.
 208. Nascimento FA, Chen PM, Cohen J, **Westover BM**. “Generalized periodic discharges with triphasic morphology: to treat or not to treat?” *Epileptic Disord*. 2021 Apr 01;23(2):435-436. doi: 10.1684/epd.2021.1272. PMID: PMC8592021.
 209. Oppersma E, Ganglberger W, Sun H, Thomas RJ, **Westover MB**. “Algorithm for automatic detection of self-similarity and prediction of residual central respiratory events during continuous positive airway pressure.” *Sleep*. 2021 Apr 09;44(4). doi: 10.1093/sleep/zsaa215. PMID: PMC8631077.
 210. Zafar SF, Rosenthal ES, Jing J, Ge W, Tabaeizadeh M, Aboul Nour H, Shoukat M, Sun H, Javed F, Kassa S, Edhi M, Bordbar E, Gallagher J, Moura V, Ghanta M, Shao YP, An S, Sun J, Cole AJ, **Westover MB**. “Automated Annotation of Epileptiform Burden and Its Association with Outcomes.” *Ann Neurol*. 2021 Aug;90(2):300-311. doi: 10.1002/ana.26161. Epub 2021 Jul 20. PMID: PMC8516549.
 211. Paixao L, Sun H, Hogan J, Hartnack K, Westmeijer M, Neelagiri A, Zhou DW, McClain LM, Kimchi EY, Purdon PL, Akeju O, **Westover MB**. “ICU delirium burden predicts functional neurologic outcomes.” *PLoS One*. 2021;16(12):e0259840. doi: 10.1371/journal.pone.0259840. Epub 2021 Dec 02. PMID: PMC8638853.
 212. van Sleuwen M, Sun H, Eckhardt C, Neelagiri A, Tesh RA, Westmeijer M, Paixao L, Rajan S, Velpula Krishnamurthy P, Sikka P, Leone MJ, Panneerselvam E, Quadri SA, Akeju O, Kimchi EY, **Westover MB**. “Physiological Assessment of Delirium Severity: The Electroencephalographic Confusion Assessment Method Severity Score (E-CAM-S).” *Crit Care Med*. 2022 Jan 01;50(1):e11-e19. doi: 10.1097/CCM.0000000000005224. PMID: PMC8678335.
 213. Zheng WL, Amorim E, Jing J, Ge W, Hong S, Wu O, Ghassemi M, Lee JW, Sivaraju A, Pang T, Herman ST, Gaspard N, Ruijter BJ, Sun J, Tjepkema-Cloostermans MC, Hofmeijer J, van Putten MJAM, **Westover MB**. “Predicting neurological outcome in comatose patients after cardiac arrest with multiscale deep neural networks.” *Resuscitation*. 2021 Dec;169:86-94. doi: 10.1016/j.resuscitation.2021.10.034. Epub 2021 Oct 24. PMID: PMC8692444.
 214. Tesh RA, Sun H, Jing J, Westmeijer M, Neelagiri A, Rajan S, Krishnamurthy PV, Sikka P, Quadri SA, Leone MJ, Paixao L, Panneerselvam E, Eckhardt C, Struck AF, Kaplan PW, Akeju O, Jones D, Kimchi EY, **Westover MB**. “VE-CAM-S: Visual EEG-Based Grading of Delirium Severity and Associations With Clinical Outcomes.” *Crit Care Explor*. 2022 Jan;4(1):e0611. doi: 10.1097/CCE.0000000000000611. Epub 2022 Jan 18. PMID: PMC8769081.

215. Amerineni R, Sun H, Lee H, Hsu J, Paterno E, **Westover MB**, Zafar SF. "Using electronic health data to explore effectiveness of ICU EEG and anti-seizure treatment." *Ann Clin Transl Neurol*. 2021 Dec;8(12):2270-2279. doi: 10.1002/acn3.51478. Epub 2021 Nov 21. PMID: PMC8670316.
216. Song JL, Kim JA, Struck AF, Zhang R, **Westover MB**. "A model of metabolic supply-demand mismatch leading to secondary brain injury." *J Neurophysiol*. 2021 Aug 01;126(2):653-667. doi: 10.1152/jn.00674.2020. Epub 2021 Jul 07. PMID: PMC8815783.
217. Chen HY, Elmer J, Zafar SF, Ghanta M, Moura Junior V, Rosenthal ES, Gilmore EJ, Hirsch LJ, Zaveri HP, Sheth KN, Petersen NH, **Westover MB**, Kim JA. "Combining Transcranial Doppler and EEG Data to Predict Delayed Cerebral Ischemia After Subarachnoid Hemorrhage." *Neurology*. 2022 Feb 01;98(5):e459-e469. doi: 10.1212/WNL.00000000000013126. Epub 2021 Nov 29. PMID: PMC8826465.
218. Ganglberger W, Bucklin AA, Tesh RA, Da Silva Cardoso M, Sun H, Leone MJ, Paixao L, Panneerselvam E, Ye EM, Thompson BT, Akeju O, Kuller D, Thomas RJ, **Westover MB**. "Sleep apnea and respiratory anomaly detection from a wearable band and oxygen saturation." *Sleep Breath*. 2022 Sep;26(3):1033-1044. doi: 10.1007/s11325-021-02465-2. Epub 2021 Aug 18. PMID: PMC8854446.
219. Salami P, Borzello M, Kramer MA, **Westover MB**, Cash SS. "Quantifying seizure termination patterns reveals limited pathways to seizure end." *Neurobiol Dis*. 2022 Apr;165:105645. doi: 10.1016/j.nbd.2022.105645. Epub 2022 Jan 29. PMID: PMC8860887.
220. Nayan M, Salari K, Bozzo A, Ganglberger W, Lu G, Carvalho F, Gusev A, Schneider A, **Westover BM**, Feldman AS. "A machine learning approach to predict progression on active surveillance for prostate cancer." *Urol Oncol*. 2022 Apr;40(4):161.e1-161.e7. doi: 10.1016/j.urolonc.2021.08.007. Epub 2021 Aug 29. PMID: PMC8882704.
221. Moura Junior V, **Westover MB**, Li F, Kimchi E, Kennedy M, Benson NM, Moura LM, Hsu J. "Hospital complications among older adults: Better processes could reduce the risk of delirium." *Health Serv Manage Res*. 2022 Aug;35(3):154-163. doi: 10.1177/09514848211028707. Epub 2021 Jul 11. PMID: PMC8748518.
222. Ong CS, Reinertsen E, Sun H, Moonsamy P, Mohan N, Funamoto M, Kaneko T, Shekar PS, Schena S, Lawton JS, D'Alessandro DA, **Westover MB**, Aguirre AD, Sundt TM. "Prediction of operative mortality for patients undergoing cardiac surgical procedures without established risk scores." *J Thorac Cardiovasc Surg*. 2023 Apr;165(4):1449-1459.e15. doi: 10.1016/j.jtcvs.2021.09.010. Epub 2021 Sep 14. PMID: PMC8918430.
223. Ng MC, Jing J, **Westover MB**. "A Primer on EEG Spectrograms." *J Clin Neurophysiol*. 2022 Mar 01;39(3):177-183. doi: 10.1097/WNP.0000000000000736. PMID: PMC8901534.
224. Ng MC, **Westover MB**. "Decoding the Spectrogram Rainbow." *J Clin Neurophysiol*. 2022 Mar 01;39(3):176. doi: 10.1097/WNP.0000000000000741. PMID: PMC8901547.
225. Baang HY, Chen HY, Herman AL, Gilmore EJ, Hirsch LJ, Sheth KN, Petersen NH, Zafar SF, Rosenthal ES, **Westover MB**, Kim JA. "The Utility of Quantitative EEG in Detecting Delayed Cerebral Ischemia After Aneurysmal Subarachnoid Hemorrhage." *J Clin Neurophysiol*. 2022 Mar 01;39(3):207-215. doi: 10.1097/WNP.0000000000000754. PMID: PMC8901442.
226. Nascimento FA, Jing J, Beniczky S, Benbadis SR, Gavvala JR, Yacubian EMT, Wiebe S, Rampp S, van Putten MJAM, Tripathi M, Cook MJ, Kaplan PW, Tatum WO, Trinka E, Cole AJ, **Westover MB**. "One EEG, one read - A manifesto towards reducing interrater variability among experts." *Clin Neurophysiol*. 2022 Jan;133:68-70. doi: 10.1016/j.clinph.2021.10.007. Epub 2021 Nov 05. PMID: PMC8926459.
227. Adra N, Sun H, Ganglberger W, Ye EM, Dümmer LW, Tesh RA, Westmeijer M, Cardoso MDS, Kitchener E, Ouyang A, Salinas J, Rosand J, Cash SS, Thomas RJ, **Westover MB**. "Optimal spindle detection parameters for predicting cognitive performance." *Sleep*. 2022 Apr 11;45(4). doi: 10.1093/sleep/zsac001. PMID: PMC8996023.
228. Waldrop G, Safavynia SA, Barra ME, Agarwal S, Berlin DA, Boehme AK, Brodie D, Choi JM, Doyle K, Fins JJ, Ganglberger W, Hoffman K, Mittel AM, Roh D, Mukerji SS, Der Nigoghossian C, Park S, Schenck EJ, Salazar-Schicchi J, Shen Q, Sholle E, Velazquez AG, Walline MC, **Westover MB**, Brown EN, Victor J, Edlow BL, Schiff ND, Claassen J. "Prolonged Unconsciousness is Common in COVID-19 and Associated with Hypoxemia." *Ann Neurol*. 2022 Jun;91(6):740-755. doi: 10.1002/ana.26342. Epub

2022 Apr 21. PMID: PMC9082460.

229. Abou Jaoude M, Jacobs CS, Sarkis RA, Jing J, Pellerin KR, Cole AJ, Cash SS, **Westover MB**, Lam AD. "Noninvasive Detection of Hippocampal Epileptiform Activity on Scalp Electroencephalogram." *JAMA Neurol.* 2022 Jun 01;79(6):614-622. doi: 10.1001/jamaneurol.2022.0888. PMID: PMC9062772.
230. Zafar SF, Rosenthal ES, Postma EN, Sanches P, Ayub MA, Rajan S, Kim JA, Rubin DB, Lee H, Patel AB, Hsu J, Patorno E, **Westover MB**. "Antiseizure Medication Treatment and Outcomes in Patients with Subarachnoid Hemorrhage Undergoing Continuous EEG Monitoring." *Neurocrit Care.* 2022 Jun;36(3):857-867. doi: 10.1007/s12028-021-01387-x. Epub 2021 Nov 29. PMID: PMC9117405.
231. Kural MA, Jing J, Fürbass F, Perko H, Qerama E, Johnsen B, Fuchs S, **Westover MB**, Beniczky S. "Accurate identification of EEG recordings with interictal epileptiform discharges using a hybrid approach: Artificial intelligence supervised by human experts." *Epilepsia.* 2022 May;63(5):1064-1073. doi: 10.1111/epi.17206. Epub 2022 Mar 07. PMID: PMC9148170.
232. Nassi TE, Ganglberger W, Sun H, Bucklin AA, Biswal S, van Putten MJAM, Thomas RJ, **Westover MB**. "Automated Scoring of Respiratory Events in Sleep With a Single Effort Belt and Deep Neural Networks." *IEEE Trans Biomed Eng.* 2022 Jun;69(6):2094-2104. doi: 10.1109/TBME.2021.3136753. Epub 2022 May 19. PMID: PMC9119908.
233. Sheikh Z, Selioutski O, Taraschenko O, Gilmore EJ, **Westover MB**, Cohen AB. "Systematic Evaluation of Research Priorities in Critical Care Electroencephalography." *J Clin Neurophysiol.* 2023 Jul 01;40(5):426-433. doi: 10.1097/WNP.0000000000000916. Epub 2022 Jan 20. PMID: PMC9296700.
234. Harid NM, Jing J, Hogan J, Nascimento FA, Ouyang A, Zheng WL, Ge W, Zafar SF, Kim JA, Alice DL, Herlopian A, Maus D, Karakis I, Ng M, Hong S, Yu Z, Kaplan PW, Cash S, Shafi M, Martz G, Halford JJ, **Westover MB**. "Measuring expertise in identifying interictal epileptiform discharges." *Epileptic Disord.* 2022 Jun 01;24(3):496-506. doi: 10.1684/epd.2021.1409. PMID: PMC9340812.
235. Kulpanowski AM, Copen WA, Hancock BL, Rosenthal ES, Schoenfeld DA, Dodelson JA, Edlow BL, Kimberly WT, Amorim E, **Westover MB**, Ning MM, Schaefer PW, Malhotra R, Giacino JT, Greer DM, Wu O. "Severe cerebral edema in substance-related cardiac arrest patients." *Resuscitation.* 2022 Apr;173:103-111. doi: 10.1016/j.resuscitation.2022.01.033. Epub 2022 Feb 08. PMID: PMC9282938.
236. Ge W, Alabsi H, Jain A, Ye E, Sun H, Fernandes M, Magdamo C, Tesh RA, Collens SI, Newhouse A, Mvr Moura L, Zafar S, Hsu J, Akeju O, Robbins GK, Mukerji SS, Das S, **Westover MB**. "Identifying Patients With Delirium Based on Unstructured Clinical Notes: Observational Study." *JMIR Form Res.* 2022 Jun 24;6(6):e33834. doi: 10.2196/33834. PMID: PMC9270709.
237. Nascimento FA, McLaren JR, **Westover MB**, Zafar SF, Stufflebeam SM. "Teaching NeuroImage: Sturge-Weber Syndrome in an Adult." *Neurology.* 2022 May 10;98(19):814-815. doi: 10.1212/WNL.0000000000200512. Epub 2022 Apr 11. PMID: PMC9141624.
238. Amorim E, Firme MS, Zheng WL, Shelton KT, Akeju O, Cudemus G, Yuval R, **Westover MB**. "High incidence of epileptiform activity in adults undergoing extracorporeal membrane oxygenation." *Clin Neurophysiol.* 2022 Aug;140:4-11. doi: 10.1016/j.clinph.2022.04.018. Epub 2022 May 06. PMID: PMC9340813.
239. Brink-Kjaer A, Leary EB, Sun H, **Westover MB**, Stone KL, Peppard PE, Lane NE, Cawthon PM, Redline S, Jennum P, Sorensen HBD, Mignot E. "Age estimation from sleep studies using deep learning predicts life expectancy." *NPJ Digit Med.* 2022 Jul 22;5(1):103. doi: 10.1038/s41746-022-00630-9. PMID: PMC9307657.
240. Thangavel P, Thomas J, Peh WY, Jing J, Yuvaraj R, Cash SS, Chaudhari R, Karia S, Rathakrishnan R, Saini V, Shah N, Srivastava R, Tan YL, **Westover B**, Dauwels J. "Time-Frequency Decomposition of Scalp Electroencephalograms Improves Deep Learning-Based Epilepsy Diagnosis." *Int J Neural Syst.* 2021 Aug;31(8):2150032. doi: 10.1142/S0129065721500325. Epub 2021 Jul 16. PMID: PMC9340811.
241. Song JL, Li Q, Zhang B, **Westover MB**, Zhang R. "A New Neural Mass Model Driven Method and Its Application in Early Epileptic Seizure Detection." *IEEE Trans Biomed Eng.* 2020 Aug;67(8):2194-2205. doi: 10.1109/TBME.2019.2957392. Epub 2019 Dec 03. PMID: PMC9371613.
242. Singh NM, Harrod JB, Subramanian S, Robinson M, Chang K, Cetin-Karayumak S, Dalca AV, Eickhoff S, Fox M, Franke L, Golland P, Haehn D, Iglesias JE, O'Donnell LJ, Ou Y, Rathi Y, Siddiqi SH, Sun H,

- Westover MB**, Whitfield-Gabrieli S, Gollub RL. "How Machine Learning is Powering Neuroimaging to Improve Brain Health." *Neuroinformatics*. 2022 Oct;20(4):943-964. doi: 10.1007/s12021-022-09572-9. Epub 2022 Mar 28. PMID: PMC9515245.
243. Torres-Lopez VM, Rovenolt GE, Olcese AJ, Garcia GE, Chacko SM, Robinson A, Gaiser E, Acosta J, Herman AL, Kuohn LR, Leary M, Soto AL, Zhang Q, Fatima S, Falcone GJ, Payabvash MS, Sharma R, Struck AF, Sheth KN, **Westover MB**, Kim JA. "Development and Validation of a Model to Identify Critical Brain Injuries Using Natural Language Processing of Text Computed Tomography Reports." *JAMA Netw Open*. 2022 Aug 01;5(8):e2227109. doi: 10.1001/jamanetworkopen.2022.27109. PMID: PMC9382443.
 244. McLaren JR, Jing J, **Westover MB**, Nascimento FA. "Journal Club: Criteria for Defining Interictal Epileptiform Discharges in EEG." *Neurology*. 2022 Sep 05;99(10):430-432. doi: 10.1212/WNL.0000000000200991. PMID: PMC9519249.
 245. Song JL, **Westover MB**, Zhang R. "A mechanistic model of calcium homeostasis leading to occurrence and propagation of secondary brain injury." *J Neurophysiol*. 2022 Nov 01;128(5):1168-1180. doi: 10.1152/jn.00045.2022. Epub 2022 Oct 05. PMID: PMC9621713.
 246. Jones DK, Eckhardt CA, Sun H, Tesh RA, Malik P, Quadri S, Firme MS, van Sleuwen M, Jain A, Fan Z, Jing J, Ge W, Nascimento FA, Sheikh IS, Jacobson C, Frigault M, Kimchi EY, Cash SS, Lee JW, Dietrich J, **Westover MB**. "EEG-based grading of immune effector cell-associated neurotoxicity syndrome." *Sci Rep*. 2022 Nov 21;12(1):20011. doi: 10.1038/s41598-022-24010-1. PMID: PMC9681864.
 247. Amidi Y, Eckhardt CA, Quadri SA, Malik P, Firme MS, Jones DK, Jain A, Danish HH, Rubin DB, Jacobson CA, Cash SS, Lee JW, Dietrich J, **Westover MB**. "Forecasting immune effector cell-associated neurotoxicity syndrome after chimeric antigen receptor t-cell therapy." *J Immunother Cancer*. 2022 Nov;10(11). doi: 10.1136/jitc-2022-005459. PMID: PMC9716920.
 248. Zheng WL, Kim JA, Elmer J, Zafar SF, Ghanta M, Moura Junior V, Patel A, Rosenthal E, Brandon **Westover M**. "Automated EEG-based prediction of delayed cerebral ischemia after subarachnoid hemorrhage." *Clin Neurophysiol*. 2022 Nov;143:97-106. doi: 10.1016/j.clinph.2022.08.023. Epub 2022 Sep 11. PMID: PMC9847346.
 249. Liang F, Baldyga K, Quan Q, Khatri A, Choi S, Wiener-Kronish J, Akeju O, **Westover MB**, Cody K, Shen Y, Marcantonio ER, Xie Z. "Preoperative Plasma Tau-PT217 and Tau-PT181 Are Associated With Postoperative Delirium." *Ann Surg*. 2023 Jun 01;277(6):e1232-e1238. doi: 10.1097/SLA.0000000000005487. Epub 2022 Jul 06. PMID: PMC9875943.
 250. Fernandes M, Donahue MA, Hoch D, Cash S, Zafar S, Jacobs C, Hosford M, Voinescu PE, Fureman B, Buchhalter J, McGraw CM, **Westover MB**, Moura LMVR. "A replicable, open-source, data integration method to support national practice-based research & quality improvement systems." *Epilepsy Res*. 2022 Oct;186:107013. doi: 10.1016/j.epilepsyres.2022.107013. Epub 2022 Aug 18. PMID: PMC9810436.
 251. Nascimento FA, Jing J, Strowd R, Sheikh IS, Weber D, Gavvala JR, Maheshwari A, Tanner A, Ng M, Vinayan KP, Sinha SR, Yacubian EM, Rao VR, Perry MS, Fountain NB, Karakis I, Wirrell E, Yuan F, Friedman D, Tankisi H, Rampp S, Fasano R, Wilmshurst JM, O'Donovan C, Schomer D, Kaplan PW, Sperling MR, Benbadis S, **Westover MB**, Beniczky S. "Competency-based EEG education: a list of "must-know" EEG findings for adult and child neurology residents." *Epileptic Disord*. 2022 Oct 01;24(5):979-982. doi: 10.1684/epd.2022.1476. PMID: PMC9812628.
 252. Nascimento FA, Jing J, Beniczky S, Olandoski M, Benbadis SR, Cole AJ, **Westover MB**. "EEG reading with or without clinical information - a real-world practice study." *Neurophysiol Clin*. 2022 Oct;52(5):394-397. doi: 10.1016/j.neucli.2022.08.002. Epub 2022 Sep 17. PMID: PMC9815944.
 253. Martinez P, Sheikh I, **Westover MB**, Zafar SF. "Implications of stimulus-induced, rhythmic, periodic, or ictal discharges (SIRPIDs) in hospitalized patients." *Front Neurol*. 2022;13:1062330. doi: 10.3389/fneur.2022.1062330. Epub 2023 Jan 23. PMID: PMC9899805.
 254. Goldenholz DM, **Westover MB**. "Flexible realistic simulation of seizure occurrence recapitulating statistical properties of seizure diaries." *Epilepsia*. 2023 Feb;64(2):396-405. doi: 10.1111/epi.17471. Epub 2022 Dec 04. PMID: PMC9905290.

255. Qu JZ, Mueller A, McKay TB, **Westover MB**, Shelton KT, Shaefi S, D'Alessandro DA, Berra L, Brown EN, Houle TT, Akeju O, MINDDS Study Team. "Nighttime dexmedetomidine for delirium prevention in non-mechanically ventilated patients after cardiac surgery (MINDDS): A single-centre, parallel-arm, randomised, placebo-controlled superiority trial." *EClinicalMedicine*. 2023 Feb;56:101796. doi: 10.1016/j.eclinm.2022.101796. Epub 2022 Dec 24. PMID: PMC9800196.
256. Fernandes MB, Valizadeh N, Alabsi HS, Quadri SA, Tesh RA, Bucklin AA, Sun H, Jain A, Brenner LN, Ye E, Ge W, Collens SI, Lin S, Das S, Robbins GK, Zafar SF, Mukerji SS, **Westover MB**. "Classification of neurologic outcomes from medical notes using natural language processing." *Expert Syst Appl*. 2023 Mar 15;214. doi: 10.1016/j.eswa.2022.119171. Epub 2022 Nov 06. PMID: PMC9974159.
257. Bucklin AA, Ganglberger W, Quadri SA, Tesh RA, Adra N, Da Silva Cardoso M, Leone MJ, Krishnamurthy PV, Hemmige A, Rajan S, Panneerselvam E, Paixao L, Higgins J, Ayub MA, Shao YP, Ye EM, Coughlin B, Sun H, Cash SS, Thompson BT, Akeju O, Kuller D, Thomas RJ, **Westover MB**. "High prevalence of sleep-disordered breathing in the intensive care unit - a cross-sectional study." *Sleep Breath*. 2023 Jun;27(3):1013-1026. doi: 10.1007/s11325-022-02698-9. Epub 2022 Aug 16. PMID: PMC9931933.
258. Chen Y, Li S, Ge W, Jing J, Chen HY, Doherty D, Herman A, Kaleem S, Ding K, Osman G, Swisher CB, Smith C, Maciel CB, Alkhachroum A, Lee JW, Dhakar MB, Gilmore EJ, Sivaraju A, Hirsch LJ, Omay SB, Blumenfeld H, Sheth KN, Struck AF, Edlow BL, **Westover MB**, Kim JA. "Quantitative epileptiform burden and electroencephalography background features predict post-traumatic epilepsy." *J Neurol Neurosurg Psychiatry*. 2023 Mar;94(3):245-249. doi: 10.1136/jnnp-2022-329542. Epub 2022 Oct 14. PMID: PMC9931627.
259. Ye EM, Sun H, Krishnamurthy PV, Adra N, Ganglberger W, Thomas RJ, Lam AD, **Westover MB**. "Dementia detection from brain activity during sleep." *Sleep*. 2023 Mar 09;46(3). doi: 10.1093/sleep/zsac286. PMID: PMC9995788.
260. Sun H, Ye E, Paixao L, Ganglberger W, Chu CJ, Zhang C, Rosand J, Mignot E, Cash SS, Gozal D, Thomas RJ, **Westover MB**. "The sleep and wake electroencephalogram over the lifespan." *Neurobiol Aging*. 2023 Apr;124:60-70. doi: 10.1016/j.neurobiolaging.2023.01.006. Epub 2023 Jan 19. PMID: PMC9957961.
261. Ganglberger W, Krishnamurthy PV, Quadri SA, Tesh RA, Bucklin AA, Adra N, Da Silva Cardoso M, Leone MJ, Hemmige A, Rajan S, Panneerselvam E, Paixao L, Higgins J, Ayub MA, Shao YP, Coughlin B, Sun H, Ye EM, Cash SS, Thompson BT, Akeju O, Kuller D, Thomas RJ, **Westover MB**. "Sleep staging in the ICU with heart rate variability and breathing signals. An exploratory cross-sectional study using deep neural networks." *Front Netw Physiol*. 2023;3:1120390. doi: 10.3389/fnetp.2023.1120390. Epub 2023 Feb 27. PMID: PMC10013021.
262. Chwalisz BK, Le VK, Cheng JR, Jain A, Brandon **Westover M**, Cheng HT. "COVID-19-induced headache in Boston and the vicinity." *J Clin Virol Plus*. 2023 Jun;3(2):100148. doi: 10.1016/j.jcvp.2023.100148. Epub 2023 Apr 07. PMID: PMC10079317.
263. Jing J, Ge W, Struck AF, Fernandes MB, Hong S, An S, Fatima S, Herlopian A, Karakis I, Halford JJ, Ng MC, Johnson EL, Appavu BL, Sarkis RA, Osman G, Kaplan PW, Dhakar MB, Jayagopal LA, Sheikh Z, Taraschenko O, Schmitt S, Haider HA, Kim JA, Swisher CB, Gaspard N, Cervenka MC, Rodriguez Ruiz AA, Lee JW, Tabaeizadeh M, Gilmore EJ, Nordstrom K, Yoo JY, Holmes MG, Herman ST, Williams JA, Pathmanathan J, Nascimento FA, Fan Z, Nasiri S, Shafi MM, Cash SS, Hoch DB, Cole AJ, Rosenthal ES, Zafar SF, Sun J, **Westover MB**. "Interrater Reliability of Expert Electroencephalographers Identifying Seizures and Rhythmic and Periodic Patterns in EEGs." *Neurology*. 2023 Apr 25;100(17):e1737-e1749. doi: 10.1212/WNL.0000000000201670. Epub 2022 Dec 02. PMID: PMC10136018.
264. Coelho LMG, Blacker D, Hsu J, Newhouse JP, **Westover MB**, Zafar SF, Moura LMVR. "Association of Early Seizure Prophylaxis With Posttraumatic Seizures and Mortality: A Meta-analysis With Evidence Quality Assessment." *Neurol Clin Pract*. 2023 Jun;13(3):e200145. doi: 10.1212/CPJ.0000000000200145. Epub 2023 Mar 30. PMID: PMC10101717.

265. Boncompte G, Sun H, Elgueta MF, Benavides J, Carrasco M, Morales MI, Calderón N, Contreras V, **Westover MB**, Cortínez LI, Akeju O, Pedemonte JC. "Intraoperative electroencephalographic marker of preoperative frailty: A prospective cohort study." *J Clin Anesth*. 2023 Jun;86:111069. doi: 10.1016/j.jclinane.2023.111069. Epub 2023 Feb 02. PMID: PMC10074446.
266. McInnis RP, Ayub MA, Jing J, Halford JJ, Mateen FJ, Brandon **Westover M**. "Epilepsy diagnosis using a clinical decision tool and artificially intelligent electroencephalography." *Epilepsy Behav*. 2023 Apr;141:109135. doi: 10.1016/j.yebeh.2023.109135. Epub 2023 Mar 03. PMID: PMC10082472.
267. Goldenholz DM, Sun H, Ganglberger W, **Westover MB**. "Sample Size Analysis for Machine Learning Clinical Validation Studies." *Biomedicines*. 2023 Feb 23;11(3). doi: 10.3390/biomedicines11030685. PMID: PMC10045793.
268. Jing J, Ge W, Hong S, Fernandes MB, Lin Z, Yang C, An S, Struck AF, Herlopian A, Karakis I, Halford JJ, Ng MC, Johnson EL, Appavu BL, Sarkis RA, Osman G, Kaplan PW, Dhakar MB, Arcot Jayagopal L, Sheikh Z, Taraschenko O, Schmitt S, Haider HA, Kim JA, Swisher CB, Gaspard N, Cervenka MC, Rodriguez Ruiz AA, Lee JW, Tabaeizadeh M, Gilmore EJ, Nordstrom K, Yoo JY, Holmes MG, Herman ST, Williams JA, Pathmanathan J, Nascimento FA, Fan Z, Nasiri S, Shafi MM, Cash SS, Hoch DB, Cole AJ, Rosenthal ES, Zafar SF, Sun J, **Westover MB**. "Development of Expert-Level Classification of Seizures and Rhythmic and Periodic Patterns During EEG Interpretation." *Neurology*. 2023 Apr 25;100(17):e1750-e1762. doi: 10.1212/WNL.0000000000207127. Epub 2023 Mar 06. PMID: PMC10136013.
269. Fernandes M, **Westover MB**, Zafar SF. "Identifying inpatient hospitalizations with continuous electroencephalogram monitoring from administrative data." *Res Sq*. 2023 May 08. doi: 10.21203/rs.3.rs-2882806/v1. PMID: PMC10197757.
270. Chen PM, Stekhoven SS, Haider A, Jing J, Ge W, Rosenthal ES, **Westover MB**, Zafar SF. "Association of Epileptiform Activity With Outcomes in Toxic-Metabolic Encephalopathy." *Crit Care Explor*. 2023 May;5(5):e0913. doi: 10.1097/CCE.0000000000000913. Epub 2023 May 05. PMID: PMC10166342.
271. Lin Z, Glass L, Brandon **Westover M**, Xiao C, Sun J. "SCRIB: Set-Classifier with Class-Specific Risk Bounds for Blackbox Models." *Proc AAAI Conf Artif Intell*. 2022 Jun 30;36(7):7497-7505. doi: 10.1609/aaai.v36i7.20714. PMID: PMC10155818.
272. Nasiri S, Ganglberger W, Sun H, Thomas RJ, **Westover MB**. "Exploiting labels from multiple experts in automated sleep scoring." *Sleep*. 2023 May 10;46(5). doi: 10.1093/sleep/zsad034. PMID: PMC10171620.
273. Chinappen DM, Xiao G, Jing J, Spencer ER, Eden UT, Kramer MA, **Westover MB**, Chu CJ. "Spike height improves prediction of future seizure risk." *Clin Neurophysiol*. 2023 Jun;150:49-55. doi: 10.1016/j.clinph.2023.02.180. Epub 2023 Mar 21. PMID: PMC10192090.
274. Adra N, Dümmer LW, Paixao L, Tesh RA, Sun H, Ganglberger W, Westmeijer M, Da Silva Cardoso M, Kumar A, Ye E, Henry J, Cash SS, Kitchener E, Leveroni CL, Au R, Rosand J, Salinas J, Lam AD, Thomas RJ, **Westover MB**. "Decoding information about cognitive health from the brainwaves of sleep." *Sci Rep*. 2023 Jul 15;13(1):11448. doi: 10.1038/s41598-023-37128-7. PMID: PMC10349883.
275. Rathmell CS, Akeju O, Inouye SK, **Westover MB**. "Estimating the number of cases of dementia that might be prevented by preventing delirium." *Br J Anaesth*. 2023 Jun;130(6):e477-e478. doi: 10.1016/j.bja.2023.03.001. Epub 2023 Apr 06. PMID: PMC10329187.
276. Wilcox DR, Rudmann EA, Ye E, Noori A, Magdamo C, Jain A, Alabsi H, Foy B, Triant VA, Robbins GK, **Westover MB**, Das S, Mukerji SS. "Cognitive concerns are a risk factor for mortality in people with HIV and coronavirus disease 2019." *AIDS*. 2023 Aug 01;37(10):1565-1571. doi: 10.1097/QAD.0000000000003595. Epub 2023 May 09. PMID: PMC10355333.
277. Sinha N, Dauwels J, Kaiser M, Cash SS, **Westover MB**, Wang Y, Taylor PN. "Reply: Computer models to inform epilepsy surgery strategies: prediction of postoperative outcome." *Brain*. 2017 May 01;140(5):e31. doi: 10.1093/brain/awx068. PMID: PMC10448005.
278. Ramantani G, **Westover MB**, Gliske S, Sarnthein J, Sarma S, Wang Y, Baud MO, Stacey WC, Conrad EC. "Passive and active markers of cortical excitability in epilepsy." *Epilepsia*. 2023 Dec;64 Suppl 3(Suppl 3):S25-S36. doi: 10.1111/epi.17578. Epub 2023 Mar 22. PMID: PMC10512778.

279. Amorim E, Zheng WL, Jing J, Ghassemi MM, Lee JW, Wu O, Herman ST, Pang T, Sivaraju A, Gaspard N, Hirsch L, Ruijter BJ, Tjepkema-Cloostermans MC, Hofmeijer J, van Putten MJAM, **Westover MB**. "Neurophysiology State Dynamics Underlying Acute Neurologic Recovery After Cardiac Arrest." *Neurology*. 2023 Aug 29;101(9):e940-e952. doi: 10.1212/WNL.0000000000207537. Epub 2023 Jul 06. PMID: PMC10501085.
280. Berger M, Ryu D, Reese M, McGuigan S, Evered LA, Price CC, Scott DA, **Westover MB**, Eckenhoff R, Bonanni L, Sweeney A, Babiloni C. "A Real-Time Neurophysiologic Stress Test for the Aging Brain: Novel Perioperative and ICU Applications of EEG in Older Surgical Patients." *Neurotherapeutics*. 2023 Jul;20(4):975-1000. doi: 10.1007/s13311-023-01401-4. Epub 2023 Jul 12. PMID: PMC10457272.
281. Ryan SL, Liu X, McKenna V, Ghanta M, Muniz C, Renwick R, **Westover MB**, Kimchi EY. "Associations between early in-hospital medications and the development of delirium in patients with stroke." *J Stroke Cerebrovasc Dis*. 2023 Sep;32(9):107249. doi: 10.1016/j.jstrokecerebrovasdis.2023.107249. Epub 2023 Aug 01. PMID: PMC10529367.
282. Leone MJ, Dashti HS, Coughlin B, Tesh RA, Quadri SA, Bucklin AA, Adra N, Krishnamurthy PV, Ye EM, Hemmige A, Rajan S, Panneerselvam E, Higgins J, Ayub MA, Ganglberger W, Paixao L, Houle TT, Thompson BT, Johnson-Akeju O, Saxena R, Kimchi E, Cash SS, Thomas RJ, **Westover MB**. "Sound and light levels in intensive care units in a large urban hospital in the United States." *Chronobiol Int*. 2023 Jun 03;40(6):759-768. doi: 10.1080/07420528.2023.2207647. Epub 2023 May 05. PMID: PMC10524721.
283. Namirembe GE, Baker S, Albanese M, Mueller A, Qu JZ, Mekonnen J, Wiredu K, **Westover MB**, Houle TT, Akeju O, Minimizing Intensive Care Unit Neurological Dysfunction with Dexmedetomidine-Induced Sleep Study Team. "Association Between Postoperative Delirium and Long-Term Subjective Cognitive Decline in Older Patients Undergoing Cardiac Surgery: A Secondary Analysis of the Minimizing Intensive Care Unit Neurological Dysfunction with Dexmedetomidine-Induced Sleep Trial." *J Cardiothorac Vasc Anesth*. 2023 Sep;37(9):1700-1706. doi: 10.1053/j.jvca.2023.04.035. Epub 2023 Apr 30. PMID: PMC10524446.
284. Parikh H, Hoffman K, Sun H, Zafar SF, Ge W, Jing J, Liu L, Sun J, Struck A, Volfovsky A, Rudin C, **Westover MB**. "Effects of epileptiform activity on discharge outcome in critically ill patients in the USA: a retrospective cross-sectional study." *Lancet Digit Health*. 2023 Aug;5(8):e495-e502. doi: 10.1016/S2589-7500(23)00088-2. Epub 2023 Jun 07. PMID: PMC10528143.
285. Eckhardt CA, Sun H, Malik P, Quadri S, Santana Firme M, Jones DK, van Sleuwen M, Jain A, Fan Z, Jing J, Ge W, Danish HH, Jacobson CA, Rubin DB, Kimchi EY, Cash SS, Frigault MJ, Lee JW, Dietrich J, **Westover MB**. "Automated detection of immune effector cell-associated neurotoxicity syndrome via quantitative EEG." *Ann Clin Transl Neurol*. 2023 Oct;10(10):1776-1789. doi: 10.1002/acn3.51866. Epub 2023 Aug 06. PMID: PMC10578889.
286. Singh SD, Oreskovic T, Carr S, Papier K, Conroy M, Senff JR, Chemali Z, Gutierrez-Martinez L, Parodi L, Mayerhofer E, Marini S, Nunley C, Newhouse A, Ouyang A, Brouwers HB, **Westover B**, Rivier C, Falcone G, Howard V, Howard G, Pikula A, Ibrahim S, Sheth KN, Yechoor N, Lazar RM, Anderson CD, Tanzi RE, Fricchione G, Littlejohns T, Rosand J. "The predictive validity of a Brain Care Score for dementia and stroke: data from the UK Biobank cohort." *Front Neurol*. 2023;14:1291020. doi: 10.3389/fneur.2023.1291020. Epub 2023 Dec 01. PMID: PMC10725202.
287. Ge W, Rice HJ, Sheikh IS, **Westover MB**, Weathers AL, Jones LK, Moura L. "Improving Neurology Clinical Care With Natural Language Processing Tools." *Neurology*. 2023 Nov 27;101(22):1010-1018. doi: 10.1212/WNL.0000000000207853. PMID: PMC10727205.
288. Yang C, Qian C, Singh N, Xiao C, **Westover MB**, Solomonik E, Sun J. "ATD: Augmenting CP Tensor Decomposition by Self Supervision." *Adv Neural Inf Process Syst*. 2022;35:32039-32052. PMID: PMC10664864.
289. Yang C, Xiao C, **Westover MB**, Sun J. "Self-Supervised Electroencephalogram Representation Learning for Automatic Sleep Staging: Model Development and Evaluation Study." *JMIR AI*. 2023;2(1):e46769. doi: 10.2196/46769. Epub 2023 Jul 26. PMID: PMC10715804.
290. Sun H, Adra N, Ayub MA, Ganglberger W, Ye E, Fernandes M, Paixao L, Fan Z, Gupta A, Ghanta M, Moura Junior VF, Rosand J, **Westover MB**, Thomas RJ. "Assessing Risk of Health Outcomes From Brain

- Activity in Sleep: A Retrospective Cohort Study.” *Neurol Clin Pract*. 2024 Feb;14(1):e200225. doi: 10.1212/CPJ.0000000000200225. Epub 2023 Dec 08. PMID: PMC10759032.
291. Ke SY, Wu H, Sun H, Zhou A, Liu J, Zheng X, Liu K, **Westover MB**, Xu H, Kong XJ. “Classification of autism spectrum disorder using electroencephalography in Chinese children: a cross-sectional retrospective study.” *Front Neurosci*. 2024;18:1330556. doi: 10.3389/fnins.2024.1330556. Epub 2024 Jan 25. PMID: PMC10850305.
 292. Fernandes M, Sun H, Chemali Z, Mukerji SS, M V R Moura L, Zafar SF, Sonni A, Biffi A, Rosand J, Brandon **Westover M**. “Brain health scores to predict neurological outcomes from electronic health records.” *Int J Med Inform*. 2023 Dec;180:105270. doi: 10.1016/j.ijmedinf.2023.105270. Epub 2023 Oct 24. PMID: PMC10842359.
 293. Wei R, Ganglberger W, Sun H, Hadar PN, Gollub RL, Pieper S, Billot B, Au R, Eugenio Iglesias J, Cash SS, Kim S, Shin C, **Westover MB**, Joseph Thomas R. “Linking brain structure, cognition, and sleep: insights from clinical data.” *Sleep*. 2024 Feb 08;47(2). doi: 10.1093/sleep/zsad294. PMID: PMC10851868.
 294. Dhakal K, Rosenthal ES, Kulpanowski AM, Dodelson JA, Wang Z, Cudemus-Deseda G, Villien M, Edlow BL, Presciutti AM, Januzzi JL, Ning M, Taylor Kimberly W, Amorim E, Brandon **Westover M**, Copen WA, Schaefer PW, Giacino JT, Greer DM, Wu O. “Increased task-relevant fMRI responsiveness in comatose cardiac arrest patients is associated with improved neurologic outcomes.” *J Cereb Blood Flow Metab*. 2024 Jan;44(1):50-65. doi: 10.1177/0271678X231197392. Epub 2023 Sep 20. PMID: PMC10905635.
 295. Goldenholz DM, Karoly PJ, Viana PF, Nurse E, Loddenkemper T, Schulze-Bonhage A, Vieluf S, Bruno E, Nasser M, Richardson MP, Brinkmann BH, **Westover MB**. “Minimum clinical utility standards for wearable seizure detectors: A simulation study.” *Epilepsia*. 2024 Apr;65(4):1017-1028. doi: 10.1111/epi.17917. Epub 2024 Feb 17. PMID: PMC11018505.
 296. Sun H, Li P, Gao L, Yang J, Yu L, Buchman AS, Bennett DA, **Westover MB**, Hu K. “Altered Motor Activity Patterns within 10-Minute Timescale Predict Incident Clinical Alzheimer’s Disease.” *J Alzheimers Dis*. 2024;98(1):209-220. doi: 10.3233/JAD-230928. PMID: PMC10977378.
 297. Barnett AJ, Guo Z, Jing J, Ge W, Kaplan PW, Kong WY, Karakis I, Herlopian A, Jayagopal LA, Taraschenko O, Selioutski O, Osman G, Goldenholz D, Rudin C, **Westover MB**. “Improving Clinician Performance in Classifying EEG Patterns on the Ictal-Interictal Injury Continuum Using Interpretable Machine Learning.” *NEJM AI*. 2024 Jun;1(6). doi: 10.1056/aioa2300331. Epub 2024 May 23. PMID: PMC11175595.
 298. Goldenholz DM, Eccleston C, Moss R, **Westover MB**. “Prospective validation of a seizure diary forecasting falls short.” *Epilepsia*. 2024 Jun;65(6):1730-1736. doi: 10.1111/epi.17984. Epub 2024 Apr 12. PMID: PMC11166505.
 299. Goldenholz D, Brinkmann BH, **Westover MB**. “How accurate do self-reported seizures need to be for effective medication management in epilepsy?.” *Epilepsia*. 2024 Jul;65(7):e104-e112. doi: 10.1111/epi.18019. Epub 2024 May 22. PMID: PMC11251847.
 300. Parikh H, Sun H, Amerineni R, Rosenthal ES, Volfovsky A, Rudin C, **Westover MB**, Zafar SF. “How many patients do you need? Investigating trial designs for anti-seizure treatment in acute brain injury patients.” *Ann Clin Transl Neurol*. 2024 Jul;11(7):1681-1690. doi: 10.1002/acn3.52059. Epub 2024 Jun 12. PMID: PMC11251465.
 301. Ouyang A, Zhang C, Adra N, Tesh RA, Sun H, Lei D, Jing J, Fan P, Paixao L, Ganglberger W, Briggs L, Salinas J, Bevers MB, Wrann CD, Chemali Z, Fricchione G, Thomas RJ, Rosand J, Tanzi RE, **Westover MB**. “Effects of Aerobic Exercise on Brain Age and Health in Middle-Aged and Older Adults: A Single-Arm Pilot Clinical Trial.” *Life (Basel)*. 2024 Jul 08;14(7). doi: 10.3390/life14070855. PMID: PMC11278044.
 302. Singh SD, Rivier CA, Papier K, Chemali Z, Gutierrez-Martinez L, Parodi L, Mayerhofer E, Senff J, Clocchiatti-Tuozzo S, Nunley C, Newhouse A, Ouyang A, **Westover MB**, Tanzi RE, Lazar RM, Pikula A, Ibrahim S, Brouwers HB, Howard VJ, Howard G, Yechoor N, Littlejohns T, Sheth KN, Rosand J, Fricchione G, Anderson CD, Falcone GJ. “The predictive validity of a Brain Care Score for late-life

- depression and a composite outcome of dementia, stroke, and late-life depression: data from the UK Biobank cohort.” *Front Psychiatry*. 2024;15:1373797. doi: 10.3389/fpsyt.2024.1373797. Epub 2024 Jul 23. PMID: PMC11301016.
303. Nascimento FA, Jing J, Traner C, Kong WY, Olandoski M, Kapur S, Duhaime E, Strowd R, Moeller J, **Westover MB**. “A randomized controlled educational pilot trial of interictal epileptiform discharge identification for neurology residents.” *Epileptic Disord*. 2024 Aug;26(4):444-459. doi: 10.1002/epd2.20229. Epub 2024 Apr 26. PMID: PMC11329359.
 304. Marcinski Nascimento KJ, **Westover MB**, Nascimento FA. “Teaching the 6 EEG Spectrogram Patterns Using an Infographic.” *Neurol Educ*. 2024 Dec 25;3(4):e200158. doi: 10.1212/NE9.0000000000200158. Epub 2024 Sep 25. PMID: PMC11436318.
 305. Luccarelli J, Smith JR, Fricchione G, **Westover MB**. “Generalized Periodic Discharges Associated With Catatonia and Delirium: A Case Series.” *J Neuropsychiatry Clin Neurosci*. 2024;36(4):340-343. doi: 10.1176/appi.neuropsych.20230174. Epub 2024 May 09. PMID: PMC11479820.
 306. Fernandes M, Cardall A, Moura LM, McGraw C, Zafar SF, **Westover MB**. “Extracting seizure control metrics from clinic notes of patients with epilepsy: A natural language processing approach.” *Epilepsy Res*. 2024 Nov;207:107451. doi: 10.1016/j.eplepsyres.2024.107451. Epub 2024 Sep 10. PMID: PMC11499027.
 307. Sanches PR, Tabaeizadeh M, Moura LMVR, Rosenthal ES, Caboclo LO, Hsu J, Patorno E, **Westover MB**, Zafar SF. “Anti-seizure medication treatment and outcomes in acute ischemic stroke patients undergoing continuous EEG monitoring.” *Neurol Sci*. 2022 Sep;43(9):5441-5449. doi: 10.1007/s10072-022-06183-9. Epub 2022 Jun 17. PMID: PMC11550088.
 308. Thangavel P, Thomas J, Sinha N, Peh WY, Yuvaraj R, Cash SS, Chaudhari R, Karia S, Jing J, Rathakrishnan R, Saini V, Shah N, Srivastava R, Tan YL, **Westover B**, Dauwels J. “Improving automated diagnosis of epilepsy from EEGs beyond IEDs.” *J Neural Eng*. 2022 Nov 24;19(6). doi: 10.1088/1741-2552/ac9c93. PMID: PMC11549972.
 309. McGraw CM, Rao S, Manjunath S, Jing J, Brandon **Westover M**. “Automated quantification of periodic discharges in human electroencephalogram.” *Biomed Phys Eng Express*. 2024 Sep 20;10(6). doi: 10.1088/2057-1976/ad6c53. PMID: PMC11580140.
 310. Ng MC, Zafar S, Foreman B, Kim J, Struck AF, **Westover MB**. “Commentary on stimulus-induced arousal with transient electroencephalographic improvement distinguishes nonictal from ictal generalized periodic discharges.” *Epilepsia*. 2024 Dec;65(12):3484-3487. doi: 10.1111/epi.18159. Epub 2024 Oct 18. PMID: PMC11649443.
 311. Chang CY, Zhang B, Moss R, Picard R, **Westover MB**, Goldenholz D. “Necessary for seizure forecasting outcome metrics: Seizure frequency and benchmark model.” *Epilepsy Res*. 2024 Dec;208:107474. doi: 10.1016/j.eplepsyres.2024.107474. Epub 2024 Nov 08. PMID: PMC11614381.
 312. Sun H, Shen S, Thomas RJ, **Westover MB**, Zhang C. “Sleep as a window to understand and regulate Alzheimer's disease: emerging roles of thalamic reticular nucleus.” *Neural Regen Res*. 2025 Jun 01;20(6):1711-1712. doi: 10.4103/NRR.NRR-D-24-00351. Epub 2024 Jul 10. PMID: PMC11688547.
 313. He H, Li C, Ganglberger W, Gallagher K, Hristov R, Ouroutzoglou M, Sun H, Sun J, **Westover MB**, Katabi D. “What radio waves tell us about sleep!.” *Sleep*. 2025 Jan 13;48(1). doi: 10.1093/sleep/zsae187. PMID: PMC11725512.
 314. Hadar PN, Westmeijer M, Sun H, Meulenbrugge EJ, Jing J, Paixao L, Tesh RA, Da Silva Cardoso M, Arnal P, Au R, Shin C, Kim S, Thomas RJ, Cash SS, **Westover MB**. “Epilepsy is associated with the accelerated aging of brain activity in sleep.” *Front Physiol*. 2024;15:1458592. doi: 10.3389/fphys.2024.1458592. Epub 2024 Nov 28. PMID: PMC11634596.
 315. Fernandes M, Gallagher K, Turley N, Gupta A, **Westover MB**, Singhal AB, Zafar SF. “Automated extraction of post-stroke functional outcomes from unstructured electronic health records.” *Eur Stroke J*. 2025 Sep;10(3):829-836. doi: 10.1177/23969873251314340. Epub 2025 Jan 22. PMID: PMC11752148.
 316. Rivier CA, Singh S, Senff J, Tack RW, Marini S, Clocchiatti-Tuozzo S, Huo S, Renedo D, Papier K, Conroy M, Littlejohns TJ, Chemali Z, Kourkoulis C, Payabvash S, Newhouse A, **Westover MB**, Lazar

- RM, Pikula A, Ibrahim S, Howard VJ, Howard G, Brouwers HB, Van Duijn CM, Fricchione G, Tanzi RE, Yechoor N, Sheth KN, Anderson CD, Rosand J, Falcone GJ. "Brain Care Score and Neuroimaging Markers of Brain Health in Asymptomatic Middle-Age Persons." *Neurology*. 2024 Aug 27;103(4):e209687. doi: 10.1212/WNL.0000000000209687. Epub 2024 Jul 25. PMID: PMC11760050.
317. Shivdat S, Zhan T, De Palma A, Zheng WL, Krishnamurthy P, Paneerselvam E, Snider S, Bevers M, O'Reilly UM, Lee JW, **Westover MB**, Amorim E. "Early Burst Suppression Similarity Association with Structural Brain Injury Severity on MRI After Cardiac Arrest." *Neurocrit Care*. 2025 Feb;42(1):175-184. doi: 10.1007/s12028-024-02047-6. Epub 2024 Jul 24. PMID: PMC11757804.
318. Nassi TE, Oppersma E, Labarca G, Donker DW, **Westover MB**, Thomas RJ. "Morphological Prediction of Continuous Positive Airway Pressure-associated Acute Respiratory Instability." *Ann Am Thorac Soc*. 2025 Jan;22(1):138-149. doi: 10.1513/AnnalsATS.202311-979OC. PMID: PMC11708763.
319. Sun S, Lomachinsky V, Smith LH, Newhouse JP, **Westover MB**, Blacker DL, Schwamm LH, Haneuse S, Moura LMVR. "Benzodiazepine Initiation and the Risk of Falls or Fall-Related Injuries in Older Adults Following Acute Ischemic Stroke." *Neurol Clin Pract*. 2025 Jun;15(3):e200452. doi: 10.1212/CPJ.0000000000200452. Epub 2025 Mar 18. PMID: PMC11936338.
320. Fernandes M, **Westover MB**, Singhal AB, Zafar SF. "Automated Extraction of Stroke Severity From Unstructured Electronic Health Records Using Natural Language Processing." *J Am Heart Assoc*. 2024 Nov 05;13(21):e036386. doi: 10.1161/JAHA.124.036386. Epub 2024 Oct 25. PMID: PMC11935650.
321. Akas Z, Jing J, **Westover MB**, Zafar SF. "Using artificial intelligence to optimize anti-seizure treatment and EEG-guided decisions in severe brain injury." *Neurotherapeutics*. 2025 Jan;22(1):e00524. doi: 10.1016/j.neurot.2025.e00524. Epub 2025 Jan 23. PMID: PMC11840355.
322. Freedman IG, Boncompagni G, Qu JZ, Khawaja ZQ, Turco I, Mueller A, Wiredu K, McKay TB, **Westover MB**, Pedemonte JC, Akeju O. "Anesthesia-induced electroencephalogram oscillations and perioperative outcomes in older adults undergoing cardiac surgery." *J Clin Anesth*. 2025 Mar;102:111770. doi: 10.1016/j.jclinane.2025.111770. Epub 2025 Feb 07. PMID: PMC11953626.
323. Goldenholz DM, Goldenholz SR, Habib S, **Westover MB**. "Inductive reasoning with large language models: A simulated randomized controlled trial for epilepsy." *Epilepsy Res*. 2025 Mar;211:107532. doi: 10.1016/j.epilepsyres.2025.107532. Epub 2025 Feb 24. PMID: PMC11908886.
324. Ge W, Godeiro Coelho LM, Donahue MA, Rice HJ, Blacker D, Hsu J, Newhouse JP, Hernández-Díaz S, Haneuse S, **Westover B**, Moura LMVR. "Automated identification of fall-related injuries in unstructured clinical notes." *Am J Epidemiol*. 2025 Apr 08;194(4):1097-1105. doi: 10.1093/aje/kwae240. PMID: PMC11978607.
325. Azarian M, Ramezani A, Sharafkhaneh A, Maghsoudi A, Kryger M, Thomas RJ, **Westover MB**, Razjouyan J. "The Association between All-Cause Mortality and Obstructive Sleep Apnea in Adults: A U-Shaped Curve." *Ann Am Thorac Soc*. 2025 Apr;22(4):581-590. doi: 10.1513/AnnalsATS.202407-755OC. PMID: PMC12005042.
326. Shaw KM, Shao YP, Ghanta M, Junior VM, Kimchi EY, Houle TT, Akeju O, **Westover MB**. "Daily Automated Prediction of Delirium Risk in Hospitalized Patients: Model Development and Validation." *JMIR Med Inform*. 2025 Apr 18;13:e60442. doi: 10.2196/60442. PMID: PMC12048784.
327. Sumsion D, Davis E, Fernandes M, Wei R, Milde R, Veltink JM, Kong WY, Xiong Y, Rao S, **Westover T**, Petersen L, Turley N, Singh A, Buss S, Mukerji S, Zafar S, Das S, Junior VM, Ghanta M, Gupta A, Kim J, Stone K, Mignot E, Hwang D, Trotti LM, Clifford GD, Katwa U, Thomas R, **Westover MB**, Sun H. "Identification of Patients With Congestive Heart Failure From the Electronic Health Records of Two Hospitals: Retrospective Study." *JMIR Med Inform*. 2025 Apr 10;13:e64113. doi: 10.2196/64113. PMID: PMC12022513.
328. Sun H, Dang R, Haack M, Hauser K, Scott-Sutherland J, **Westover MB**, Parthasarathy S, Redline S, Thomas RJ, Mullington JM. "Facility-measured nocturnal hypoxemia and sleep among adults with long COVID versus age- and sex-matched healthy adults: a preliminary observational study." *Sleep Adv*. 2025 Apr;6(2):zpaf017. doi: 10.1093/sleepadvances/zpaf017. Epub 2025 Mar 22. PMID: PMC12070477.

329. Zhou S, Song G, Sun H, Zhang D, Leng Y, **Westover MB**, Hong S. "Continuous sleep depth index annotation with deep learning yields novel digital biomarkers for sleep health." NPJ Digit Med. 2025 Apr 11;8(1):203. doi: 10.1038/s41746-025-01607-0. PMCID: PMC11992070.
330. Ganglberger W, Nasiri S, Sun H, Kim S, Shin C, **Westover MB**, Thomas RJ. "Refining sleep staging accuracy: transfer learning coupled with scorability models." Sleep. 2024 Nov 08;47(11). doi: 10.1093/sleep/zsae202. PMCID: PMC12104150.
331. Baizer L, Bures R, Nadkarni G, Reyes-Guzman C, Ladwa S, Cade B, **Westover MB**, Durmer J, de Zambotti M, Desai M, Parekh A, Si B, Fernandez-Mendoza J, Minor K, Mazzotti DR, Lee S, Katabi D, Kiss O, Spira AP, Morris J, Seixas A, Kioumourtzoglou MA, Bridges JFP, Brown M, Hale L, Purcell S. "Big data approaches for novel mechanistic insights on sleep and circadian rhythms: a workshop summary." Sleep. 2025 Jun 13;48(6). doi: 10.1093/sleep/zsaf035. PMCID: PMC12163129.
332. Krumm L, Kranz DD, Halimeh M, Nelde A, Amorim E, Zafar S, Jing J, Thomas RJ, **Westover MB**, Meisel C. "Toward a Universal Map of EEG: A Semantic, Low-Dimensional Manifold for EEG Classification, Clustering, and Prognostication." Ann Neurol. 2025 Aug;98(2):357-368. doi: 10.1002/ana.27260. Epub 2025 Jun 20. PMCID: PMC12278028.
333. **Westover AM, Westover TM, Westover MB**. "Interrater Reliability Estimation via Maximum Likelihood for Gwet's Chance Agreement Model." Open J Stat. 2024 Oct;14(5):481-491. doi: 10.4236/ojs.2024.145021. Epub 2024 Oct 28. PMCID: PMC12061032.
334. Li J, Goldenholz DM, Alkofer M, Sun C, Nascimento FA, Halford JJ, Dean BC, Galanti M, Struck AF, Greenblatt AS, Lam AD, Herlopian A, Nwankwo C, Weber D, Maus D, Haider HA, Karakis I, Yoo JY, Ng MC, Selioutski O, Taraschenko O, Osman G, Katyal R, Schmitt SE, Benbadis S, Cash SS, Tatum WO, Sheikh Z, Kong WY, Bayas G, Turley N, Hong S, **Westover MB**, Jing J. "Expert-Level Detection of Epilepsy Markers in EEG on Short and Long Timescales." NEJM AI. 2025 Jul;2(7). doi: 10.1056/aioa2401221. Epub 2025 Jun 26. PMCID: PMC12276842.
335. Leng Y, He Y, Amini S, Magdamo C, Paschalidis I, Mukerji SS, Moura LMVR, **Westover MB**, Vranceanu AM, Ritchie CS, Blacker D, Dickson JR, Das S. "A GPT-4o-powered framework for identifying cognitive impairment stages in electronic health records." NPJ Digit Med. 2025 Jul 03;8(1):401. doi: 10.1038/s41746-025-01834-5. PMCID: PMC12229571.
336. McDevitt WM, Jones TJ, Quinn L, Easter CL, Jing J, **Westover MB**, Scholefield BR, Seri S, Drury NE. "Surgery on the aortic arch and feasibility of electroencephalography (SAFE) monitoring in neonates: protocol for a prospective observational cohort study." BMJ Open. 2025 Jul 10;15(7):e106423. doi: 10.1136/bmjopen-2025-106423. PMCID: PMC12258354.
337. Alkofer M, Yang C, Ganglberger W, Beal J, Hegde M, Kang JY, Yoo JY, Gelfand MA, Thio LL, Kutluay E, Campbell Z, Schmitt S, Gleichgerricht E, Waterhouse E, Lopez MR, Eisenschenk S, Galanti M, Singh RK, Wills KE, Meulenbrugge EJ, Dlugos D, Dean B, Halford JJ, Goldenholz D, Jing J, Thomas R, **Westover MB**. "Automated detection of interictal epileptiform discharges with few electroencephalographic channels." Epilepsia. 2025 May 03;66(7):e114-20. doi: 10.1111/epi.18431. Epub ahead of print. PMCID: PMC12291011.
338. Li J, Aguirre AD, Moura V, Jin J, Liu C, Zhong L, Sun C, Clifford G, **Westover MB**, Hong S. "An Electrocardiogram Foundation Model Built on over 10 Million Recordings." NEJM AI. 2025 Jul;2(7). doi: 10.1056/aioa2401033. Epub 2025 Jun 26. PMCID: PMC12327759.
339. Yuan D, Katyal R, Sheikh I, Karakis I, Benbadis S, Amin U, Vinayan KP, Barot N, Weber D, Greenblatt A, Beniczky S, **Westover MB**, Nascimento FA. "Utility of the IFCN criteria for identifying interictal epileptiform discharges by experts: A decision hygiene approach to improve inter-rater reliability." Clin Neurophysiol. 2025 May;173:138-146. doi: 10.1016/j.clinph.2025.02.275. Epub 2025 Mar 08. PMCID: PMC12335889.
340. Tesh RA, Zahoor A, Banks J, Gallagher K, Eckhardt CA, Sun H, Karakis I, Katyal R, Williams J, Nayak C, Herlopian A, Ng MC, Greenblatt AS, Meyers E, Westmeijer M, Harrison DS, Ganglberger W, Galina Gheihman, Fan T, Struck AF, Sheikh IS, Nascimento FA, **Westover MB**. "Inter-Rater Reliability of EEG-Based Encephalopathy Grading." J Clin Neurophysiol. 2026 Mar 01;43(3):237-244. doi: 10.1097/WNP.0000000000001185. Epub 2025 Jul 02. PMCID: PMC12335890.

341. Caffarelli M, Simmons R, Tolokh I, Karukonda V, Guterman EL, Smith W, Fox CK, **Westover MB**, Amorim E. "A Quantitative Electroencephalographic Index for Stroke Detection in Adults." *J Clin Neurophysiol*. 2026 Jan 01;43(1):23-31. doi: 10.1097/WNP.0000000000001151. Epub 2025 Mar 05. PMID: PMC12360042.
342. Sun C, Jing J, Turley N, Alcott C, Kang WY, Cole AJ, Goldenholz DM, Lam A, Amorim E, Chu C, Cash S, Junior VM, Gupta A, Ghanta M, Nearing B, Nascimento FA, Struck A, Kim J, Sartipi S, Tauton AM, Fernandes M, Sun H, Bayas G, Gallagher K, Wagenaar JB, Sinha N, Lee-Messer C, Silvers CT, Gunapati B, Rosand J, Peters J, Loddenkemper T, Lee JW, Zafar S, **Westover MB**. "Harvard Electroencephalography Database: A comprehensive clinical electroencephalographic resource from four Boston hospitals." *Epilepsia*. 2025 Sep;66(9):3411-3425. doi: 10.1111/epi.18487. Epub 2025 Jun 04. PMID: PMC12455399.
343. Nasiri S, Ganglberger W, Nassi T, Meulenbrugge EJ, Moura Junior V, Ghanta M, Gupta A, Stone KL, Kjaer MR, Sum-Ping O, Mignot E, Hwang D, Trotti LM, Clifford GD, Katwa U, Sun H, Thomas RJ, **Westover MB**. "CAISR: achieving human-level performance in automated sleep analysis across all clinical sleep metrics." *Sleep*. 2025 Aug 14;48(8). doi: 10.1093/sleep/zsaf134. PMID: PMC12341759.
344. Nassi T, Amidi Y, Oppersma E, Donker DW, Redeker NS, **Westover MB**, Thomas RJ. "Unraveling sleep apnea dynamics: quantifying loop gain using dynamical modeling of ventilatory control." *Sleep*. 2026 Feb 10;49(2). doi: 10.1093/sleep/zsaf213. PMID: PMC12398341.
345. Meulenbrugge EJ, Sun H, Ganglberger W, Nasiri S, Thomas RJ, **Westover MB**. "Ordinal Sleep Depth: A Data-Driven Continuous Measurement of Sleep Depth." *J Sleep Res*. 2026 Feb;35(1):e70074. doi: 10.1111/jsr.70074. Epub 2025 Apr 25. PMID: PMC12353260.
346. Singh A, Sartipi S, Sun H, Milde R, Turley N, Quinn C, Harrold GK, Gillani RL, Turbett SE, Das S, Zafar S, Fernandes M, **Westover MB**, Mukerji SS. "A Machine Learning Approach for Identifying People With Neuroinfectious Diseases in Electronic Health Records: Algorithm Development and Validation." *JMIR Med Inform*. 2025 Aug 29;13:e63157. doi: 10.2196/63157. PMID: PMC12396800.
347. Huang GB, **Westover MB**, Tan EK, Wang H, Cui D, Ma WY, Wang T, He Q, Wei H, Wang N, Tian Q, Lam KY, Yao X, Wong TY. "Artificial intelligence without restriction surpassing human intelligence with probability one: Theoretical insight into secrets of the brain with AI twins of the brain." *Neurocomputing (Amst)*. 2025 Feb 28;619. doi: 10.1016/j.neucom.2024.129053. Epub 2024 Dec 16. PMID: PMC12656881.
348. Kalkach-Aparicio M, Fatima S, Selte A, Sheikh IS, Cormier J, Gallagher K, Avagyan G, Cespedes J, Krishnamurthy PV, Elazim AA, Khan N, Hussein OM, Maganti R, Larocque J, Holla S, Desai M, **Westover B**, Hirsch LJ, Struck AF. "Seizure Assessment and Forecasting With Efficient Rapid-EEG: A Retrospective Multicenter Comparative Effectiveness Study." *Neurology*. 2024 Jul 23;103(2):e209621. doi: 10.1212/WNL.0000000000209621. Epub 2024 Jun 14. PMID: PMC12530332.
349. Siddiquee AT, Hwang YH, Kim S, Lee SK, Lee MH, Kim HJ, Kim YJ, Kim BJ, Hadar PN, **Westover MB**, Thomas RJ, Kim NH, Shin C. "Obstructive Sleep Apnea and Cerebral Microbleeds in Middle-Aged and Older Adults." *JAMA Netw Open*. 2025 Oct 01;8(10):e2539874. doi: 10.1001/jamanetworkopen.2025.39874. PMID: PMC12569715.
350. Tripathi A, Nasiri S, Ganglberger W, Nassi T, Meulenbrugge EJ, Sun H, Stone KL, Mignot E, Hwang D, Trotti LM, Reyna MA, Clifford GD, Katwa U, Thomas RJ, **Westover MB**. "Automated analysis of the American Academy of Sleep Medicine Inter-Scorer Reliability gold-standard polysomnogram dataset." *J Clin Sleep Med*. 2025 Nov 01;21(11):1821-1829. doi: 10.5664/jcsm.11848. PMID: PMC12582208.
351. Donahue MA, Brooks JD, Hsu J, Price M, Blacker D, Schwamm LH, Newhouse JP, **Westover MB**, Haneuse S, Moura LMVR. "Differences in patterns of outpatient epilepsy-specific medication initiation after acute ischemic stroke in the Medicare population." *Epilepsia*. 2025 Aug;66(8):2728-2742. doi: 10.1111/epi.18396. Epub 2025 Apr 04. PMID: PMC12353908.

352. Bayas G, Bayas G, Bayas A, Bayas Cardenas EO, Franco N, Sigcha CM, Shapiro K, Moskowitz S, Espinosa PS, Dey RK, Camia CS, Coates S, Stow C, Turley N, Jing J, **Westover MB**. "Decreasing the Epilepsy Treatment Gap in Tena, Ecuador: A Report From 2021 to 2023." *Cureus*. 2025 Oct;17(10):e94581. doi: 10.7759/cureus.94581. Epub 2025 Oct 14. PMCID: PMC12614661.
353. Pathmanathan J, Little D, Sadeghi K, Di Marco T, Kleinschmidt D, Nerrie J, Tjijartanto N, Arslan A, Hubbard J, Olivieri A, Puryear CB, Brandon **Westover M**, Donoghue J. "The Insomnia EEG Score: a new tool for the classification of people with poor sleep." *Sleep*. 2026 Apr 16;49(4). doi: 10.1093/sleep/zsaf353. PMID: 41217192.
354. **Westover MB**, Kaplan PW, Husain AM. "Controversies: Periodic discharges in critically ill patients -" urgent treatment is essential". *Clin Neurophysiol Pract*. 2025;10:286-291. doi: 10.1016/j.cnp.2025.06.010. Epub 2025 Jul 05. PMCID: PMC12284540.
355. Vlasac IM, Bormes GW, Do E, Benkhokha SS, Diallo N, Fryou NL, Gioia S, Akeju O, Joseph C, Kuan A, Lapan J, Oluwadara D, Team TP, Rahman SA, Saxena R, Scheer FAJL, **Westover MB**, Winkelman JW, Woodson F, Lane JM. "A Novel Home-Based Study of Circadian Rhythms: Design, Rationale, and Methods for the Circadia Study." *Sleep*. 2023 Aug 09. doi: 10.1093/sleep/zsad197. Epub ahead of print. PMID: 37555446.
356. Wang Z, Kulpanowski AM, Copen WA, Rosenthal ES, Dodelson JA, McCrory DE, Edlow BL, Taylor Kimberly W, Amorim E, Brandon **Westover M**, Ning M, Zabihi M, Schaefer PW, Malhotra R, Giacino JT, Greer DM, Wu O. "Automated detection of severe cerebral edema using explainable deep transfer learning after hypoxic ischemic brain injury." *Resuscitation*. 2025 Sep;214:110652. doi: 10.1016/j.resuscitation.2025.110652. Epub 2025 May 23. PMCID: PMC12715856.
357. Sun H, Ganglberger W, Nasiri S, Nassi TE, Meulenbrugge EJ, Lam AD, Zafar S, Gupta A, Ghanta M, Moura Junior VF, Shin C, Au R, Cash SS, Thomas RJ, **Westover MB**. "Health-oriented sleep states: making sleep states reflect health conditions." *Sleep*. 2026 Feb 10;49(2). doi: 10.1093/sleep/zsaf229. PMID: 40795277.
358. Tripathi A, Ganglberger W, Sun H, Alcott C, Turley N, Fitzgerald R, Mitra A, Waters S, Gupta A, Gupta A, Ghanta M, Junior VM, Nasiri S, Nearing B, Stone KL, Mignot E, Hwang D, Reyna MA, Koscova Z, Robichaux C, Zhang Z, Li Q, Ganjoo G, Trotti LM, Clifford GD, Silvers CT, Gunapati B, Thomas RJ, **Westover MB**, Maski K, Katwa U. "The Boston Children's Hospital Sleep Corpus: A Collection of 15,695 Annotated Pediatric Polysomnograms." *Sleep*. 2025 Sep 16. doi: 10.1093/sleep/zsaf273. Epub ahead of print. PMID: 40971987.
359. Ning MH, Rodionov A, Ross JM, Ozdemir RA, Burch M, Lian SJ, Alsop D, Cavallari M, Dickerson B, Fong T, Jones RN, Libermann T, Marcantonio E, Santarnecchi E, Schmitt EM, Touroutoglou A, Travison T, Acker L, Smith MR, Sun H, **Westover MB**, Pascual-Leone Á, Berger M, Inouye SK, Shafi M. "Prediction of postoperative delirium in older adults from preoperative cognition and occipital alpha power from resting-state electroencephalogram." *Age Ageing*. 2025 Oct 30;54(11). doi: 10.1093/ageing/afaf330. PMCID: PMC13171208.
360. Cheng D, Fernandes MB, **Westover MB**, Moura L, Zafar SF. "Real-World Antiseizure Medication Prophylaxis and Outcomes in Hospitalized Adults with Acute Brain Injuries." *Neurocrit Care*. 2026 Apr;44(2):656-665. doi: 10.1007/s12028-025-02345-7. Epub 2025 Oct 01. PMID: 41034682.
361. Chang CY, Moss R, **Westover MB**, Goldenholz DM. "Rigorous evaluation of five models for e-diary-only seizure forecasting: Retrospective and prospective datasets do not outperform the Napkin method." *Epilepsia*. 2026 Feb;67(2):753-761. doi: 10.1111/epi.18677. Epub 2025 Oct 14. PMCID: PMC12959853.
362. Habib M, Cazé de Medeiros R, Ahsan SM, McDonald Wojciechowski A, Donahue MA, Blacker D, Newhouse JP, Schwamm LH, **Westover MB**, Moura LMVR. "Claims-Based Machine Learning Classifier of Modified Rankin Scale in Acute Ischemic Stroke." *J Am Heart Assoc*. 2025 Oct 21;14(20):e041635. doi: 10.1161/JAHA.125.041635. Epub 2025 Oct 14. PMCID: PMC12684556.
363. Nascimento FA, Yuan D, Sheikh IS, Hirsch LJ, Beniczky S, **Westover MB**. "The EEG Talk experience: Lessons in e-teaching EEG." *Epileptic Disord*. 2025 Dec;27(6):1337-1339. doi: 10.1002/epd2.70096. Epub 2025 Sep 04. PMID: 40904123.

364. Kong WY, Nascimento FA, Struck A, Duhaime E, Kapur S, Amorim E, Kapinos G, Rodriguez A, Thomas B, Desai M, Lee JW, **Westover MB**, Jing J. "Evaluating crowdsourcing for ICU EEG annotation: A comparison with expert performance." *Epilepsia*. 2025 Nov;66(11):4366-4380. doi: 10.1111/epi.18547. Epub 2025 Aug 06. PMID: 40768345.
365. Suen C, Ma J, Lin X, Thomas RJ, **Westover MB**, Stone K, Yaffe K, Ensrud KE, Redline S, Tanzi RE, Sun H, Zhang C. "Linking sleep microstructure, blood markers of inflammation and metabolism, and cognition: mediation analysis in the osteoporotic fractures in men study." *Geroscience*. 2026 Feb;48(1):1277-1289. doi: 10.1007/s11357-025-01730-6. Epub 2025 Jun 14. PMCID: PMC12972494.

Peer-reviewed publications (accepted, in press)

1. Nascimento FA, Katyal R, Kass NR, Yuan D, Sirven JI, Westover MB, Beniczky S. "Electroencephalography in clinical practice: results from an AAN Practice Current survey." *Journal of Clinical Neurophysiology*.**
2. Sartipi S, Godfrey DS, Tauţan AM, Fernandes MP, Ghanta M, Gupta A, Nearing B, Kim J, Struck AF, Loddenkemper T, Peters J, Lee JW, Westover MB, Zafar SF. Automated Extraction of Seizures and Ictal-Interictal Continuum Patterns from EEG Reports to Enable Large-Scale Neurophysiology and Neurocritical Care Research. Available at SSRN. doi: 10.2139/ssrn.5489575. SSRN.**

Peer-reviewed publications (other - 8 total)

1. Westover MB. "Review of Peter Dayan & L. F. Abbott's Theoretical Neuroscience: Computational and Mathematical Modeling of Neural Systems." *Philosophical Psychology*. 2002. 15: 563-567.
2. Bianchi MT and Westover MB. (2009) "Comment on: Analysis of SPARCL trial data for reduction of future stroke risk in older stroke patients with recent stroke or TIA." *Neurology* 73(10):817-818.
3. Westover MB, Gaspard N. "Not a simple plumbing problem: Updating our understanding of delayed cerebral ischemia in aneurysmal subarachnoid hemorrhage." *J Clin Neurophysiol*. 2016 Jun;33(3):171-3. PMCID: PMC4894327. Invited review as Guest Editor for a special issue of JCN on the neurophysiology of subarachnoid hemorrhage.
4. Dreier J, Fabricius M, Cenk A, Sakowitz O, Shuttleworth C, Dohmen C, Graf R, Vajkoczy P, Helbok R, Suzuki M; Schiefecker A.; Major S, Winkler M, Kang EJ, Milakara D, Oliveira-Ferreira A, Reiffurth C, Revankar G, Sugimoto K, Dengler N, Hecht N, Foreman B, Feyen B, Kondziella D, Friberg C, Piilgaard H, Rosenthal ES, Westover MB, Maslarova A, Santos E, Hertle D, Sanchez-Porrás R, Jewell S, Balança B, Platz J, Hinzman J, Luckl J, Schoknecht K, Schöll M, Drenckhahn C, Feuerstein D, Eriksen N, Horst V, Bretz J, Jahnke P, Scheel M, Böhner G, Rostrup E, Pakkenberg B, Heinemann U, Claassen J, Carlson A, Kowoll C, Lublinsky S, Chassidim Y, Shelef I, Friedman A, Brinker G, Reiner M, Kirov S, Andrew R, Farkas E, Guerresir, E, Vatter H, Chung L, Brennan K, Lieutaud T, Marinesco S, Maas A, Sahuquillo J, Dahlem M, Richter F, Herreras O, Boutelle M, Okonkwo D, Bullock R, Witte OW, Martus P, Van den Maagdenberg A, Ferrari M, Dijkhuizen R, Shutter L, Andaluz N, Schulte A, MacVicar B, Watanabe T, Woitzik J, Lauritzen M, Strong A, Hartings J. "Recording, analysis, and interpretation of spreading depolarizations in neurointensive care: review and recommendations of the COSBID research group." *Journal of Cerebral Blood Flow and Metabolism*. 2016 Jun 17. pii: 0271678X16654496. [Epub ahead of print] PMID: 27317657.
5. Westover MB, Halford JJ, Bianchi MT. What it should mean for an algorithm to pass a statistical Turing test for detection of epileptiform discharges. *Clin Neurophysiol*. 2017 Jul;128(7):1406-1407. doi: 10.1016/j.clinph.2017.02.026. Epub 2017 Mar 16. PMID: 28495216; PMCID: PMC9359665.
6. Bianchi MT, Thomas RJ, Westover MB. An open request to epidemiologists: please stop querying self-reported sleep duration. *Sleep Med*. 2017 Jul;35:92-93. doi: 10.1016/j.sleep.2017.02.001. Epub 2017 Feb 17. PMCID: PMC9371612.

7. Singla S, Garcia GE, Rovenolt GE, Soto AL, Gilmore EJ, Hirsch LJ, Blumenfeld H, Sheth KN, Omay SB, Struck AF, Westover MB, Kim JA. "Detecting Seizures and Epileptiform Abnormalities in Acute Brain Injury." *Curr Neurol Neurosci Rep*. 2020 Jul 27;20(9):42. doi: 10.1007/s11910-020-01060-4. Review. PMC9353690.
8. Nasiri S, Ganglberger W, Sun H, Thomas RJ, Westover MB. Exploiting Labels from Multiple Experts in Automated Sleep Scoring. *Sleep*. 2023 Feb 16;zsad034. doi: 10.1093/sleep/zsad034. Epub ahead of print. PMID: 36795078.

9.

Non-Peer-reviewed Articles

1. Westover MB, O'Sullivan JA. "Towards an information theoretic framework for object recognition." IEEE 2004 International Symposium on Information Theory, Chicago, Illinois, June 27-July 2, 2004.
2. Westover MB, O'Sullivan JA. "Achievable rates for pattern recognition: Binary and Gaussian cases." IEEE 2005 International Symposium on Information Theory, Adelaide, Australia, Sept 4-9, 2005.
3. O'Sullivan JA, Singla N, and Westover MB. "Successive Refinement for Pattern Recognition," IEEE Information Theory Workshop, Punta del Este, Uruguay, Feb 2006.
4. Liddle S, Zhang R, Grover L, Khitrov M, Brown J, Cobb JP, Goldman J, Chou J, Westover MB, Yagoda D, Reisner A, "Safety Evaluation of a Medical Device Data System". International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC'12), San Diego, California, USA on August 28 - September 1, 2012.
5. Bianchi MT, Thomas RJ, Westover MB. "Response." *Sleep Med*. 2017 Aug 3. pii: S1389-9457(17)30316-7. doi: 10.1016/j.sleep.2017.07.017. [Epub ahead of print] PMID: 2884338
6. Amorim E, Zheng WL, Ghassemi MM, Aghaeavali M, Kandhare P, Karukonda V, Lee JW, Herman ST, Sivaraju A, Gaspard N, Hofmeijer J, van Putten MJAM, Sameni R, Reyna MA, Clifford GD, Westover MB. "The International Cardiac Arrest Research (I-CARE) Consortium Electroencephalography Database." *medRxiv*. 2023 Aug 28:2023.08.28.23294672. doi: 10.1101/2023.08.28.23294672. Preprint. PMID: 37693458.
7. Sun H, Ganglberger W, Westover MB, Thomas RJ. "Sleep as a Window into Brain Health: Clinical Implications from the Sleeping Brain Wave-Structure Associations." *SLEEP*. 2025 May. (Editorial).**

8.

Book Chapters

1. Westover MB, Thomas TT, Greer DM, "Delirium" in *Pocket Neurology*, ed. Westover MB, Choi E, Awad KM, and Greer DM, Lippincott Williams & Wilkins, 2010, 86-89.
2. Westover MB, Choi E, Awad KM, Greer DM, Editors. *Pocket Neurology*. Lippincott Williams & Wilkins, 2010.
3. Westover MB, Peters JM, Bromfield EB, "Seizures and Other Spells" in *Pocket Neurology*, ed. Westover MB, Choi E, Awad KM, and Greer DM, Lippincott Williams & Wilkins, 2010, 57-79.
4. Westover MB, Xia, Z, Cho TA, "Meningitis, Encephalitis, & Brain Abscesses" in *Pocket Neurology*, ed. Westover MB, Choi E, Awad KM, and Greer DM, Lippincott Williams & Wilkins, 2010, 136-149.
5. Thomas TT, Westover MB, "Rapidly Progressive Dementia" in *Pocket Neurology*, ed. Westover MB, Choi E, Awad KM, and Greer DM, Lippincott Williams & Wilkins, 2010, 89-96.
6. Kummer TT, Westover MB, "Central Nervous System Vasculitis" in *Pocket Neurology*, ed. Westover MB, Choi E, Awad KM, and Greer DM, Lippincott Williams & Wilkins, 2010, 175-177.
7. Westover MB, Awad K, "Ischemic Stroke, Muscular Dystrophy and Friedreich's Ataxia" in *MGH Cardiology Board Review*. Springer-Verlag, London, 2013.

8. Westover MB, Edlow BL, Greer DM, “Coma After Cardiac Arrest: Management and Neurological Prognostication” in MGH Cardiology Board Review. Springer-Verlag, London, 2013.
9. Westover MB, Shafi MM, “Seizures” in Pocket ICU, Lippincott Williams & Wilkins, 2013.

VI. GRANT FUNDING

Current:

09/2021 – 05/2026	Funder: NIH-NINDS (R01 AG073410) Title: Comparative Safety of Seizure Prophylaxis within the Medicare Program Role: Co-Investigator (PI: Lidia Maria Moura)
05/2021 – 04/2026	Funder: NIH-NINDS (R01 NS120947) Title: Establishing a Brain Health Index from the Sleep Electroencephalogram Role: PI
09/2022 – 08/2027	Funder: NIH-NHLBI (R01 HL161253) Title: Data-Driven Sleep Biomarkers of Brain Health, Heart Health, and Mortality Role: PI
08/2022 – 04/2027	Funder: NIH-NIA (R01 AG073598) Title: Low Neurophysiologic Resistance to Anesthetics as a Marker of Preclinical/Prodromal Alzheimer’s Disease and Neurovascular Pathology, Delirium Risk and Inattention Role: Co-Investigator (PI: Miles Berger, via Duke subcontract)
01/2023 – 12/2027	Funder: NIH-NINDS (R01 NS126282) Title: Prospective Validation of Neurophysiologic Outcome Prediction in Acute Brain Injury Role: PI
09/2023 – 05/2028	Funder: NIH-NINDS (R01 NS130119) Title: Late-Onset Unexplained Epilepsy as a Risk Factor for Dementia Role: Co-PI (with Alice D. Lam)
12/2023 – 11/2028	Funder: NIH-NINDS (R01 NS131347) Title: Comparative Effectiveness of EEG-Guided Anti-Seizure Treatment in Acute Brain Injury Role: Co-Investigator (PI: Sahar F. Zafar)
11/2023 – 10/2028	Funder: NIH-NIA (R01 AG078261) Title: Delirium Dynamics: Understanding Causes and Effects Role: Co-Investigator (PI: Eyal Kimchi, via Northwestern subcontract)
09/2023 – 09/2025	Funder: NIH-NIA (R21 AG085495) Title: Development of Novel Polysomnography-Based Digital Biomarkers to Predict Alzheimer’s Disease and Parkinson’s Disease Role: Co-Investigator (PI: Yue Leng, via UCSF subcontract)
02/2025 – 02/2026	Funder: Takeda (PO8001791544) Title: Analysis of Risk of Narcolepsy through Clinical Notes

Role: PI

Submitted:

12/2025 – 11/2030	Funder: NIH (Pending) Title: Leveraging Clinical Data and Automation to Predict Epilepsy After TBI Role: Co-Investigator (PI: Jennifer Kim)
12/2025 – 11/2030	Funder: NIH (Pending) Title: De-Biasing Effect Assessment of Seizure Effects on Neurologic Outcomes (DEBIAS) Role: Co-Investigator (PI: Adrian Dalca)
12/2025 – 11/2030	Funder: NIH-NINDS (Pending) Title: Automating Critical Care Electroencephalography Using AI Foundation Models Role: PI
12/2027 – 11/2030	Funder: NIH (Pending) Title: *Tana Mu Na?* Feasibility of Mobile EEG and Automated Intelligence to Diagnose Epilepsy in Children in Guinea-Bissau Role: Co-Investigator (PI: Farrah Mateen)
07/2025 – 06/2030	Funder: NIH-NINDS (1UG3NS143749-01) Title: Continuous Monitoring of Pain Using Facial Expressions in the ICU Environment (COMPANION) Role: Co-Investigator (PI: Hao Deng)

Completed:

2012 – 2014	Funder: American Brain Foundation Clinical Research Training Fellowship Title: Early Detection of Delayed Cerebral Ischemia After Subarachnoid Hemorrhage Using Context-Sensitive Computational EEG Role: PI
2014 – 2019	Funder: NIH-NINDS (K23 NS090900) Title: Quantitative Neurophysiological Monitoring and Control of Sedation-Analgesia in the ICU Environment Role: PI
2017 – 2019	Funder: SAGE Therapeutics Title: Automated Detection of Seizure-Spectrum Patterns and Quantitative Neurological Outcome Prediction Role: PI
2018 – 2019	Funder: Partners Healthcare Innovation Title: Artificial Intelligence-Based Seizure Detection and Classification Role: PI
2018 – 2019	Funder: MGH-MIT Grand Challenge Title: Big Data and Deep Learning for the Interictal-Ictal-Injury Continuum Role: PI

- 2018 – 2021 Funder: Glenn Foundation for Medical Research
Title: Electroencephalogram-Based Brain Age and Its Relationship with Cognitive Function and Sleep Quality
Role: PI
- 2018 – 2023 Funder: NIH-NINDS (R01 NS102190)
Title: Investigation of Sleep in the Intensive Care Unit
Role: PI
- 2019 – 2024 Funder: NIH-NINDS (R01 NS107291)
Title: Deep Learning for the Ictal-Interictal-Injury Continuum
Role: PI
- 2018 – 2023 Funder: NIH-NINDS (R01 NS102574)
Title: Multimodal Network Connectivity Architecture (MOCA) of the Brain and Its Role in Recovery of Consciousness in Comatose Cardiac Arrest Patients
Role: Co-Investigator (PI: Ona Wu)
- 2019 – 2024 Funder: NIH-NIA (R01 AG060987)
Title: Treating Hyperexcitability in Alzheimer’s Disease with Levetiracetam to Improve Brain Function and Cognition
Role: Co-Investigator (PI: Mouhsin Shafi)
- 2019 – 2024 Funder: NIH-NIA (RF1 AG064312)
Title: Integrated Motor Activity Biomarker for Risk of Alzheimer’s Dementia
Role: Co-Investigator (PI: Kun Hu)

VII. PROFESSIONAL ACTIVITIES

Editorial Service

- 2010 Editor, Pocket Neurology, Lippincott Williams & Wilkins
2016 Editor, Pocket Neurology, 2nd Edition; Wolters Kluwer
2019-2024 Editorial Board Member, Journal of Clinical Neurophysiology

Ad Hoc Reviewer

- 2010- present Neurology
2010- present JAMA Neurology (formerly Archives of Neurology)
2006- present Philosophical Psychiatry
2010- present Journal of Clinical Epidemiology
2010- present BMJ Case Reports
2010- present IEEE Transactions on Information Theory
2010- present IEEE Transactions on Signal Processing
2010- present Journal of Medical Engineering
2010- present Journal of Clinical Neurophysiology
2010- present Journal of Neuroscience
2010- present European Journal of Neurology
2010- present Critical Care Medicine
2010- present Annals of Clinical and Translational Neurology
2010- present Journal of Neural Engineering
2010- present Journal of Cerebral Blood Flow & Metabolism
2010- present Computers in Biology and Medicine

2010- present	Proceedings of the National Academy of Sciences (PNAS)
2010- present	Brain
2010- present	Epilepsy Research
2010- present	American Journal of Respiratory and Critical Care Medicine
2010- present	SLEEP

Service as Grant Reviewer

2018-2020	Ad Hoc Member, Acute Neural Injury & Epilepsy (ANIE), NIH
2021- 2025	Permanent Member, Acute Neural Injury & Epilepsy (ANIE), NIH

University Administrative Service

2011-2017	Critical Care Informatics Committee Massachusetts General Hospital
2011-2022	Neurocritical Care Clinical Review Committee Massachusetts General Hospital
2011-2022	Co-director, MGH Neuromonitoring Committee Massachusetts General Hospital
2010-2016	Member, MGH EEG Reporting System Redesign Committee, MGH
2013-2017	MGH Infusion Pump Committee Massachusetts General Hospital
2014-2022	MGH Cooling Committee Massachusetts General Hospital
2015-2018	Member, PhD Thesis Committee (Candidate: Mohammad Ghassemi), MIT
2015-2019	Member, PhD Thesis Committee (Candidate: Jing Zhi An), MIT
2017-2019	Chair, Data Safety Monitoring Board (DSMB) for R01 (PI: Ken Solt), MGH
2018-2020	Member, DSMB for R01 (PI: Alvaro Pascual Leone), BIDMC

Service to Professional Organizations

2011-2014	TRENDs (Treatment of Recurrent Nonconvulsive SeizureS) Clinical Trial Investigators Committee, Duke University / Duke Clinical Research Institute (DCRI)
2015-2017	Member at Large Critical Care EEG Monitoring Research Consortium (CCEMRC)
2016	Co-Chair, 7th International Conference on Extreme Learning Machines (ELM2016). Singapore, December 13-15, 2016.
2017-2019	Vice-Chair, Critical Care EEG Monitoring Research Consortium (CCEMRC)
2017	Co-Chair, The 8th International Conference on Extreme Learning Machines (ELM2017). Yantai, China, October 4-7, 2017.
2019-2021	Chair, Critical Care EEG Monitoring Research Consortium (CCEMRC)
2025-2029	Chair, International League Against Epilepsy (ILAE) – Commission on Big Data and AI
2025-	Committee Member, NINDS Common Data Elements (CDE) in Epilepsy Research Oversight Committee.

Community Service

2012 Christ Lutheran Nursery School, Belmont, MA	Educational Speaker
2014 Boy Scout Troop 66, Belmont, MA. Belmont, MA	Educational Speaker
2014 Butler Elementary School, Belmont, MA,	Educational Speaker
2016 Butler Elementary School, Belmont, MA,	Educational Speaker

VIII. INVITED PRESENTATIONS (SINCE 2020)

Grand Rounds:

1. “Little; Yellow; Different: Hyperbilirubinemia in preterm infants of lactating depressed female Green Monkeys receiving CACKLE-modified dairy supplements.” Neonatology Grand Rounds, Stanford University, Stanford, CA (2022)

2. "Monitoring and Closed-Loop Control of Burst Suppression." Neurology Grand Rounds, Massachusetts General Hospital, Boston, MA (2013)
3. "Continuous Brain Monitoring in Critical Care." Anesthesia, Critical Care and Pain Medicine Grand Rounds, Massachusetts General Hospital, Boston, MA (2013)
4. "Why Brains Need Computers." Neurology Grand Rounds, Massachusetts General Hospital, Boston, MA (2016)
5. "Towards Personalized Critical Care Brain Monitoring." Neurology Grand Rounds, Stanford University, Palo Alto, CA (2017)
6. "Machine Learning in Neurology." Neurology Grand Rounds, University of Wisconsin, Madison, WI (2019)
7. "Early Ischemia Detection Using EEG in Subarachnoid Hemorrhage." Neurology Grand Rounds, University Hospitals Cleveland Medical Center, Cleveland, OH (2019)
8. "Automating Clinical Neurophysiology." Neurology Grand Rounds, Stanford University, Palo Alto, CA (2019)
9. "AI in Clinical Neurophysiology." Neurology Grand Rounds, Massachusetts General Hospital, Boston, MA (2020)
10. "Covid-19 Outcome Risk Prediction." Neurology Grand Rounds, Massachusetts General Hospital, Boston, MA (2020)
11. "Automated EEG Analysis in Clinical Care." Neurology Grand Rounds, University of California San Diego, San Diego, CA (2020)
12. "Denoising the Diagnosis of Epilepsy." Neurology Grand Rounds, Einstein Medical Center Philadelphia, Philadelphia, PA (2022)
13. "Quantifying the Ictal-Interictal-Injury Continuum." Neurocritical Care Grand Rounds, University of Washington, Seattle, WA (2022)
14. "Superhuman Management of Seizures in the ICU." Neurology Grand Rounds, Yale School of Medicine, New Haven, CT (2023)
15. "Optimizing Detection and Treatment of Nonconvulsive Status Epilepticus." Geschwind Pan-Harvard Grand Rounds, Harvard Medical School, Boston, MA (2023)
16. "Big Data and AI to Achieve Superhuman Detection and Treatment of Seizures in ICU Patients." Healthcare Delivery Science Grand Rounds, Beth Israel Deaconess Medical Center / Harvard Medical School, Boston, MA (2023)
17. "Toward Optimal Treatment of Harmful Brain Activity in Critical Illness." Neurology Grand Rounds, University Hospital Cleveland Medical Center, Cleveland, OH (2024)
18. "Reducing Noise in the Diagnosis of Epilepsy." Neurology Grand Rounds, University of Texas Houston, Houston, TX (2024)
19. "Reducing Noise in the Diagnosis of Epilepsy." Neurology Grand Rounds, University of Pennsylvania, Philadelphia, PA (2024)
20. "Reducing Diagnostic Noise in Neurology: Lessons from Epilepsy." Neurology Grand Rounds, Beth Israel Deaconess Medical Center / Harvard Medical School, Boston, MA (2025)
21. "Reducing Noise in Neurology: Lessons from Epilepsy." Neurology Grand Rounds, University of Michigan, Ann Arbor, MI (2025)

Local Invited Presentations:

1. "What is monkey cortex layer four for?." Neuroanatomy & Neurobiology Department, Washington University School of Medicine, St. Louis, MO (2002)
2. "EEG Patterns in the Neurological ICU." MGH Neurosciences ICU, Boston, MA (2010)
3. "Should Statins Be Avoided After Intracerebral Hemorrhage? Implications of SPARCL." MGH Stroke Research Center, Boston, MA (2010)
4. "Fundamentals of ICU EEG Monitoring." Nursing Skills Day, MGH Neurosciences ICU, Boston, MA (2012)

5. "Status Epilepticus and the Ictal–Interictal Continuum: Introduction to ICU EEG." MGH Critical Care Center, Boston, MA (2012)
6. "Neurological Prognostication in Coma After Cardiac Arrest." MGH Critical Care Center, Boston, MA (2013)
7. "Monitoring and Closed-Loop Control of Burst Suppression." Department of Brain & Cognitive Sciences, MIT, Cambridge, MA (2014)
8. "EEG-Related Skin Lesions: Statistical Process Monitoring & Risk Reduction." MGHfC Quality & Safety Operations Lead Team Meeting, Boston, MA (2014)
9. "Fundamentals of Continuous EEG Monitoring in Neurocritical Care; Quantitative Clinical EEG Fundamentals." MGH Neurosciences ICU & BWH/MGH Clinical Neurophysiology Fellows Didactic Conference, Boston, MA (2014)
10. "Automated Surveillance for Ventilator-Associated Pneumonia." MGH Infection Control Committee, Boston, MA (2015)
11. "The Role of Continuous EEG in Coma After Cardiac Arrest: Prognostic Value." MGH EEG Technologist Training Meeting, Boston, MA (2015)
12. "Quantitative Electroencephalography." Epilepsy Fellows Training Conference, MGH Neurology, Boston, MA (2017)
13. "CDAC: Bringing Big Clinical Data to Life." MGH Neurosurgery Faculty Meeting, Boston, MA (2018)
14. "Landmark Project #4: Clinical Time Series Data." Partners HealthCare EDMT Meeting, Boston, MA (2018)
15. "Validating the Brain Age Index: Modifying Brain Aging Through Exercise." McCance Center for Brain Health Faculty Meeting, Boston, MA (2019)
16. "Automatic Detection of Nonconvulsive Seizures." BIDMC Research Day, Boston, MA (2023)

National and Regional Meetings:

1. "Automatic Detection of Beta Amyloid in Digital Images Using Visual Texture." Alzheimer's Disease Research Center, Washington University, St. Louis, MO (2005)
2. "Early Detection of Delayed Cerebral Ischemia Using EEG." Greater Boston Epilepsy Society, Boston, MA (2015)
3. "Polysomnographic Signatures of Brain Age." Massachusetts Sleep Society, Westborough, MA (2017)
4. "Artificially Intelligent Brain Monitoring." World Medical Innovation Forum, Boston, MA (2019)
5. "Expert-Level Automatic Detection of Epileptiform Discharges." American Neurological Association, St. Louis, MO (2019)
6. "EEG Monitoring for Early Detection of Cerebral Ischemia." American Clinical Neurophysiology Society Annual Meeting, New Orleans, LA (2020)
7. "Automating Clinical Neurophysiology." Duke Comprehensive Epilepsy Center Quarterly Research Symposium, Durham, NC (2020)
8. "Big Data and Machine Learning to Advance Status Epilepticus Research." American Neurological Association, Virtual Meeting (2020)
9. "Automated Detection of Seizures and Ictal–Interictal Continuum Patterns Using Deep Learning." American Clinical Neurophysiology Society Annual Meeting, Virtual (2021)
10. "Sleep as a Window into Brain Health." Harvard Sleep Grand Rounds, Virtual (2021)
11. "Avoiding Pitfalls in EEG Interpretation." American Academy of Neurology Annual Meeting, Seattle, WA (2022)
12. "Artificial Intelligence in Sleep Medicine." American Neurological Association Annual Meeting, Chicago, IL (2022)
13. "Addressing Problems in Critical Care Neurophysiology Using Machine Learning." Machine Learning in Medicine Seminar Series, Carnegie Mellon University, Pittsburgh, PA (2022)
14. "Epileptiform and Non-Epileptiform Findings in ICU Electroencephalography." American Academy of Neurology Annual Meeting, Boston, MA (2023)

15. "Neurophysiological Grading of Encephalopathy." American Academy of Neurology Annual Meeting, Boston, MA (2023)
16. "Identifying and Overcoming Barriers for Clinical Use of Artificial Intelligence in Sleep Medicine." American Academy of Sleep Medicine Annual Meeting, Indianapolis, IN (2023)
17. "Optimizing Treatment of Nonconvulsive Status Epilepticus." American Epilepsy Society Annual Meeting, Orlando, FL (2023)
18. "Collaborating with Industry." American Clinical Neurophysiology Society Annual Meeting, Orlando, FL (2024)
19. "Quantitative EEG." American Clinical Neurophysiology Society Annual Meeting, Orlando, FL (2024)
20. "The EEG Map of Everything (MOE)." Critical Care EEG Monitoring Research Consortium Meeting, Orlando, FL (2024)
21. "AI in ICU EEG." American Epilepsy Society Annual Meeting, Los Angeles, CA (2024)
- 22.

International Meetings:

1. "Opportunities for Intelligent Computing in Clinical Neurophysiology." Nanyang Technical University Symposium on Novel Technologies for Clinical Practice in Neurology, Singapore (2012)
2. "Real-Time Measurement and Closed-Loop Control of Burst Suppression." IEEE Engineering in Medicine and Biology Society Conference, Osaka, Japan (2013)
3. "Optimizing the Treatment of Refractory Status Epilepticus." Networks in Epilepsy Conference, Beijing, China (2015)
4. "Engineering in Neurocritical Care: Modeling & Control of Refractory Status Epilepticus." International Symposium on EEG Signal Analysis, Xi'an, China (2015)
5. "Quantitative Electroencephalography for Detection of Seizures and Ischemia." Beijing ICU Neuroelectrophysiological Monitoring Forum, Beijing, China (2015)
6. "Why Brains Need Computers: How Machine Learning and Big Data Can Help Neurology." Extreme Learning Machines Annual Conference, Singapore (2016)
7. "Machine Learning in Neurocritical Care." Northwestern University International Forum on Big Data in Healthcare, Xi'an, China (2017)
8. "Quantitative EEG: Detecting Ischemia and Seizures." International Congress of Clinical Neurophysiology, Washington, DC (2018)
9. "Removing the Confusion from Toxic-Metabolic Encephalopathy." University of Twente Mini-Symposium, Enschede, Netherlands (2018)
10. "Mapping the Ictal–Interictal–Injury Continuum." Neurocritical Care Society Annual Meeting, Vancouver, Canada (2019)
11. "Superhuman Detection of Seizures, Rhythmic and Periodic Patterns in ICU EEG." International Congress on Clinical Neurophysiology (ICCN), Geneva, Switzerland (2022)
12. "Quantifying the Ictal–Interictal Continuum." International Conference on Technology and Analysis for Seizures (ICTALS), Bern, Switzerland (2022)
13. "Denoising the Diagnosis of Epilepsy." AI in Epilepsy Conference, Breckenridge, CO (2025)
14. "MORGOTH: Automating the Interpretation of Clinical Neurophysiology Tests." International Conference on Technology and Analysis for Seizures (ICTALS), Montreal, Canada (2025)
15. "Data-Driven Biomarkers for Brain Health from Sleep." SLEEP 2025 Annual Meeting, Seattle, WA (2025)
16. "Artificial Intelligence in Neurology: Promise and Peril." Donald Baxter Lecture, Montreal Neurological Institute, Montreal, Canada (2025)

IX. TEACHING

Teaching during Stanford appointment:

None.

Teaching prior to Stanford:

Courses

2001–2002	Instructor for Computational Neuroscience, Washington University School of Medicine, Department of Anatomy and Neurobiology.
2003	Instructor for Detection and Estimation Theory, Washington University, Department of Electrical and Systems Engineering.
2004	Instructor for Information Theory, Washington University, Department of Electrical and Systems Engineering.
2008–2009	Instructor for Neuroanatomy Laboratory, Harvard Medical School.
2012–2025	Lecturer for Medical Decision Analysis and Diagnostic Test Interpretation (HST-192.0), Harvard–MIT Program in Health Sciences and Technology.
2013–2017	Course Director and Lecturer for Medical Decision Analysis and Diagnostic Test Interpretation (HST-192.0), Harvard–MIT Program in Health Sciences and Technology.

Tutorials and Lectures

10/2009	Granger Causality Analysis Techniques for EEG Data. Clinical Neurophysiology Research Group, MGH, Boston, MA.
5/2010	Synchronization Likelihood for Functional Connectivity Analysis of EEG Data. Clinical Neurophysiology Research Group, MGH, Boston, MA.
7/2008	Probability in Medical Reasoning. Neurology Residents, Brigham and Women's Hospital, Boston, MA.
9/2009	Invited lecturer, Bayes' Rule for Physicians. Partners Neurology, MGH/BWH, Boston, MA.
6/2010	EEG Alphabet Soup: GCSE, NCSE, PLEDs, BIPLEDs, GPEDs, & DEADs. Partners Neurology, MGH/BWH, Boston, MA.
8/2011	Status Epilepticus: Diagnosis and Management. Partners Neurology, MGH/BWH, Boston, MA.
9/2012	Introduction to ICU EEG: Seizures, Spikes, PEDs, Ischemia, and Coma. Partners Neurology, MGH/BWH, Boston, MA.
7/2018	Quantitative Electroencephalography. Neurophysiology and Epilepsy Fellows, MGH Epilepsy Division, Boston, MA.
7/2019	Recognizing Seizures and Ischemia in Critical Care EEG Monitoring. Neurophysiology and Epilepsy Fellows, MGH Epilepsy Division, Boston, MA.
6/2020	Emergency Electroencephalography. Emergency Medicine Residents, Massachusetts General Hospital, Boston, MA.
7/2020	Introduction to Spectrograms in ICU EEG. Neurophysiology and Epilepsy Fellows, MGH Epilepsy Division, Boston, MA.

Clinical Supervisory and Training

2012–	Supervisor, MGH Adult Epilepsy Clinic (½-day session per week).
2012–	Attending Physician, MGH Inpatient Neurology Service (2-week block per year), supervising neurology residents and Harvard medical students.
2012–	Attending Physician, MGH General Neurology Consult Service (2-week block per year), supervising neurology residents.
2012–	Director, MGH Critical Care EEG Monitoring Service (≈10 weeks per year), supervising Epilepsy and Clinical Neurophysiology Fellows.
2012–	Attending, MGH Epilepsy Monitoring Unit (≈6 weeks per year), supervising Epilepsy and Clinical Neurophysiology Fellows.
2012–	Faculty, Outpatient EEG Reading Service (≈2 days per month), supervising Epilepsy and Clinical Neurophysiology Fellows.

Research Supervisory and Training Responsibilities

2012– Supervisor of postdoctoral research fellows (average 5–10 fellows per year), Massachusetts General Hospital.

2012– Supervisor of graduate students (average 2–4 per year), Massachusetts General Hospital and MIT.

X. TRAINEES

Postdoctoral Fellows

2013-2014 Jing Jin, PhD from NTU in Singapore. postdoctoral fellow at MGH

2013-2014 Zhang Rui, PhD, visiting scholar from NTU in Singapore, mentoring for 2 months through present. Professor of Mathematics at Northwestern University, Xi'An, China

2014-2018 Sunil Belur Nagaraj, PhD, electrical engineering postdoctoral fellow. an Assistant Professor at University of Groningen, Department of Clinical Pharmacology, Groningen, Netherlands

2014- Lauren McClain, PhD. Research Fellow. a PhD student at Fordham University, New York

2019- Eline Oppersma, PhD. Visiting scholar. Assistant Professor of Technical Medicine at University of Twente, Netherlands

2019- WenDong Ge, PhD. Postdoctoral fellow.

2015–present Sahar Zafar, MD (K23).

2014–2020 Jennifer Kim, MD, PhD (K23).

2014–2019 Edilberto Amorim, MD (K23).

2019–present Marta Priscila Bento Fernandes, PhD.

2020–2022 Rajesh Amerineni, PhD.

2019–2022 Wendong Ge, PhD.

2019–2022 Weilong Zheng, PhD.

2022–2024 Yalda Amidi, PhD.

2021–2023 Samaneh Nasiri, PhD.

2019–present Wolfgang Ganglberger, PhD.

2024–present Shadi Sartipi, PhD.

2024–present Alexandra Tautan, PhD.

2024–present Ayush Tripathi, PhD.

2024–present Samuel Waters, PhD.

2024–present ChenXi Sun, PhD.

2024–present Charlotte Maschke, PhD.

2024–present Bruce Nearing, PhD.

2024–present Ayan Mitra, PhD.

2016–2018 Sungtae An, MS / PhD.

2016–present Haoqi Sun, MD.

2013–present Jing Jin, PhD.

Graduate Students

2014- Mohammed Ghassemi, MIT PhD candidate (now graduated), mentorship resulted in Advanced Multimodal Neuroimaging Training Program (AMNTP) fellowship award, 3 published manuscripts. CEO of a startup company, Ghamut

2014- Jingzhi An, PhD candidate at Harvard/MIT HST program.

2014-2019 Moura, Junior Valdery, MS (computer science), programmer analyst / research assistant at MGH. Project: establishing a database for the Clinical Data Animation Center (CDAC). 3 peer-reviewed publications to d. Project Manager for Landmark Project #4 at Partners Healthcare

2014-2017	Apeksha Shenoy, MS (electrical engineering), programmer analyst / research assistant. a Senior Consultant at Boston Strategic Partners
2015-	Elham Bagheri, MS (electrical and systems engineering). Nanyang Technical University (NTU), Singapore, PhD candidate, co-supervised with Prof.
2015-	John Thomas, MS (electrical and systems engineering). Nanyang Technical University (NTU), Singapore, PhD candidate, co-supervised with Prof.
2015-2016	Siddharth Biswal, MS, computer programmer / research assistant. a PhD candidate at Georgia Institute of Technology
2015-	Gladia Hotan, PhD candidate at MIT. Mentorship resulted in 2 publications to date.
2015-2016	Ellis Wickering, MS, University of Twente, Netherlands. working as an computer programmer at Prolira
2016-	Jefferson T, Oliva, PhD candidate in University of Sao Paulo, Brazil, Institute of Mathematics and Computer Sciences.
2016-	Jord Vink, MS, University of Twente, Netherlands, 3 month research internship with me at MGH, co-mentored with Dr. working at Brain Science Tools, Netherlands
2016-	Wei-Hung Sun, MD, masters student in the Harvard Biomedical Informatics program. a PhD candidate at MIT in computer science
2016-	Manohar Ghanta, MS (computer science). Computer engineer / research assistant.
2016-	Yu-Ping Shao, MS (computer science). Computer engineer / data scientist / research assistant.
2016-	Sungtae An, MS (computer science). Georgia Tech PhD candidate, co-supervised with Dr.
2016-	Kunal Malhotra, MS (computer science). Georgia Tech (GAT) PhD candidate, co-supervised with Dr.
2016-	Haoqi Sun, MD (computer science). Nanyang Technical University (NTU), Singapore, PhD candidate, co-supervised with Prof.
2016-	Senan Ebrahim, MD, PhD student in the Harvard Medical School HST program; co-supervised with Dr.
2016-2017	Sophia Bechek, research assistant. a PhD candidate at Stanford
2016-2018	Balaji Goparaju, MSc. a Data Scientist at Novartis Institutes for BioMedical Research (NIBR)
2016-	Jiang Ling Song, MS (mathematics), Northwestern University, Xian, China.
2017	Dennis Rebergen, Research internship.
2017-	Emile d'Angremont, Research internship.
2017-2018	Peiyun Ni, MD/PhD candidate at Harvard Medical School / MIT. Project: Antiepileptic drug treatment after an unprovoked first seizure: A decision analysis.
2016-2017	Nishant Sinha, PhD candidate (now graduated) at NTU, Singapore, department of computer science.
2018-	Qiang Li, PhD candidate in Mathematics, Northwestern University, Xi'An, China.
2018-2019	Kicky van Leeuwen, Masters student in Technical Medicine from U. Twente, Netherlands. a Medical Machine Learning Scientist at Medix
2018-	Michael Leone, MSc.
2014-2020	Jingzhi An, PhD.
2014-2019	Mohammad Ghassemi, PhD.
2016-2017	Jefferson T. Oliva, PhD.
2018-2020	Qiang Li, PhD.
2024-present	Arnav Gupta.
2021-2025	Thijs Nassi, MSc.

Medical Students / Residents / Fellows

	Mentorship for 4 months (2 summers), resulted in 4 scientific publications.
2011-2012	Sarah Wahlster, MD, Neurology resident. Assistant Professor of Neurology at University of Washington
2011-2012	Craig Williamson, MD, Neurology resident, Mentorship for 1 year, resulted in 1 published scientific manuscript. Assistant Professor of Neurology at University of Michigan
2013-2014	Lidia Moura, MD, Epilepsy Fellow at MGH. Two abstracts, 1 published scientific manuscript. Assistant Professor at Harvard Medical School
2013-2014	Marcus Ng, MD, Epilepsy Fellow at MGH. One published manuscript. Assistant Professor at University of Manitoba, Winnipeg, Canada
2013-2014	Deirdre O'Rourke, MD, Epilepsy Fellow at MGH, mentorship for 1 year, resulted in 1 abstract and 1 published manuscript. Neurologist at Connolly Hospital Blanchardstown/ Beaumont Hospital Dublin
2013-2014	Sandipan Pati, MD, Epilepsy Fellow at MGH. 1 abstract, 2 published scientific manuscripts. Assistant Professor of Neurology at UAB Birmingham
2013-2014	Aaron Berkowitz, MD, PhD, neurology resident at BWH/MGH/Partners. Assistant Professor of Neurology at Harvard Medical School / Brigham and Women's hospital
2014	Vincent Alvarez, MD, visiting Research Fellow from Universite de Lausanne (Switzerland). 5 published manuscripts. an Attending Neurologist at Hopital du Valais, Sion, Switzerland
2014	Sanjay Menon, MD, Neurosciences ICU Fellow at MGH, mentorship for 6 months.
2014	Ammar Kheder, MD, Epilepsy Fellow at MGH, mentorship for 1 year. an Assistant Professor Of Clinical Neurology at U
2014-	Aaron Struck, MD, Epilepsy Fellow at MGH. 2 abstracts, 8 published manuscripts. Assistant Professor of Neurology at University of Wisconsin
2014-2016	Gina Deck, MD, Epilepsy Fellow at MGH, mentorship for 1 year. an Assistant Professor of neurology at The Warren Alpert Medical School of Brown University
2014-	Edilberto Amorim, MD, neurointensive care fellow at MGH. Research on the prognostic value quantification of EEG reactivity in coma in patients with anoxic brain injury.
2014-	Jennifer Kim, MD, PhD, neurology resident at MGH/BWH. Project: "Identifying EEG Hallmarks for Increased Seizure Risk in SAH: A Quantitative EEG Analysis." Starting in July 2019, will be Assistant Professor of Neuro.
2014-2017	Katherine O'Connor, Project: Predicting delayed cerebral ischemia in patients with aneurysmal subarachnoid hemorrhage.
2015-	Sahar Zafar, MD, neurointensive care fellow at MGH. Research on the impact of pathological brain activity and its treatments on neurological outcome in ICU patients.
2015-2017	Carlos Muniz, MD. Epilepsy / Neurophysiology Fellow. a practicing neurologist in Towson, Maryland
2016-	Ilya Toloch, MD, PhD, neurology resident at BWH/MGH/Partners.
2016-2016	Eva Postma, MD, University of Twente, Netherlands, visiting scholar for 6 months in my laboratory, co-mentored by Dr.
2016-2017	Jia Jian, PhD (mathematics), Northwestern University (NWU), Xian, China. an Associate Professor of Mathematics at NWU
2016	Honghui Zhang, PhD (mathematics), Northwestern University, Xian, China.

2014-2018	Durga Jonnalagadda, MD, Projects: Diagnosing severity of delirium in hospitalized patients from EEG recordings.
2017-	Aline Herlopian, MD. Fellow in Clinical Neurophysiology and Epilepsy at Harvard Medical School, MGH. Projects: Characterizing EEG patterns in patients with CAR-T cell encephalopathy, and Measuring expert inter-rater a. Assistant Professor of Neurology at Yale
2017-	Hassan Aboul Nour, MD. Project: Estimating the effect of exposure to inadvertent pharmacological burst suppression during critical illness on neurologic outcomes.
2017-2018	Elahe Bordbar, MD. Research Fellow. a Resident in Neurology at Temple University
2017-	Muhammad Edhi, MD. Research Fellow.
2017-	Farrukh Javed, MD. Research Fellow. a Neurology Resident at University of Virginia
2017-	Solomon Kassa, MD. Research Fellow.
2017-2019	Maryum Shoukat, MD. Research Fellow.
2017-	Mohammad Tabaeizadeh, MD. Research Fellow.
2017-2018	Erik Bao, MD candidate in the Harvard/MIT Health Sciences & Technology.
2017-2018	Monica (Ling-Ya) Chao, MD candidate in Harvard HST program.
2018-	Kendrick Shaw, MD, PhD. Anesthesia Resident and incoming faculty (Instructor) at Harvard Medical School / MGH. Project: Predicting delirium using electronic health record data.
2018-	Chris McGray, MD, PhD. Fellow in Epilepsy and Clinical Neurophysiology.
2018-	Sophia Ryan, MD. Harvard Medical School/CRICO Patient Safety and Quality Fellow at Harvard Medical School.
2018-	Benjamin Grannan, MD. Resident in Neurosurgery at Harvard Medical School / MGH. Project: Deep learning analysis of CT images to detect cases in need of surgical intervention.
2019-	Eliezer Sternberg, MD. Fellow in Epilepsy and Clinical Neurophysiology.
2014–2015	Elliott Akama-Garren.
2014–2020	Jennifer Kim, MD, PhD.
2018–present	Eyal Kimchi, MD, PhD (K23).
2022–present	Daniel Goldenholz, MD, PhD (K23).
2022–present	Peter Hadar, MD, PhD.
2022–2025	Wan Yee Kong, MD, MSc.
2022–present	James Lucarelli, MD, PhD.
2023–present	Callison Alcott, MD, PhD.
2011–2012	Sarah Wahlster, MD.
2011–2012	Craig Williamson, MD.
2013–2014	Lidia Moura, MD.
2013–2014	Marcus Ng, MD.
2013–2014	Patrick Chen, MD.
2013–2014	Deirdre O’Rourke, MD.
2013–2014	Sandipan Pati, MD.
2013–2014	Ammar Kheder, MD.
2013–2014	Aaron Berkowitz, MD, PhD.
2017–2019	Luis Paixao, MD.
2017–present	Aline Herlopian, MD.
2020–2021	Nitish Harid, MD.

Undergraduate Students

High School Students

2018
2016–2017
2023–2024

Alek Westover, High school student.
Sophia Bechek.
Samvrit Rao.